

Troubleshooting Control Circuits

Edited by Mike Eby

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Systematic methodology is the key to quickly and effectively troubleshooting control circuits. Even the most experienced troubleshooter must rely on a systematic troubleshooting process to solve problems on today's complex control circuits. At a high level, a good troubleshooting process is simple. First, you must investigate the symptoms. Then, try to identify the possible causes. The next step is

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Even the most experienced troubleshooter must rely on a systematic troubleshooting process to solve problems on today's complex control circuits. At a high level, a good troubleshooting process is simple. First, you must investigate the symptoms. Then, try to identify the possible causes. The next step is to test the system and verify possible causes. After correcting the problem follow through by monitoring the operation to make sure you've pinpointed the root cause, and completing any required documentation. Let's take a closer look at each of these steps.

Investigate the symptoms. Make sure you understand the system. Pull any available documentation, whether online or hardcopy. Look for schematics and piping and instrumentation diagrams, as well as loop sheets. Talk to the operators and anyone else familiar with the operation. Look up operations and maintenance records and control and configuration parameters. Some of this information may be available from the PLC or DCS or other online databases.



Because you often won't know where the problem lies, keep the big picture in mind. Start by breaking down even the most complex system into the following five elements:

- Process controller—most often involving a microprocessor.
- Input field devices—sensors of some type that monitor the process.
- Output field devices— drives, valves, and alarms that receive a command signal from a control element.
- Connectivity elements—wires, cables, and buses.
- Process material(s).

Don't forget the sixth element: the people who can affect the process and its control system.

Since the wiring and the inputs and outputs (I/Os) are the most vulnerable elements in a system, you'll want to examine them first. As you talk to people and review information, look for a reoccurrence or pattern. If you see a pattern, is it related to shift changes, process changes, or any other reoccurring event? Use your judgment on when to quit gathering information, but make sure the data displayed by the human machine interface (HMI) match what the operator tells you.

Understand everything the operator did in response to the problem. Walk down the system or process to make sure the field conditions match those reported by the operator HMI.

Identify possible causes. Analyze the system with an open mind, systematically eliminating components and functional elements from the overall process as unlikely trouble spots. Start by following the logic through from input to output. What happens in the cause-and-effect chain? Compare the current symptoms with the action that the specified decision logic or control algorithm should produce. As you eliminate some process elements as possible causes, you can also start building and prioritizing your list of most likely causes, keeping in mind that you'll want to test the system to eliminate these possibilities.

You can usually eliminate simultaneous, unrelated problems as being too unlikely. If you can link a problem to one likely cause, do so. At this stage, don't look for interrelated, multiple causes. Your first priority should be to get the operation back up and running. Tackle complex situations after a quick fix gets things going. Just don't forget to use your company's work procedures to highlight the open job. Operations people sometimes confuse a quick fix with a problem solution, so be very clear that your fix is temporary.

As you prioritize possible causes, go back to your sources of information. Maintenance records can help you decide that one component has been much more trouble-prone than another. For example, construction work in the area might lead you to suspect damaged cabling rather than an I/O board failure, because cabling running through the plant is more likely to suffer damage than is an I/O board inside a cabinet.

Test possible causes. When you've narrowed your probable cause list down to a manageable size, you can begin testing. Once the process is back up and running, first do those tests that don't interrupt operations. Quick and easy tests can save you time in eliminating potential causes, so do those early in your troubleshooting. In many cases you need to look, listen, or feel specific components. When working around or with energized equipment, don't take chances with safety. In all cases, follow established and required safety procedures.

As stated before, inputs and outputs are usually the first place you should look for problems. Most inputs and outputs fall into one of two categories: discrete devices with two states (on or off), or analog devices that can send and receive continuously varying signals.

Common discrete devices include limit switches, solenoid valves, indicators, and alarms. When PLCs send signals to a master PLC or DCS, they count as discrete devices. Common analog devices include resistance temperature devices; thermocouples; transmitters for pressure, level, temperature, and flow; valves; analytical field devices like pH sensors; and variable speed drives.

Discrete field devices typically use low-voltage DC. A variation in these voltages usually indicates a problem. Some drift is acceptable, but anything more than 5% to 10% in either direction, at either end of the range, calls for a close look.

Use a scope to check a discrete signal. Rise and fall times that aren't instantaneous usually indicate a fault in the sensor itself, which can typically be attributed to sticking contacts in a mechanical switch or an impending failure in a solid-state device. High signals that aren't flat usually indicate loose ground connections, ground loops, or improper shield connections. Low signals that aren't flat are often noisier than the high signals and usually indicate a grounding or shield problem. Noisy low signals can also indicate an improperly wired field device.

If a measurement suddenly dips to a minimum or maximum, you've most likely got a sensor, wiring, or other I/O problem that should be relatively easy to find. Often the best place to check is the field termination assembly because you can divide a process loop in half.

More gradual changes could indicate much more complex and hard-to-pin down problems like a change in valve stiction (static friction), a subtle change in the process materials, or a drift in instrument calibration. Your job will be much easier if you're working with a DCS because you can pull up the history of each signal loop and look for changes over time.

Sometimes the only way to test a circuit is to see how the system reacts to a manual input. When working with PLCs you're forcing the contacts. When working with continuous process control loops, you're bumping the system. If you can't manually force the system to respond to your input, you probably have a problem with the outputs. If the outputs respond properly to manual inputs, you can probably eliminate outputs and look more closely at the field transmitters, proximity switches, and other related input devices.

Be careful testing the process this way. Forcing contacts, adjusting timers and counters, changing set points, and tinkering with loop tuning parameters or the control program is risky business that can have disastrous results.

Coordinate closely with the process operator. Be sure you know what limits the process can tolerate so you don't destabilize the system or crash it.

Follow through. Follow through with careful replacement of faulty parts, a period of monitoring the operation, and documentation of what you did according to your plant's requirements. If your action was a quick fix to get equipment up and running, follow your plant's root-cause analysis procedure to get to the bottom of the problem.

All the sophisticated equipment and software in the world is useless if the troubleshooters who use it fail to follow a systemic process and make full use of the tools available. Take the time to understand what you're doing. Don't be afraid to ask for training if you need it. Then conduct your troubleshooting methodically all the way through to root cause and you'll have the respect of management and your peers.

<http://m.ecmweb.com/contractor/troubleshooting-control-circuits>

Troubleshooting Your VFDs

[Stan Turkel](#)

Stan Turkel

Aug 01, 2000

Variable-frequency drives (VFDs), also known as adjustable-speed drives (ASDs) have become the preferred method of controlling speed to meet load requirements. The most common drives use a pulse width modulation (PWM) design, which is affordable, reliable, and cost effective for most applications. While simple in their design, they can give you problems when it comes to taking operational and troubleshooting

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Aug 1, 2000 12:00 PM, By Stan Turkel

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While simple in their design, they can give you problems when it comes to taking operational and troubleshooting measurements. Knowing what measurements to take will save you time and money. You'll need this type of testing for troubleshooting and diagnosing a defective unit as well as when performing routine maintenance.

Where do you start taking measurements? The four main aspects for testing a drive system are the building's power supply, the drive unit itself, the motor, and the load. Each item in a drive system works together and becomes the entire drive application. This creates a potential problem: Any one item can shut down the drive. Knowing what measurements to take at each section is critical for the troubleshooting and maintenance of the drive.

The facility's power supply. In today's power-hungry society, it's getting hard to obtain a good source of clean non-distorted power. Over and under voltage conditions greater than plus-or-minus 10% will trip most drives. A voltage unbalance between phases of 3% to 5% can cause tripping of the drive's overload fault protection device.

With the drive in operation and carrying a load, measure the incoming line voltage at the input side of the drive itself. Using safety precautions, measure the incoming line voltage between A-B Phase, B-C Phase, C-A Phase. Sometimes, you may want to also measure A, B, and C, to ground. What you want to look for is over and under voltage conditions as well as unbalance between the phases. It's important to take the above readings during peak loads on the drive as well as during off-hours to see if you're experiencing voltage swings as a result of the dynamics of your facility. Then, take a current reading of each of the three phases on the line side of the drive. Again, look for any unbalance between phases.

The following formula will help you calculate the percentage of unbalance between phases.

Percent Voltage Unbalance = Maximum deviation from the average voltage x 100 / Average voltage

Taking readings in the VFD. There are several measurements you should take at the drive that involve getting into close quarters while the drive is in operation. Use caution when testing a powered drive unit. Remember to adhere to all safety procedures while taking these measurements.

PWM type drives take incoming AC line voltage and rectify this to a constant DC voltage that is then supplied to the switching, or inverter section, to create an adjustable alternating frequency and variable-voltage source to the motor. Measure DC bus voltage in the drive for over and under voltage conditions, which can be generated by line power changes or load regenerative conditions.

There are two important DC bus voltage measurements you should take. The first is the actual DC bus voltage, which should be equal to the line-side peak voltage (rms voltage x 1.41). Once the capacitors are charged, the reading should remain constant. On a 480V system, the DC bus voltage will be about 676VDC. You should take the second DC bus voltage measurement to determine the amount of AC ripple found on the DC bus. This reading helps pinpoint capacitor breakdown and reduced filtering of the DC bus, which can cause current trips.

DC bus voltage measurements. With the drive in operation and carrying a load, take a voltage reading at the DC bus. You take this measurement at the connections to the drive capacitor or capacitor bank. (See drive manual for exact location). Set the meter on DC volts and measure the positive and negative sides of the DC bus. This should be equal to the line voltage x 1.41.

Now, remove the meter from the circuit and set it on AC volts and take the same measurement. The meter should show very low AC voltage ripple, as this is a filtered DC source. You should discuss readings above 5VAC with the drive manufacturer, as this may indicate a possible breakdown of the capacitor filtering.

All drives maintain a constant volts-per-Hertz ratio to the motor. This ratio is kept constant, regardless at what speed the drive operates. Thus, as the frequency (Hz) changes the motor speed, so does the voltage. The only exception to this rule comes with flux vector drives. These types of drives may change the ratio, depending on special torque requirements. One of the most effective ways of troubleshooting a drive is to verify that the volts-per-Hertz ratio is being maintained at different speed settings.

Drive output volts-per-Hertz ratio measurement. Using safety precautions, set the analog meter for the maximum AC volts. With the drive running at full speed (60 Hz), measure the voltage to the motor at the drive motor terminals. For example, a 460V motor operating at 60 Hz should have a ratio of 7.6V applied to the motor for every Hertz applied. The voltage should be equal to the nameplate voltage for the motor.

Now, set the drive to 50% speed (30 Hz), and take the same motor terminal voltage reading. It should now be half of the last reading, or 230V for a 460V motor. Then, adjust the drive to 25% speed (15 Hz), and the motor voltage should now be 25% of the full voltage reading, or 115V.

What about leakage current? Drive problems can also appear if the leakage current from the drive's power transistors is excessive. A transistor does not actually open up like a mechanical switch when it turns off, it just reduces the amount of current it lets through. Sometimes a transistor that starts to become defective will show signs of excessive leakage current when turned off.

Transistor leakage current measurement. With the drive energized and a run command given, set the drive to zero speed (0 Hz) and measure the voltage to the motor between phases. In this state, the drive should not be firing any of the transistors, and there should be 40V or less leakage, depending on the manufacturer. You should discuss voltages above 60V with the manufacturer. This higher reading may indicate a pending transistor failure.

The motor. Even though you took the voltage and current readings at the motor terminals in the drive itself, you will also want to take the same measurements at the motor. Your meter should be the same as used at the drive. The analog meter gives you smoothed readings and should match the expected volts-per-Hertz ratio.

Essentially, the voltage and current values should be the same. A voltage drop or poor connections may be cause for concern. Also check for any unusual vibrations (vibration is usually a sign of excess bearing wear).

Taking temperature readings. We all know many electrical failures are the result of excessive heat, which breaks down insulation of conductors and windings. Temperature readings of motors, conductors, and heatsinks of electronic components are valuable diagnostic measurements, and you should consider them part of a yearly

maintenance program. The key is to be consistent in the location of the readings, and to take them under similar loading conditions. There is much information available from the vendors of temperature probes and meters.

The load. If it turns, then it must be okay, right? Depending on your application, there are still a couple of things you may consider measuring. If you're concerned about speed regulation for a process, then a tachometer reading at full load conditions is in order. Always verify rotation direction. I have uncovered instances where equipment was simply operating backward; and unknown to those using the equipment.

The torque stresses the shaft and drive train components. Having started up hundreds of drive applications over the years, I still can't understand why folks need to ramp up these large loads in 10 sec or faster. It just makes sense to adjust the ramp speed of a load to as long as possible to reduce all kinds of stresses; both mechanical and electrical.

There are two important areas not included in this article: harmonics and overvoltage reflections at the motor windings. Although harmonics is a concern, it's not normally associated with the tripping of drives and shutdowns. You can usually control overvoltage reflections at the motor windings by keeping the motor leads as short as possible. Several motor manufacturers can provide special windings to withstand such overvoltage conditions.

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Sidebar: What Type of Testing Meter Should You Use?

You can use any clamp-on true digital multimeter (DMM) Cat. 3 rated for testing motor drives. The best recommendation is a clamp-on 1000V Cat. 3 unit. A true rms clamp-on DMM, or an AC-only clamp-on attachment for a DMM will work. If your meter is not rated Cat. 3, then don't use it.

Sidebar: Use an Analog Meter to Test the Load Side Output of the Drive

You would expect to use a true rms DMM to test the load side output of the drive. But for this measurement, the DMM might give you an incorrect reading. Why?

The output of a VFD is a series of very high transmittal oscillating positive and negative voltages. Using IGBT transistors, the oscillations approach 20 kHz, which vary in base frequency and duration (width of pulse, hence the name pulse width modulated drives). A digital multimeter takes many samples per second and converts this analog information into digital information for display. Using a digital meter for output readings causes a problem: It will attempt to follow the high-frequency switching of the IGBT transistors, giving false information. The analog meter has a smoothing effect and ends up reading a voltage equal to what the motor actually sees.

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Troubleshooting VFD Problems

Paul Frank

Nov 01, 2000

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When a motor drive goes down, production often grinds to a halt. How do you find and resolve the problem quickly? When that variable frequency drive (VFD) goes down, you're under pressure to get it back online. Don't let this pressure make you take even longer to resolve the problem. Instead, remember the VFD troubleshooting checkpoints: check the basics (the controller display, connections, and temperatures),

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check the basics (the controller display, connections, and temperatures), check the motor, and check the drive - then check a little closer.

Check the controller display. Most VFD controllers include an interface to set up the drive for operation and to display information about its operation, once it's underway. Although the information displayed varies, most controllers tell you about high current (usually including blown fuses and overload trips), high and low voltages on the input and output sides, high temperatures, internal faults, and even offer advanced power diagnostics.

Check the connections. If the fault codes can't help you track down the problem, then check the connections. Loose connections are among the most common causes of faulty operation in VFD applications. Just eyeballing a connection is sometimes enough to know it's loose. You can also check for a voltage drop across the connection if you're still powered up - or resistance through a connection if you're powered down. Don't forget to isolate the connection to get a reliable reading.

Check temperatures. Checking the temperature of connections with a temperature probe or IR-thermometer is one way to tell if they're loose. They should never be hotter than the connecting wires. You can check temperatures in the drive and at the motor. For example, if the motor insulation is unsuitable for VFDs, it'll gradually degrade until it develops a short. Such shorts are often too small to blow a fuse, and too intermittent to trip an overload - but enough to shut down a controller. An IR thermometer can show what's going on. Also, use your nose: If a motor smells hot, it is.

You can do more. But usually, just checking the basics will be enough to uncover any problems you may have and get the system running again. That can give you the time you need for a permanent fix.

Sidebar: Check a Little Closer

In new installations, apparent problems with drive performance are often due to improper application, drive selection, setup, or installation of the motor as well as the drive. Sometimes "drive problems" are due to process control logic and not the drive at all.

In the case of frequent breaker trips, you may need to examine protection coordination, to ensure your breakers are the right size from the drive back to the service. You may also need to check other branch and feeder circuits. But first, see if your drive can reduce inrush current with a "soft start" function. Also, you may need to check:

- Current to the motor (ammeter),
- Voltage notching (oscilloscope),
- Inductive noise in signal, control, or power wiring (oscilloscope),
- Cable routing (visual inspection),
- Damaged signal, control, or power wiring (insulation resistance, TDR), and
- Current through the controller during sudden load changes, or during speed ramps (controller display).

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Troubleshoot with PQ Interviewing Techniques

[Stan Turkel](#)

Stan Turkel

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The majority of power quality problems are relatively easy to troubleshoot, provided you ask a lot of questions and you don't lose focus. Learning the art of acquiring and documenting information is a part of the overall troubleshooting

process - especially for problems that are transient in nature and occur infrequently. A good example of this is when a main feeder breaker in a manufacturing plant

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Learning the art of acquiring and documenting information is a part of the overall troubleshooting process - especially for problems that are transient in nature and occur infrequently. A good example of this is when a main feeder breaker in a manufacturing plant trips unexpectedly. While this may only occur two or three times a year, the cost in lost production and materials can be in the thousands of dollars.

All power quality troubleshooting starts at the load. Because users or operators notice something has occurred at the load, this should be the beginning of the troubleshooting trail. The problem is while many people are technically inclined to perform the troubleshooting task, they may lack the interviewing skills needed to get to the source of the problem.

Having a troubleshooting document at hand (based on interviews) helps you to keep a record of the problems. You should use this document each time an event occurs. Over a period of time, this document will almost always lead you in the right direction.

Having a troubleshooting document is just as important as having the correct meter for the job. But just like any tool, you need to use it correctly to obtain maximum benefit. During the question and answer period, your job is to collect the most accurate data possible. While this may sound like a simple task, most people are not experienced in collecting data to troubleshoot a power quality problem. Creating a standard document to make certain you leave nothing unnoticed is a good start.

While filling out your power quality document, a valuable technique is to always ask the person you're interviewing: "How do you know?" For example, if the operator tells you the machine started acting funny around 2:00 p.m., you should ask: "How do you know it was 2:00 p.m.?" They will now have to give more thought to the question, and you will most likely get a more exact answer the second time. It's also important to interview several individuals about the same problem. You'll have less of a chance of overlooking a well-concealed culprit when interviewing many sources. Gathering as much past and present information about the event or occurrence as you can will help you discover all of the possibilities when troubleshooting a problem.

After finding and correcting the problem, you should always follow up. In some cases, you may have only reduced the frequency or severity of a problem, yet the problem still exists. A good follow-up is just as important as making the first discovery of the cause of a problem.

The goal of a PQ troubleshooter is to find the sources of a malfunction in the least amount of time with accuracy. This is pretty easy, if you think and act like a detective.

Sidebar: Three Ps of Troubleshooting

Past Information. Gather as much past information about the event or occurrence as you can. Our tendency is to focus on the immediate situation, but in troubleshooting it's important to connect the present situation to any and all past events. Using your detective skills, you will search out past events from many sources. This includes, installers, manufacturers, plant personnel, and operators.

Present information. It would seem that gathering present information is relatively easy, yet this is where much information is overlooked. Having an interviewing guide is important to gather even some of the smallest details, which are most often overlooked. It's common to have one small detail or fact about an event that becomes the thread connecting many other dissimilar pieces of information. You can see why details are so important at this stage.

Possibilities. The act of troubleshooting is connecting past with present information and coming up with possibilities. Many novice troubleshooters will jump at one or two possibilities too early and make them fit the facts. With experience comes the knowledge that you should leave all possibilities open until you exhaust all avenues.

<http://m.ecmweb.com/content/troubleshoot-pq-interviewing-techniques>

The Basics of Variable-Frequency Drives

Peter Novak, Fluor, Inc.

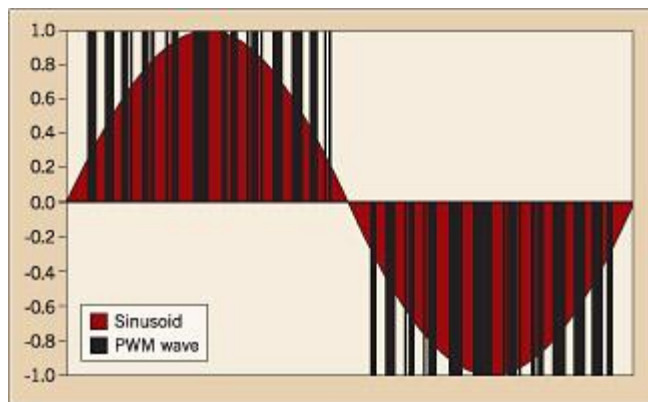
May 01, 2009

How to use various types of VFDs and harmonic mitigation tactics to combat the heat loss that accompanies these devices Find more articles on: [Variable Frequency Drives](#)

How to use various types of VFDs and harmonic mitigation tactics to combat the heat loss that accompanies these devices

When Tesla first introduced the 3-phase alternating current (AC) induction motor in 1888, he knew that his invention was more efficient and reliable than Edison's direct current (DC) motor. However, AC motor speed control requires either varying the magnetic flux or changing the number of poles on the motor. Even decades after the induction motor gained widespread use, changing the frequency for speed control remained an extremely difficult task — and the physical construction of the motor prevented manufacturers from creating motors with more than two speeds.

As a result, DC motors were necessary where accurate speed control and significant power output were required. In contrast to AC motor speed control requirements, DC motor speed control was achieved by inserting a rheostat into the low-power DC field circuit, which was feasible with available technology. These simple motor controls varied the speed and torque, and were the most economical way to do so for a number of decades.



By the 1980s, AC motor drive technology became reliable and inexpensive enough to compete with traditional DC motor control. These variable-frequency drives (VFDs) accurately control the speed of standard AC induction or synchronous motors. With VFDs, speed control with full torque is achieved from 0 rpm through the maximum rated speed and, if required, above the rated speed at reduced torque. VFDs manipulate the frequency of their output by rectifying an incoming AC current into DC, and then using voltage pulse-width modulation to recreate an AC current and voltage output waveform. However, this frequency conversion process causes 2% to 3% loss as heat in the VFD — caloric energy that must be dissipated. The process also yields overvoltage spikes and harmonic current distortions.

Variable-frequency types

There are three common types of VFDs. Current source inversion (CSI) has been successfully used in signal processing and industrial power applications. CSI VFDs are the only type that has regenerative power capability. In other words, they can absorb power flow back from the motor into the power supply. CSI VFDs give a very clean current waveform but require large, expensive inductors in their construction and cause cogging (pulsating movement during rotation) below 6 Hz.

Voltage source inversion (VSI) drives have poor power factor, can cause motor cogging below 6 Hz, and are non-regenerative. Consequently, CSI and VSI drives have not been widely used.

Pulse-width modulation (PWM) VFDs are most commonly used in industry because of excellent input power factor due to fixed DC bus voltage, no motor cogging, higher efficiencies, and lower cost. A PWM VFD uses a series of voltage pulses of different lengths to simulate a sinusoidal wave (**Fig. 1** on page 8). Ideally, the pulses are timed so that the time average integral of the drive yields a perfect sinusoid. The current method of choice to produce this waveform runs a triangle wave and sine wave through a comparator, and outputs a voltage pulse whenever the sine

wave's value is greater than the triangle wave. The current electric component of choice to generate the voltage pulse is the insulated gate bipolar transistor (IGBT), although silicon-controlled rectifiers (SCRs) can work as well. In the near future, injection-enhanced gate transistors (IEGTs) will be used to perform this task. Much more long term, memristors will probably become the component of choice for this task.

Memristors are the fourth passive circuit element, linking electric charge and magnetic flux. Memristors have been hypothesized to exist for more than 30 years, but were not fabricated until April 2008 by Hewlett Packard Labs. Hewlett Packard hopes to use these devices as a passive transistor, reducing their heat generation compared to other types of memory. Regardless of the component used to form the sine wave, the switching action causes problems.

Heat, power losses, and harmonics

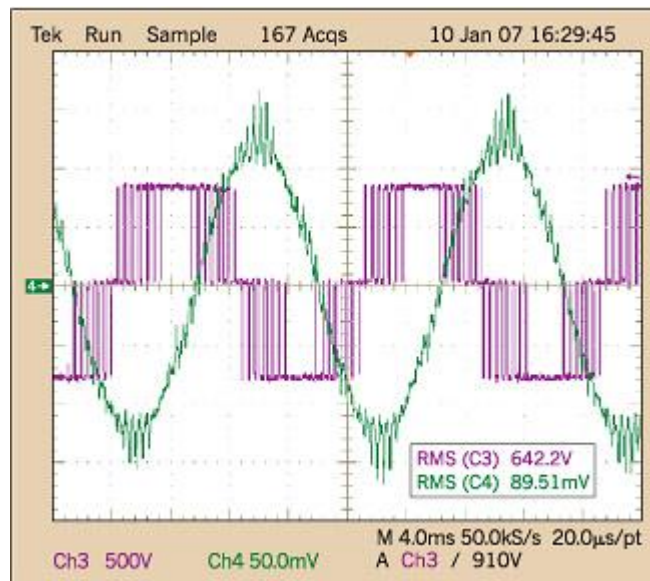
The first problem a VFD manufacturer needs to address is heat. Although VFDs are highly efficient devices, manufacturers are unable to produce an ideal set of components. The heat lost in the drive is governed by the following equation:

$$H_{\text{loss}} = P_i (1-\eta)$$

Where H_{loss} is the power lost (W), P_i is the power through the drive (W), and η is the efficiency of the drive. Usually, VFDs have an efficiency rating between 95% and 98%. This means the amount of air that must be moved through the drive is governed by the equation:

$$m = H_{\text{loss}} \div (C_p \Delta T) = P_i (1-\eta) \div (C_p \Delta T)$$

Where m is the mass flow rate (kg/s), C_p is the specific heat of air [$\text{kJ} \div (\text{kg} \times \text{K})$], and ΔT is the difference in temperature between the incoming air and the outgoing air (K). This heat can cause significant cooling costs to be added into the design, especially if the drive is unable to be placed in an unclassified location (area free of flammable gases or particles). If the drive must be placed in a classified location, then the airflow going to the drive will need to be purged and pressurized.



Heating is only one of the problems with VFDs. The other major problem lies with system harmonics. A picture of the PWM and the harmonics they cause is shown in **Fig. 2**. The irregularities in the sine wave are called harmonics. In an ideal power circuit world, these harmonics should not exist. They do nothing but cause problems. Fortunately, there are a number of ways to mitigate harmonics.

One of the simplest methods of dealing with harmonics is to place a sine wave filter on either side of the VFD. On the line side, these are typically called line reactors and have reactance values anywhere between 1.5% and 5.0% impedance. Higher impedance not only stops more harmonics, but it also limits the power going to the VFD.

Another tactic that can be used on the line side of the VFD is to place capacitors at a common bus. Because the impedance of a capacitor is inversely proportional to the frequency of a signal, the harmonics see a short through the

capacitor and travel through the capacitor to ground, hopefully ignoring the other loads on the bus. VFDs may also use an active front end to limit the harmonics that the line side sees. An active front end has another IGBT switching at an inverse voltage as the main IGBT, but it is placed through a high pass filter so that the fundamental power signal goes to ground. The summation of the two harmonic signals ideally should be zero. If an active front-end drive is not suitable for some reason, a passive front-end VFD might be procured. Passive front-end VFDs use multiple phase-shifting transformers and diode bridges to mitigate harmonics.

The more pulses a passive front-end VFD has, the fewer problems with harmonics exist. The trade-off is that the line voltages must be well balanced, and with each additional phase shifting transformer there is increased cost and a loss in efficiency. In extreme cases, an isolation transformer might be procured. Although this is one of the most effective ways to prevent harmonics from spreading, it's also one of the most costly.

If harmonics are not sufficiently mitigated on the line side of the VFD, crosstalk and overheating could become issues. Overheating could either cause bus sizes to be derated or increase cooling costs. Crosstalk is defined as the signal from one circuit interfering with another circuit. Generally speaking, it is a larger issue than overheating. An example of this is a radio just slightly out of tune. Although it is possible to hear the music through the static, the static is annoying. Crosstalk is an annoying thing in telecommunication circuits. In power circuitry, crosstalk will cause overheating and frequency relay trips.

Just as harmonics left unchecked on the line side can cause problems, they can create issues on the load side as well. This is because of the nature of waves. For example, a small force exerted on a Slinky at either end will cause a high amplitude sine wave. Electromagnetic waves act in the same fashion, meaning a small amount of reactance can cause large voltage spikes. Because this reactance is inductive in nature, most output filters are capacitors connected in a delta configuration. Ideally, this should make the reactance portion of the impedance go to zero. If the impedance is matched properly, then this does not occur.

A note of caution: Capacitors connected on the load side of the VFD can create a large number of problems, up to and including destroying a drive. Therefore, it's wise to check with the drive manufacturer before installing a sine wave filter on the load side of the VFD. On rare occasions, an active filter may be used. Although these tend to work well, they are rather expensive and usually have to be custom designed.

VFD benefits

Despite the fact that VFDs generate a large amount of harmonics and heat, they would not be as widely used and popular as they are today if they did not have significant economic benefits.

Electrically, VFDs run at a high power factor. Any class of induction motors usually has a low power factor at half and three-quarters load (0.75 to 0.85). This actually decreases the life of the motor, because the unnecessary increase in current overheats the winding insulation. VFDs bypass this problem by running the load at a frequency below the fundamental.

The most obvious reason to procure a VFD is speed control. This is usually done for process, operation, and economic benefits. One economic benefit comes from the reduction of maintenance when using a VFD, especially not having to deal with the DC motor carbon brushes or mechanical speed-control gearboxes (transmissions). The most obvious economic benefits of VFDs occur with fans and pumps. The power that a pump or fan consumes is directly proportional to the cube of the velocity. This means if an operator can run a fan at 80% of full speed, it theoretically uses 51% of full load power.

VFDs also optimize motor starting characteristics. VFDs bring motors up to full speed quickly and by drawing only 100% to 150% of full load amps (FLAs). This ability to start at normal FLA is very important if the power supply cannot withstand the normally six times FLA across-the-line starting draw, or even the 350% FLA soft-start device current. VFDs do this by managing the magnetic flux of an induction motor. Magnetic flux is directly proportional to the voltage and inversely proportional to the frequency. By keeping the flux constant, the inrush current does not exceed the FLA rating of the motor, and full torque is maintained. This is a significant improvement on a soft-start, which has significant voltage drop problems and cannot start under full load.

Another potentially useful aspect of VFDs is demonstrated in **Fig. 3**, ([click here to see Fig. 3](#)) which shows the output of a constant torque VFD. Notice the two regions, constant torque, and constant horsepower. The constant torque region is fairly self explanatory; the VFD is regulating the flux so that the current is constant. Once the VFD surpasses the rated system frequency, the voltage cannot increase due to the physical constraints of the system. Because the voltage is static — and the frequency is increasing — the flux is forced to decrease. When this occurs, the current and

torque are forced to decrease as well. This is called field weakening. Although not necessarily a good thing, it can be useful if there is a need to power a partial torque load above the rated speed. In addition to this capability, VFDs can also take any form of input power whether it's single-phase AC, 3-phase AC, or DC. VFDs fed from a DC source still power an AC load without an internal rectifier.

VFDs also have some applications on the power grid. One classic example of this is a doubly fed induction generator, in which the VFD can force a fixed frequency and voltage signal out of a variable-speed (frequency) input. This is commonly seen in wind turbines and other small hydroelectric generation projects that will be connected to the power grid. Other renewable energy sources, such as photovoltaic cells, can use VFDs to act as an inverter before connecting to the power grid, although inverters with buck-boost technology are more common. While there are many potential uses for VFDs on the commercial power grid, they are beyond the scope of this article.

In summary, whenever a load has either a variable torque or a variable speed, a VFD should be considered. A VFD might be considered if a large motor has a problem with voltage drop, torque, or inrush current during start-up. Even though VFDs undoubtedly solve their fair amount of problems and provide substantial energy savings, the heat they generate must be dissipated — and the harmonics they produce must be mitigated.

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<http://m.ecmweb.com/power-quality/basics-variable-frequency-drives>

Behind the Scenes with VFDs

Doug Weber, Rockwell Automation

Sep 01, 2006

VFDs can help control maintenance costs-not just motor speed and torqueFind more articles on: [Variable Frequency Drives](#)

Ask any system designer to name the top three reasons for specifying a variable-frequency drive (VFD), and you probably won't hear a lot of maintenance cost reduction responses in return. Instead, you'll likely get answers ranging from range of precision and/or control to ease of installation to energy reduction the latter of which tops everyone's list. That's because using a VFD to control the speed

Ask any system designer to name the top three reasons for specifying a variable-frequency drive (VFD), and you probably won't hear a lot of "maintenance cost reduction" responses in return. Instead, you'll likely get answers ranging from "range of precision and/or control" to "ease of installation" to "energy reduction" — the latter of which tops everyone's list. That's because using a VFD to control the speed of a centrifugal fan or pump at 80% of rated speed, for example, can cut energy costs in half. This dramatic energy savings is one reason designers like to use VFDs in today's commercial and residential HVAC systems.

So where does the savings come from? Doesn't adding another piece of equipment increase the costs of maintenance by providing more equipment to maintain? Not necessarily. In fact, converting a process from fixed speed to variable speed can significantly reduce wear and tear on mechanical systems by reducing start/stop cycles. VFDs can also eliminate the need for such active components as vanes, dampers, and valves. Ultimately, you have less equipment to maintain, and longer runtime between failures.

Simply slapping a drive into an existing system, however, isn't going to cut it. To get the desired cost savings, first you need to understand the features offered by the particular drive you have or are planning to buy. Then, you need to think about how to implement those features in a way that will reduce system maintenance and overall operating costs.

Smoothing acceleration. When a load transitions from steady-state speed to accelerating or decelerating, the transition is usually instantaneous. However, the mass of the load doesn't instantly follow. This difference causes a jerking action that puts considerable stress on mechanical components.

VFDs can control acceleration and deceleration along the torque/speed curve to eliminate the jerkiness — and thus reduce the stress on components. This method has long been recognized as an aid in the handling of very light conveyor loads (e.g., a bottling line), extending the life of mechanical components in any application that has fast transitions.

Avoiding overcurrent conditions. Controlling a motor that is already spinning (commonly called a flying start) creates overcurrent challenges. Avoiding the maintenance downtime that could result from an overcurrent trip in those flying-start situations requires an AC drive that can reconnect the drive to a motor already spinning as quickly as possible to resume normal operation with minimal impact on load or speed.

When a drive executes a normal start, it initially applies 0 Hz and ramps up to the commanded frequency. Starting the drive in this mode with the motor already spinning will generate large currents. This can result in an overcurrent trip, if the current limiter does not react quickly enough. The likelihood of an overcurrent trip is even greater, if there is a residual flux on the spinning motor when the drive starts.

Simply preventing an overload trip isn't enough. If done incorrectly, the deceleration and subsequent reacceleration can place extreme mechanical stress on the application. And, of course, this creates a potential for causing premature equipment failure with the attendant downtime and repair costs. This is why a VFD needs a flying start mode.

In flying start mode, the drive responds to a start command by identifying the motor speed and then beginning its output synchronized in frequency, amplitude, and phase to the spinning motor. The motor will then be reconnected at its existing speed, and be smoothly accelerated to the commanded frequency. This process eliminates overcurrent tripping and significantly reduces the time for the motor to reach its desired frequency. Since the motor is “picked up” smoothly at its rotating speed and ramped to the proper speed, little or no mechanical stress occurs.

Skip frequency. All rotating machinery — from motorcycles to industrial fans and pumps — have mechanical resonance points. These are the frequency points at which vibration can rapidly damage that specific equipment. If you're aware of these points and avoid them — by either accelerating beyond or decelerating below them so the motor doesn't run at those points — you can prevent the rapid damage. VFDs with skip features allow you to do exactly that. In fact, most drives offer multiple skip frequency parameters to mitigate different resonance points.

The skip frequencies do not affect normal acceleration and deceleration. The drive output will ramp through the band, uninterrupted. When the operator issues a command to operate continuously inside the established band; however, the drive will alter the output to remain outside the band until a new command is issued.

If you know the mechanical resonant frequencies of your equipment, you can program the drives to “skip” through operation at those frequencies. That is, your equipment will run at those frequencies only momentarily, rather than continuously — just long enough to arrive at a safe frequency of operation. How can you determine what these resonant frequencies are? You may find this information in the equipment manual. A more common method is simply observing the equipment for noticeable changes in heat (for example, at bearings), noise, or motion when the operating frequency changes.

Monitoring the system. While drives don't possess the extensive monitoring capabilities of devices designed specifically for predictive maintenance or monitoring, they do monitor motor current and speed. You can put that information out on your industrial network. A distributed control system or PLC can provide reminders, warnings, and alarms to maintenance personnel.

Using just motor current and speed, a control system can determine a load problem is occurring. It can then call a designated cell phone for intervention before failure occurs. Such a system can also call alternate numbers and take backup actions, which may include more notifications or corrective action.

Overloads and current limits. Almost all drives have a built-in electronic motor thermal overload. When a motor runs outside its safe operating limits, the overload can reduce the output current (or shut off the motor) to prevent thermal damage or outright failure.

Overload software uses an algorithm incorporating motor current, speed, and time as inputs to model the temperature of the motor. This may also be done with thermister feedback directly from devices buried in the motor windings, using actual temperature readings to determine motor stress.

Multi-motor applications (those using one AC drive and more than one motor) require the motor overload to be disabled. The drive can't distinguish the current of each individual motor to provide individual protection. These

applications require more advanced monitoring devices that can accept data from multiple sources to alert personnel of impending faults and failures.

A VFD has control of the amount of current it supplies to a motor. By limiting current or shutting down the motor, VFDs can reduce mechanical damage.

Many drives have a feature called an electronic shear pin. This feature is based on the proven concept of the mechanical shear pin. Snow blowers, for example, are equipped with mechanical shear pins — including one on the main driveshaft. If an object — such as your kid's skateboard buried under a foot of a spring snow — jams the rotating blades, the driveshaft shear pin breaks. When it breaks, it disconnects the drive train to the motor. This obviously protects the blower motor. Using shear pins to mechanically disconnect the rotating blades from the motor saves the expense of replacing a damaged motor.

Similarly, an electronic shear pin can define a current limit level that would cause damage. If the torque in the motor ever exceeds the set limit, the drive will automatically shut off the motor.

By limiting torque to a set level, AC drives provide good protection for systems that can become jammed. A common application for this is the chain conveyor. By not allowing a motor to power through the jam, you can use the VFD to prevent chain breakage and damage.

You probably have unused features in your existing motor drives, which means you have untapped cost savings. By taking advantage of the wide array of techniques already available, you can minimize stress placed on valuable plant machinery, increase equipment uptime, and reduce maintenance costs. One last bit of advice when calculating your return on investment for VFDs: Be sure to quantify the maintenance cost savings — especially if you need to submit a capital request. Ask your drive vendor for assistance in obtaining realistic numbers.

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<http://m.ecmweb.com/contractor/behind-scenes-vfds>

How to Keep Variable-Frequency Drives and Motors Running

Dan Orchard, Intigral, Inc.

Feb 01, 2009

How to keep variable frequency drives (VFDs) and electric motors running smoothly and practical tips for a more effective preventive maintenance program Find more articles on: [Variable Frequency Drives](#)

When applied to blowers and pumps, variable-frequency drives (VFDs) offer energy savings. In mechanical applications, they allow fine adjustments that wouldn't be possible by other methods. Despite their popularity in multiple applications, they are not simple plug-in-and-forget devices. Because VFDs are full of electronics, they're susceptible to all sorts of problems from incoming power disturbances

When applied to blowers and pumps, variable-frequency drives (VFDs) offer energy savings. In mechanical applications, they allow fine adjustments that wouldn't be possible by other methods. Despite their popularity in multiple applications, they are not simple plug-in-and-forget devices. Because VFDs are full of electronics, they're susceptible to all sorts of problems — from incoming power disturbances to environmental hazards to wrong operation, not to mention one or two unexpected issues that may arise. The motors they drive present their own set of challenges.



I work with motors (from 300 hp down to 1/16 hp) and variable-speed drives on a daily basis. During my 24 years of practical experience, I've learned a great deal about troubleshooting different situations. Here's a list of practical tips to keep your motors up and running as well as common errors to avoid.

Start with baseline readings

Don't just take a motor out of the box, throw it in place, and hope for the best. Before putting a motor into full operation, take insulation resistance readings from phase-to-phase and phase-to-ground. Measure the insulation resistance of the windings using an insulation multimeter to determine what a good reading is. In addition, measure the starting and running amperage, the running voltage, and the leg-to-leg balance.

Measure the temperature at first startup (unloaded, loaded) and after a period of use. A motor may run hot because it's been used hard, is in a high-temperature area, or has a problem. Without knowing its normal temperature, it's difficult to tell which is the case. It's nice to know whether motors are running hot or not. A lot of times, you won't see any problems until the heat really builds up. For example, the inside temperature near the glass tempering ovens in my plant in the summertime is normally about 130°F.

You should also measure temperature using an infrared thermometer or a thermocouple connected to an insulation multimeter. It's good to compare results between the various methods.

Make other measurements periodically

Depending on your preventive maintenance (PM) schedule and the cost of unscheduled downtime, take additional amperage, resistance, and insulation resistance readings. Compare these readings to previous readings. If the measurements deviate by more than 5% to 10%, start looking for bad electrical connections or loose/ill-fitting mechanical connections. Has the load increased, the frequency of use changed, or have ambient temperatures increased/decreased?

Find out if the motor matches the application and was specified for the system, or if upgrades are needed.

Check the protection

Look at the protection systems, the overload contactors, and fusing. Is the overload set for full load amperes, or is it set too high or low? Is the fusing correct for the application? Overload contactors are designed to take care of overloads, while fuses and circuit breakers are intended for short circuit protection. Are they sized according to the load? Do the fuses blow without tripping the overload? Are the fuses rated properly?

If the fuses blow repeatedly, there's a temptation to replace them with higher-rated fuses. But if some time later the overload decides to short across itself and doesn't trip any more, suddenly those fuses that are too high will make the motor cook. That means a lot of back-checking, pulling out the manuals (if they're available), or looking at the nameplate data.

Don't change parts instead of troubleshooting

Some technicians will change out parts until the trouble goes away. This is an expensive way to troubleshoot, because most motors and drives start at \$500 and up. It's not unusual to find that the same motor/drive that had failed may start working in another application.

This makes the job of finding the original problem harder, because the failure was only temporary. Was it the loading, application, or a combination of things that led up to the failure?

Cabling can be an issue, too

Check the line at the motor, not just at the panel on the wall, which may be 100 ft away. Power lines in hostile (e.g., high-temperature) areas, even when protected by conduit, may fail. Checking the voltage at the panel and not at the motor may result in replacing a perfectly good motor when the problem is in the wiring.

Look at the drive's setup and parameters. Check the acceleration and deceleration times. Are you running at line frequency (higher or lower)?

Make sure it's the right motor

Sometimes, motors are put into applications for which they are not designed. Inverter-rated motors make a big difference in the longevity of the system. Running a standard-duty motor at 50 Hz, for example, often leads to overheating. Similarly, running it at 90 Hz or 120 Hz may work for a while, but the motor can't accept that as a steady diet.

The duty cycle of the motors will help determine where and what application they are suited. A motor designed to run 8 hr a day, five days a week, will fail prematurely if it has to run 24/7.

Nameplate data is an important troubleshooting tool. It will tell the motor's service factor, the duty cycle, and more. It will also provide useful information about the protection circuits and fusing.

Continue on Page 2

Check for power problems

Many drive failures come from power spikes, phase loss, or undervoltages. After a power issue, it's important to measure the power to see if the problem has been corrected or is still happening. If you don't check the power after an outage, the drives will pay the price. When the power comes back after an outage, the machine operators may just automatically restart and try to run again. Suddenly, you start popping drives and burning up motors because of single-phase conditions.

Most newer drives have settings that will not let the system restart after a fault has occurred. I set up mine so that something like a missing leg of 480VAC is not overlooked, using phase loss indicators to help the maintenance staff look for problems.

Editors note: A similar version of this article originally appeared in Fluke**Electrical** News (Volume 7, Number 1).

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<http://m.ecmweb.com/training/how-keep-variable-frequency-drives-and-motors-running>

Troubleshoot with PQ Interviewing Techniques

[Stan Turkel](#)

Stan Turkel

Dec 01, 2000

The majority of power quality problems are relatively easy to troubleshoot, provided you ask a lot of questions and you don't lose focus. Learning the art of acquiring and documenting information is a part of the overall troubleshooting process - especially for problems that are transient in nature and occur infrequently. A good example of this is when a main feeder breaker in a manufacturing plant

The majority of power quality problems are relatively easy to troubleshoot, provided you ask a lot of questions and you don't lose focus.

Learning the art of acquiring and documenting information is a part of the overall troubleshooting process - especially for problems that are transient in nature and occur infrequently. A good example of this is when a main feeder breaker in a manufacturing plant trips unexpectedly. While this may only occur two or three times a year, the cost in lost production and materials can be in the thousands of dollars.

All power quality troubleshooting starts at the load. Because users or operators notice something has occurred at the load, this should be the beginning of the troubleshooting trail. The problem is while many people are technically inclined to perform the troubleshooting task, they may lack the interviewing skills needed to get to the source of the problem.

Having a troubleshooting document at hand (based on interviews) helps you to keep a record of the problems. You should use this document each time an event occurs. Over a period of time, this document will almost always lead you in the right direction.

Having a troubleshooting document is just as important as having the correct meter for the job. But just like any tool, you need to use it correctly to obtain maximum benefit. During the question and answer period, your job is to collect the most accurate data possible. While this may sound like a simple task, most people are not experienced in collecting data to troubleshoot a power quality problem. Creating a standard document to make certain you leave nothing unnoticed is a good start.

While filling out your power quality document, a valuable technique is to always ask the person you're interviewing: "How do you know?" For example, if the operator tells you the machine started acting funny around 2:00 p.m., you should ask: "How do you know it was 2:00 p.m.?" They will now have to give more thought to the question, and you will most likely get a more exact answer the second time. It's also important to interview several individuals about the same problem. You'll have less of a chance of overlooking a well-concealed culprit when interviewing many sources. Gathering as much past and present information about the event or occurrence as you can will help you discover all of the possibilities when troubleshooting a problem.

After finding and correcting the problem, you should always follow up. In some cases, you may have only reduced the frequency or severity of a problem, yet the problem still exists. A good follow-up is just as important as making the first discovery of the cause of a problem.

The goal of a PQ troubleshooter is to find the sources of a malfunction in the least amount of time with accuracy. This is pretty easy, if you think and act like a detective.

Sidebar: Three Ps of Troubleshooting

Past Information. Gather as much past information about the event or occurrence as you can. Our tendency is to focus on the immediate situation, but in troubleshooting it's important to connect the present situation to any and all past events. Using your detective skills, you will search out past events from many sources. This includes, installers, manufacturers, plant personnel, and operators.

Present information. It would seem that gathering present information is relatively easy, yet this is where much information is overlooked. Having an interviewing guide is important to gather even some of the smallest details,

which are most often overlooked. It's common to have one small detail or fact about an event that becomes the thread connecting many other dissimilar pieces of information. You can see why details are so important at this stage.

Possibilities. The act of troubleshooting is connecting past with present information and coming up with possibilities. Many novice troubleshooters will jump at one or two possibilities too early and make them fit the facts. With experience comes the knowledge that you should leave all possibilities open until you exhaust all avenues.

<http://m.ecmweb.com/content/troubleshoot-pq-interviewing-techniques>

Speed Control of Motors

[Steve Vidal](#)

Steve Vidal, P.E., Joseph J. Vidal & Sons, Inc.

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An introduction to variable-frequency drives Find more articles on: [Variable Frequency Drives](#)

Speed, torque, and horsepower are three inter-related parameters in motor control. The speed of a motor, measured in revolutions per minute (rpm), defines a motor's ability to spin at a rate per unit time. The torque of a motor, measured in foot-pounds (ft-lb), is a rotational characteristic of the motor that is the algebraic product of force multiplied by distance. Electrically, one horsepower is

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An electric motor is a device that converts electrical energy into mechanical energy. An electrical signal is applied to the input of the motor, and the output of the motor produces a defined amount of torque related to the characteristics of the motor. It's important to understand speed-torque characteristic curves as they show the relationship between speed as a percent of rated speed, versus load torque as a percent of full rating. Motors are available in multi-speed configurations that can provide constant torque variable horsepower, constant horsepower variable torque, and variable torque variable horsepower.

Traditionally, DC motors have been used in precise speed control applications because of their ability to provide acceleration and deceleration from a dead stop position to full speed fairly easily. You control the speed of a DC series motor (the field is in series with the armature) by increasing or decreasing the applied voltage to the circuit. In a DC shunt motor (the field is in parallel with the armature), the speed is controlled by increasing or decreasing the applied voltage to the shunt field or armature by means of a field rheostat or an armature rheostat. Silicon-controlled rectifiers (SCRs) have replaced rheostats as they can control large blocks of power without the heat dissipation problems of carbon- or wire-wound variable resistors. Additionally, SCRs are much smaller in size than their earlier counterparts and interface well with programmable logic controllers.

The AC squirrel cage induction motor is essentially a constant speed device. The speed of the rotating magnetic field is referred to as synchronous speed. The synchronous speed (S) of a motor is defined as: $S = 120(F) \div P$, where (F) is the incoming line frequency and (P) is the number of poles the machine is constructed of. Here's an example to help illustrate this point.

In the United States, the AC line frequency is 60 Hertz. A 4-pole AC squirrel cage induction motor would therefore have a synchronous speed of 1,800 rpm $[(120 \times 60) \div 4]$. In practice, the motor will run at less than 1,800 rpm as load is placed on the rotor. This difference in speed between synchronous speed and full load speed is referred to as slip, usually expressed as a percentage. Note that the only two variables in this equation that define speed are the incoming line frequency and the number of poles in the machine. Because the number of poles in a machine is fixed, the only variable that's left to change is the incoming line frequency — this is the basis for operation of a variable-frequency drive (VFD).

It's important to understand the difference between the AC and DC machine at this point. Earlier, we mentioned a DC machine could have its speed changed by increasing or decreasing the applied voltage. This is not the case for an AC motor. In fact, you can damage an AC squirrel cage induction motor if you vary the incoming supply voltage.

The term VFD is often used interchangeably with AC drive, inverter, or adjustable-frequency drive (AFD). The two most common circuits for adjusting the speed of an AC squirrel cage induction motor are the inverter and the cycloconverter.

Using an inverter, the VFD does two things: First, it takes the incoming AC signal and converts it to a DC signal through a process known as rectification; next, it takes the rectified DC signal and inverts it back to a variable voltage and variable-frequency AC signal. An inverter takes a waveform like a rectified DC signal and generates an equivalent time-varying waveform resembling a sinusoid. A block diagram for an inverter type VFD is shown in the **Figure** ([click here to see figure](#)).

The VFD using a cycloconverter is a device that produces an AC signal of constant or controllable frequency from a variable-frequency AC signal input. The output frequency is usually one-third or less than the input frequency. The cycloconverter type of VFD is normally used with larger motors or groups of motors.

Typical specifications you might encounter with an inverter-type VFD are listed below.

- Horsepower: 1 to 10 hp @ 230V
- Input frequency: 50/60Hz
- Output frequency: 0 to 120Hz standard, 0 to 400Hz jumper selectable
- Frequency setting potentiometer: 10k 1/2W
- Ambient temperature: 0 to +40°C
- Control method: PWM (pulse width modulation)
- Transistor type: IGBT (insulated gate BJT)
- Analog outputs: assignable
- Digital outputs: opto-isolated assignable

Terminal strips present on the VFD allow the device to interface to the outside world with familiar switching devices such as start, stop, forward run, and reverse run. Instead of using a 3-wire control circuit to start and stop a motor with momentary contact devices, the electronics of the drive control all those familiar operations.

Normally, the VFD also has a backlit liquid crystal display that shows a variety of motor operational parameters that are fully programmable by the user. Solid-state devices, such as the silicon-controlled rectifier, triac, and insulated gate bipolar junction transistor, have allowed the VFD to become the method of choice for AC motor speed control.

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<http://m.ecmweb.com/content/speed-control-motors>

Correspondence Lesson 1: The Application of Controls

[Paul Rosenberg](#)

Paul Rosenberg, Consulting Datacom Editor

Feb 01, 1999

The real genius of control work is being able to combine the many different devices to get a very complex and difficult job done. This first lesson is a step in that direction. Control work covers a lot of territory. The term itself conjures up different

images in almost everyone's mind. When this writer thinks of the term "control work," he usually remembers the dreaded Duplex Pump circuit he had

The real genius of control work is being able to combine the many different devices to get a very complex and difficult job done. This first lesson is a step in that direction.

Control work covers a lot of territory. The term itself conjures up different images in almost everyone's mind. When this writer thinks of the term "control work," he usually remembers the dreaded Duplex Pump circuit he had to learn in trade school. Sometimes an old electrician named Mr. Clark comes to mind: He's doing an emergency reroute of the controls for a sewage treatment plant, and stopping the several-million gallons of raw sewage that were spilling onto 34th St.

Some of you will think of assembly lines, limit switches, timing relays, motor starters, variable speed drives, and a hundred other applications of controls. We'll try to cover the more common control technologies here. But before we attempt to go through the technical explanations of equipment and control schemes, we should lay a good foundation as to why controls are important, who uses them, how they are used, and so on. You need a global understanding of how and why these systems work, not just how to connect "A" to "B."

Why we use controls. Electrical controls have been a big part of the electrical industry since it began: when the first electricians found out they could use circuitry and electromechanical devices to do things humans could do. Rather than employ a man to turn a machine on and off at appropriate times, electricians arranged a control circuit and a few special switches to do the same thing. The result: the same job done more reliably and at a much lower cost.

Since that time, we've expanded the use of controls continually, and for the same reason: They provide for reliable, inexpensive, automatic control of equipment, while allowing us to control machines in a variety of ways.

Types of control systems we use. For the purpose of clarity, we'll specify three basic types of control systems. Keep in mind, however, all of these systems can (and frequently do) operate together as a single control system. We're breaking them apart here for easier understanding.

- *Electrical controls.*
These operate by starting, stopping, directing, or regulating the flow of electricity in circuits. In other words, you're allowing electricity to turn a machine on, stop the machine by opening the circuit that controls it, or by redirecting the current in a different way.
- *Electronic and computerized controls.*
These operate by conveying complex messages from one place to another; changing, coding, or conditioning electrical currents; or performing complex and interactive routines.
- *Pneumatic and hydraulic controls.*
These allow for control of equipment in response to the pressures of air and fluids, respectively.

Types of control devices we use. The most basic control devices are:

Mechanical switches. Whether operated by hand or some other object pressing, pulling, or twisting it, hundreds of types of mechanical switches exist. They're used to stop, start, or redirect the electrical current passing through them.

Solenoids. These are coils of wire (acting as electromagnets) that pull a plunger (iron bar) into the core when energized. You can use the action of the solenoid (movement of the plunger) for anything. In most cases, it's used to operate some type of switch. Notice that a relay is a combination device, using a solenoid combined with mechanical switching, and building them into a single unit.

Signaling devices. Pilot lights, warning lights, bells, and the like are signaling devices. These alert someone to a pre-specified condition. LED and LCD displays are also signaling devices.

Sensors. There are hundreds of sensors. Some sense visible light, infrared, ultra sound, current level, and many others. All of them are designed to respond to a specific condition and pass along that information or respond in some way. The simplest and most common type of response is to throw a switch that's packaged in the same unit as the sensor.

Logic devices. These (usually microprocessors) can make some type of intelligent decision. The basic intelligent operation is an IF/THEN operation. For example, a logic device could operate as follows: If the occupancy sensors show a human presence, and if it is a weekday, and if it is between 6:30 and 7:00 am, then turn on the high bay lights.

This capability allows logic devices to do much more complex tasks than simply turning things on and off at preset times. This also requires the device listen, as it must keep track of which devices are being used. In other words, use the logical device with some type of sensor (whether it be internal or external to the logical device), so it can tell whether or not a specific control device is being used (the if function). These devices also require specific programming, so they can respond appropriately to a unique application.

Logic devices can be as simple as a single microchip in a cheap controller, or as complex as a mainframe computer with multiple peripheral devices.

Clocks and timers. These provide a time-reference for controlling a wide variety of devices.

Transformers. Use these to manipulate voltage in a variety of applications.

Hardware devices. These devices include lugs, harnesses, enclosures, and other supplementary items.

Valves. Use these devices in pneumatic and hydraulic controls as well as in process controls. They basically control some type of gaseous, liquid, or granular flow, usually combined with a solenoid-type of electric activation.

Communications media. This high-tech term refers to some method of getting information or current from one place to another in the newer electronic control systems. Perhaps the most common communications media is a twisted-pair cable, transferring binary signals between a computer and a programmable controller. But in other cases, it could be radio signals, infrared light, microwaves, or even power line carrier signaling. The Internet is a form of communications media, allowing for control of complex control processes from almost anywhere on the planet.

Transformative devices. These are devices that change one thing into another. An example is a thermocouple: It changes heat energy into electrical current. It generally serves as a sensor, but it's different in the physics of operation. A modem would be a computerized transformative device because it changes sound into electrical pulses and vice versa.

Electronic devices. By this we don't mean microchips and logic devices. Instead, we mean basic electronic devices: such as transistors, capacitors, choke coils, and other devices that act as electronic valves, amplifiers, or storage devices. They condition or change electrical currents in some way.

Wiring. This term includes conductors, raceways, fuses, connectors, etc. Really, this is supplementary equipment, but since it's absolutely necessary for control systems to function, we include it here.

Power sources. Again, this is usually supplementary equipment (as in the case of the power generating station down the road), but not always. Backup batteries or specific-use power sources are valuable and frequently used control devices. The above category of devices doesn't cleanly cover everything in the controls field. (For example, is a thermal overload a sensor or a mechanical switch?

Really, it's both.) However, this list gives you a good overview of control devices and their essential functions. Note: This list does not cover the things being controlled (such as motors and pumps), only the devices you would use to do the controlling.

How we combine devices and systems. The real genius of control work is being able to combine all these devices to get a complex and difficult job done. Someone who does this well (as in the case of Mr. Clark, the electrician) is both a skilled technician and an artist. Achieving this competence requires:

- *A thorough understanding of all the items you want controlled and all of the control devices.*
It's not enough to know how "A" connects to "B." You must understand the principles that make "A" and "B" work. If all you know about single-phase motors is that you connect 240V to Terminals 4 and 7, you won't be able to modify the motor for a custom use. You must understand the motor's primary principles of operation: rotating magnetic fields, current lead and lag, and so on.
- *Coordination of specifications.*
Some control devices operate on 12VAC; most computerized devices at 5VDC; and most motors at 240VAC or 480VAC, 3-phase. When combining devices, the most common problem is coordinating current and voltage

requirements of each device. To combine a 480V motor and a 120V timer, you'll need an appropriately sized transformer to connect between them.

- *Coordination of duty cycles.*

Devices used together must be able to operate together over long periods. For example, some coils will burn out if you use them continuously. If you need a solenoid for continuous use, you probably need one with the mechanically held feature. In any event, be sure to properly match all control devices.

- *Environmental factors.*

Be sure all your devices function well in areas where they're installed. This can be as simple as using NEMA 3R enclosures on an outside wall, or as difficult as determining which types of contactors will operate correctly in the cargo bay of the space shuttle. The new motors you're going to control may vibrate so much that you need to isolate your control devices with rubber mounts.

- *Effects upon existing devices and systems.*

Don't forget the effects of the equipment you're installing upon existing equipment. Many control devices create vibrations, which may not be acceptable. Others produce magnetic fields, which can interfere with operation of sensitive equipment in the area.

- *Safety.*

Safety is an important issue. There's probably no way to avoid every possible injury where machinery is used, but you must get as close to that goal as possible. This requires the intelligent use of warning signs, overloads and resets, barriers and guards, interlocks, lockouts, clutches, and anything else you can use to avoid injuries.

- *Maintenance.*

When putting a control system together, consider ease of maintenance and accessibility of critical items for repair. A few changes in the beginning can save a lot of hassle over the next 30 years. In most cases, you'll have to combine items expected to last for 20 years with items reasonably expected to last only five or 10 years. As such, it's a good idea to make notes of such things for whoever will be maintaining the system.

- *Management.*

You're installing control systems to serve a productive purpose. It follows that somebody will need to control the system from a distance, or at least determine the system's state of operation without difficulty.

- *Modification.*

In many instances, you'll build into your control system the ability to modify it for a different use at a later date. Again, a little forethought here can save money.

- *Ergonomics.*

Consider the use of your equipment by real people. For example, you can mount a switch used only once a year at an inconvenient location. Think about using the equipment before you specify its location.

Coordinating this list isn't easy; it's a job for a professional. With some thought, this course will give you what you need to reach that goal.

<http://m.ecmweb.com/content/correspondence-lesson-1-application-controls>

Correspondence Lesson 2: Basic Wired Control Devices

[Paul Rosenberg](#)

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Our coverage of specific control devices begins with basic wired controls- the most commonly used control items. You can find them in thousands of locations worldwide. Push button switches. Most of these switches consist of two parts: a contact unit and an operating mechanism. This allows for many combinations. You can get contact units in blocks that contain one normally open (NO) and one normally

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Push button switches. Most of these switches consist of two parts: a contact unit and an operating mechanism. This allows for many combinations.

You can get contact units in blocks that contain one normally open (NO) and one normally closed (NC) contact. (The designation "normal" is the position of the contact before it's acted upon.) You can also assemble multiple contact blocks to get up to four NO and four NC contacts. With some switches, you can add even more units.

The contact rating for a typical heavy-duty, oil-tight unit is: AC Volts: 110V to 125V Amperes Normal: 6.0A Amperes Inrush: 60.0A

In some cases, you can find the contact block mounted in a push button enclosure. Thus, it may be pre-wired, and you can install the operating mechanism later. The alternate method is panel mounting, where you mount the base, with the contact block attached, through an opening in the panel. You then secure it with a threaded ring, installed from the front of the panel. The ring is part of the operator assembly. This arrangement provides a space for a terminal block installation in the base of the enclosure. Thus, you can terminate all connecting circuits at a convenient location.

You may have to modify the hole when substituting one make of unit for another. Many types of operations are available to suit most applications.

Some people consider the easy identification of switches important, like how most of us associate certain functions with a specific color. For example, in the machine tool industry, we assign the following functions for the colors red, yellow, and black:

- Red: Stop or emergency stop
- Yellow: Return or emergency return
- Black/Green: Start motors or cycle.

While this is a good pattern to follow, you should never assume anything. When working on an existing circuit, always verify which button is for which operation. This may slow you down, but it will also keep you safe.

Selector switches. You can get selector switches with up to four positions. They can be the maintained contact-type, a three-position switch arranged for spring return from the right, left, or both right and left. Units are available with up to eight contacts per device.

Available operators include standard knob, knob lever, or wing lever types. You can use a cylinder lock to lock the switch in any one position or all positions. The arrangement for opening or closing contacts in any one position or more depends on a cam in the operator.

A simpler form, called a double-pole, double-throw selector switch, comes with two contacts arranged for two positions. Similarly, a double-pole, double-throw with neutral selector switch has two contacts arranged for three positions. As the number of positions increases to four and the number of contacts (poles) increases to eight, the manufacturer generally codes its reference to a specific operation through a symbol chart or function table. Heavy-duty switches and special application switches are also available.

Indicating lights. Three basic types of indicating or pilot lights are available: full voltage, resistor, and transformer.

Due to the vibration normally present in machines, most usually prefer the low-voltage bulb. It operates at 6V to 8V, obtained through a resistance or transformer unit.

The lenses are either plastic or glass and come in a variety of colors. Again, as for push button operators, you would use colors to increase safety of operation. For example:

- Red: Danger, abnormal conditions
- Amber: Attention
- Green: Safe condition
- White or clear: Normal condition

The push-to-test indicating light provides another feature. Suppose the indicating lamp isn't lighted. It may be the bulb isn't energized, or is burned out. Depressing the lens unit connects the bulb directly across the control voltage source, providing a check on the condition of the bulb.

Like the standard line of devices, manufacturers make miniature oil-tight push buttons, selector switches, and pilot lights. This line covers about the same selection as the standard line, while offering advantages where space is limited.

The oil-tight designation. All oil-tight operators mount on the panel of an enclosure with some type of sealing ring, for the oil-tight rating. Nameplates identify the unit, and are available in different sizes, generally depending on the amount of information required to properly describe the function of the unit. Manufacturers will custom-engrave nameplates for special applications (for an additional fee, of course). Almost all control devices of this type are rated "oil-tight."

Solenoids. Solenoids are extremely important for controlling machines. Like the relay and contactor, the solenoid is an electromechanical device, using electrical energy to magnetically cause mechanical movement.

When you first energize a solenoid, you partially remove the iron core from the core of the coil, and the inductive reactance of the unit is very low. The initial current (called inrush current) is relatively high. The holding current is the current at the closed position: It's sometimes called sealed current. The ratio of inrush current to holding current generally varies from approximately 5:1 in small solenoids to as much as 15:1 in large solenoids.

A solenoid is made up of three basic parts: the frame, plunger, and coil. The frame and plunger are made up of laminations of high-grade silicon steel. The coil is wound of an insulated copper. Solenoids for AC use are available as oil-immersed types. You get better heat dissipation and wear conditions with this design. They're also available with a plug-in base.

When energizing the coil of a solenoid, you produce a magnetic field about the coil. This field produces a force called pull, which acts on the solenoid plunger. Due to this force, the plunger moves into the coil. The pull in solenoids varies, from as low as a fraction of an oz. to as high as nearly 100 lb.

There are two primary considerations in the application of solenoids:

- Make sure the pull of the solenoid exceeds the load. If the pull is a little less, you'll get sluggish action, and the solenoid may not complete the stroke. You also have to consider conditions somewhat out of your control, such as low voltage or increased loading through friction or pressure. That's why it's generally advisable to overrate the solenoid by 20% to 25%. But don't oversize it too much; that would cause the plunger to slam, resulting in damage to the plunger and frame.
- Make sure you know the duty cycle of the workload. Some applications require a duty cycle of only an occasional operation. Others require several hundred operations per minute. You have to be careful with the latter application because operating a solenoid above its maximum cycling rate will result in excessive heating and mechanical damage.

The pull-in force of a solenoid decreases rapidly as the voltage decreases below the coil nominal rating. On the other hand, as the voltage increases over the nominal value, the pull-in force increases. But, the solenoid temperature may also rapidly increase.

From a low-voltage standpoint, you should allow for adequate force at some arbitrary low-voltage level when selecting a solenoid size. This will prevent failure to pull out (as low voltage leads to minimal current and consequent coil burnout).

Design practices vary, but low-voltage levels are usually set at 85% or 90% of rated or nominal levels.

Relays. As mentioned in Lesson 1, the relay is a combination of a solenoid and one or more switches. A relay connects two circuits with differing characteristics. Every relay is connected to a controlling circuit that activates its coil, either leaving it in its normal position, or moving the coil and attached switch contacts to its second position. This opens or closes the contacts, as the case may be. This controlling circuit must be of the correct voltage, amperage, and characteristics for the relay coil. A relay can control one or more additional circuits, being opened or closed by the solenoid action of the relay's coil.

Relays vary from small panel-mounted units to huge high-voltage relays. However, the essential operations of all are the same. Manufacturers supposedly design coils on electromechanical devices (such as relays, contactors, and motor starters), so they don't drop out (de-energize) until the voltage drops to about of 85% of the rated voltage. Also, coils shouldn't pick up (energize) until the voltage rises to approximately 85% of the rated voltage. However, there is some "fudge factor" built into these numbers, and most electromechanical devices will not drop out until they "see" a lower voltage level. Also, most electromechanical devices will pick up at a lower rising voltage level. Generally, coils on electromechanical devices will operate continuously at 110% of the rated voltage without damage to the coil.

The two important factors in a relay are the coil and contacts. Of these, the contacts generally require greater consideration in practical circuit design. Most relays used in machine control have double-break contacts. The three ratings generally published are:

- Inrush or "make contact" capacity;
- Normal or continuous carrying capacity; and
- Opening or break capacity.

A typical industrial relay may have the following contact ratings: 10A noninductive continuous load (AC); 6A inductive load at 120 V (AC); and 60A make and 60 A break, inductive load at 120V (AC).

A DC resistance is an example of a noninductive load. This may be a resistance unit used as a heating element. An inductive load is a coil (usually a solenoid, contactor coil, or motor starter coil). The important thing is: When determining the contact rating, you must have a clear understanding of which rating is given.

Latching relays. Manufacturers add mechanical latching attachments to control relays, with the resulting devices called mechanically held relays. These have two control coils, one to pull the relay into one position, and one to move it to the other position.

The latching relay operates electromagnetically: It's held by means of a mechanical latch. And by energizing a coil (called the latch coil), the relay operates. This results in the relay's NO contacts closing and the NC contacts opening. When you remove the electrical energy from the relay's coil (de-energize it), the contacts remain in their operated condition. To return the contacts to original condition, you must energize a second coil on the relay, the unlatch coil. The industry refers to this arrangement as a memory relay.

The latching relay has several advantages in electrical circuit design. For example, suppose you want to open or close contacts early in a cycle. At the same time, you may want to de-energize a section of the circuit responsible for the initial energizing of the relay latch coil. Later in the cycle, you can energize the unlatch coil to return the contacts to their original or non-operated condition. The circuit is then set up for the next cycle.

Another use for the latching-type relay involves power failure. Here, you may want the contacts to remain in their operated condition during the power-off period. Conditions in this case are the same after the power failure as they were before.

Quietness of operation is another feature of the latching relay. Since the coil is only energized momentarily, you don't have the usual hum.

Plug-in relays. Some industrial machine relays are available in plug-in types and designed for multiple switching applications at or below 240V. Coil voltages cover standard levels from 6V to 120V. These relays are available for AC or DC operation. Mountings include tube-type socket, square-base socket mounting, or flange mounting using slip-on connectors.

The plug-in relay has a distinct advantage when you want to change relays without disturbing the circuit wiring. In critical operations (where relay service is very hard and downtime is a premium), the plug-in relay may have some advantages. You'll have to assess the actual operating conditions in specific cases to determine their need.

Contactors. A contactor is a large relay, often designed for a specific use (as in controlling lighting circuits). The major difference is in the size range available with contactors. They're capable of carrying current in the range of 9A through approximately 2250A. For example, a size 00 contactor is rated at 9A (200V to 575V). A size 9 contactor is rated at 2250A (200V to 575V).

Like the relay, it's an electromechanical device. The same coil conditions exist in that a high inrush current is available when you energize its coil (generally at 120V). The current level drops to the holding or sealed level when you operate the contacts. Generally, you find contactors in 2-, 3-, or 4-pole arrangements. Manufacturers generally supply one NO auxiliary contact as standard on most contactors. You would use this contact as a holding contact in the circuit, for example, around a NO push button switch. You can get additional NO and NC auxiliary contacts as an option from the manufacturer or ordered as a separate unit and mounted in the field.

You use a contactor for switching power to resistance heating elements, lighting, magnetic brakes, or heavy industrial solenoids. You can also use them to switch motors, if you supply separate overload protection.

Control circuits. The circuits we use to connect all of our controls are obviously important. Three things are essential for control circuits:

- They must be properly designed for their jobs. (To control equipment properly.)
- They must be safe to operate.
- They must be reliable in the long term.

Control circuits usually use standard wiring methods, in normal circumstances-the most common being type THHN conductors in EMT (electrical metallic tubing or "thinwall" conduit). In more difficult environments, such as in or around industrial machines, you will find THHN conductors in heavy-wall (GRC) conduit.

Probably the three most basic applications of control circuitry are:

- You operate a motor start push button switch, which energizes a motor starter coil, closes motor starter contacts, and then energizes a motor.
- A NO thermostat contact closes, energizing the coil of a contactor, which in turn, closes contacts, and then energizes heating elements.
- A NO limit switch contact connected to energize the coil of a relay is held closed (operated) at the start of a cycle. A NO relay contact then closes to energize a solenoid.

Obviously, these are very simple applications. Other control circuits can be very complex, requiring skilled professionals to design, install, and especially troubleshoot them.

Circuit diagrams. In the schematic diagrams used to depict control circuits, symbols represent each component, and every wire is either shown by itself or included in an assembly of several wires that appear as one line on the drawing. However, each wire in the assembly is numbered when it enters and keeps the same number when it emerges to be connected to some electrical component in the system.

Refer to the schematic and wiring diagrams (Fig. 2 and Fig. 3, original article) with this lesson. They show the various devices (in symbol form, as shown in Fig. 1, on page 59, original article) and indicate the connections of all wires between the devices.

The holding circuit. If you refer to the first motor starter control circuit drawing (Fig. 2, original article), you'll see one of the most basic yet important control techniques: the holding circuit (also called the stick circuit). Follow the current through this circuit, starting at the left (at L1). The line L1 represents one of the two power conductors that provide voltage for the control circuit's operation. Current in this control circuit flows from L1 to L2. Presumably, L1 and L2 are two of the three conductors feeding the 3-phase motor this circuit controls. In this case, the control circuit is operating at the same voltage as the motor. Otherwise, you would have a small transformer (commonly called a control transformer) installed to bring the voltage down to an appropriate level for the control circuit.

The current flows first through a STOP button. Notice two things about this device: First, it's an NC switch; second, all the current in the circuit must flow through it. (It's wired in series.) This arrangement ensures when you press the stop button, all current in the control circuit will stop.

The next component in the circuit is the START button. Notice this is a NO contact, and there is a NO set of contacts in parallel with the START button. This is the main component of the holding circuit. This set of contacts is marked "M," which means the "M" coil (further to the right in this control circuit) controls them. So, when there's current flowing through this circuit, the M coil is energized and the M contacts close, providing a path for current flow through the circuit, even without the START switch remaining in the closed position. You must push the START switch to start the circuit's operation, but once you have, the M contacts close and keep the circuit going without the START button remaining closed. This holds the circuit in its energized state.

The M contacts are separated from the M coil in this diagram. These two components, which are not connected electrically, are connected mechanically: The contacts make and break when you energize and de-energize the coil.

Just before this circuit reaches L2, you see three overload contacts. These are NC contacts located in the motor starter. If one of the three conductors exceeds its limit, the overload contact associated with it will open. This will cause the control circuit to open, the M coil to de-energize, the M contacts to open, and the main motor contacts (which are not shown in this drawing, but are operated by the M coil) to open, stopping the motor. Notice that the overload contacts are connected in series, so any one of them opening will de-energize the control circuit. Three overload contacts are shown in this drawing, which is required on all new installations. Keep in mind, however, there are old installations that have only two overload contacts.

A final note on this circuit: If you look at the START and STOP buttons, you'll see "1," "2," and "3" near them. These points represent the three wiring terminals inside the common START/STOP station. They're shown on some schematic diagrams, but are not fully necessary.

Schematic diagrams show the various components as they relate to each other electrically. Wiring diagrams (as shown in Fig. 3, original article) show the components as they relate to each other physically. Most electricians consider schematic diagrams better for understanding the circuit, its components, and its operation. Wiring diagrams, while generally less preferred, are helpful during installation and troubleshooting. This is especially true when trying to plan conductor routings for complex circuits.

<http://m.ecmweb.com/content/correspondence-lesson-2-basic-wired-control-devices>

Correspondence Lesson 3: Process Controls

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Industrial process control can be difficult, complex, and problematic. Yet, a smooth running process line is a site to behold; thanks to technological advances in control devices. In our discussion, the word "process" implies any industrial manufacturing, testing, or assembly operation in which an operator controls, depending on the position or state of the items made, tested, or packaged. Let's take

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Motion control. One of the most important aspects of process control is controlling the motion of the items you're making or modifying. In most industrial processes, the position of an item dictates the operation. Think of an assembly line with dozens of operations. Each operation occurs in a certain location, with the item moving from one machine or process to the next. The goal is to supply accurately and reliably position information by providing an appropriate electrical signal.

Limit switches. The most commonly used process control device is the limit switch. It functions by opening or closing when an item reaches a pre-set location. We subdivide mechanical-type limit switches into those operated by linear and rotary motion. There are large and small switches that operate from linear motion. The precision limit switch is an example of a small switch. It varies from the larger size mainly in a lower operating force and shorter stroke. The operating force may be as low as 1 lb, and the stroke only a few thousandths of an inch.

We generally call limit switches operated by rotary motion rotating cam limit switches. These are control-circuit devices used with machinery having a repetitive cycle of operation; one correlated to shaft rotation. They're used to limit and control the movement of a rotating machine.

The switch assembly consists of one or more snap-action switches. Cams assembled on a shaft activate the snap-action switches. The shaft, in turn, is either direct- or gear-driven by a rotary motion on the machine.

The cams are independently adjustable for operating at different locations within a complete 360 degrees rotation. In some cases, the number of total rotations available is limited. In others, rotation can continue at speeds up to 600 revolutions per minute (rpm).

When selecting a limit switch, you must determine its application in the electrical circuit and consider these factors: contact arrangement; current rating of the contacts; slow or snap action; isolated or common connection; spring return or maintained; and number of normally open (NO) and normally closed (NC) contacts required.

Contacts. In most cases, the switch consists of double-break, snap-action, silver-tipped, or solid silver contacts. The contact current rating will vary from 5A to 10A at 120VAC, continuous. The make-contact rating will be much higher, and the break-contact rating will be lower. Isolated NO and NC contacts are available.

Action. Here the operator has the major decision, to dictate the type of mechanical action available to operate the switch. Length of travel, speed, force available, accuracy, and type of mounting are some considerations.

In discussing the action of limit switches, you should become familiar with the following industry terms: Operating force: The amount of force applied to the switch to cause the "snap over" of the contacts. Release force: The amount of force still applied to the switch plunger at the instant of "snap back" of the contacts to the unoperated condition. Pre-travel or trip travel: The distance traveled in moving the plunger from its free or unoperated position to the operated position. Over-travel: The distance beyond operating position to the safe limit of travel (usually expressed as a minimum value). Differential travel: The actuator travel from the point where the contacts snap over to the point where they snap back. Total travel: The sum of the trip- and over-travel.

Most limit switch manufacturers list this information for certain switches in their specifications. Be careful: The accuracy of switch operators at the point of snap-over varies with different types and manufacturers. In general, it's in the range of 0.001 in. to 0.005 in.

The operator probably having greatest use is the roller lever. It's available in a variety of lever lengths and roller diameters. The next most frequently used operator is the push rod. It can come only as a rod or with a roller at the end. In most cases, particularly with the oil-tight machine-tool limit switch, you can rotate the head carrying the operator to four positions, each 90 degrees apart. You can also mount it either at its top or side. Two other operators used in machine control are the fork lever and the wobble stick.

Symbols. You use the limit switch symbol in applying the limit switch to the schematic or elementary circuit diagram. The symbol represents the switch in four different conditions. Usually, the switch has a NO contact and/or a NC contact. In some switches, there are two NO and two NC contacts.

Show the switch symbol as the switch being in the unoperated condition of the machine. This may result in the operated or unoperated condition.

There are times when you may use both contacts on a given limit switch in a circuit. Under this condition, it helps to join the two contact symbols with a broken line, which indicates they are contacts on the same limit switch.

Proximity switches. One of the most common control switches, the proximity switch is a device capable of acting as an electronic switch when in the presence or close proximity of an object. What differentiates it from a mechanical switch is it does not require physical contact with anything else to operate. The methods of achieving this operation are many: changes in RF fields, magnetic fields, capacitive fields, acoustic fields, and light rays. Objects having ferrous (iron) material content usually alter RF fields.

Magnetic fields can close reed switches. Bringing a magnet up close to the switch or introducing a magnetic material between the magnet and the switch does the activation. Magnets also can alter electrical fields in devices through a phenomenon called the Hall Effect.

Capacitive devices make use of a change in capacitance. This occurs when the sensed object (acting as one plate of a capacitor) passes by the sensor (acting as the other). The sensor "sees" an alternation in the dielectric between the "plates" when detecting nonmetallic objects.

Sonic devices use sound fields interrupted by a detected object. These also detect reflection of sound from objects.

Photoelectric devices work in a similar manner except they detect beams of light rather than sound waves.

All these methods have strengths and weaknesses in actual application. Certain processes use two-wire DC proximity switches because they eliminate the potential for wiring mistakes during installation. Basically, they're not sensitive to voltage polarity.

Inductive sensors. Proximity switches that use RF (radio frequency magnetic wave) field usually employ one half of a ferrite core, with the coil being part of an oscillator circuit. When a metallic object enters this field, the object at some point will absorb enough energy from the field to cause the oscillator to stop oscillating. It's this difference between oscillating and not oscillating that the switch detects as the difference between an object being present or not present. Several variables determine the distance at which this detection takes place, including:

- The diameter of the core (distance varies directly with the core diameter);
- Size of object to be sensed (distance varies directly with size);
- Kind of metal (distance is greatest with iron, less for other metals); and
- Circuit sensitivity (distance is set by circuit design).

Capacitive sensors. These sensors also have oscillators. However, their oscillators usually begin oscillating in the presence of the detected object. This happens when the object creates enough capacitance in a critical part of the oscillator circuit to cause oscillation. In both methods (capacitive and inductive), detection is in the form of the presence or absence of oscillation. You then use this information to do useful work by operating a load directly through a solid-state output circuit or indirectly through a relay.

Other types. The magnet-operated proximity limit switch operates by passing an external magnet near the face of the sensing head. This actuates a small, hermetically sealed reed switch. The 120VAC pilot-duty model also includes an epoxy-encapsulated triac output.

A mercury switch operates by passing a permanent magnet of sufficient strength past the switch.

The vane-operated limit switch actuates by passage of a separate steel vane through a recessed slot in the switch. You attach either vane or switch to the moving part of the machine. As the vane passes through the slot, it changes the balance of the magnetic field, causing the contacts to operate. The switch is available with either a NO or NC contact and can detect very high speeds of vane travel without detrimental effects, such as wear and breakage. There is no physical contact between vane and switch. Therefore, the upper limit on vane speed is governed by factors other than the switch. The vane switch offers excellent accuracy and response time. Repeatability is constant within plus or minus 0.0025 in. or less, provided the path of the vane through the slot is constant. Response time (after the vane has reached the operating point) is less than a millisecond.

Solid-state devices. These devices respond differently than electromechanical output devices, especially to excess currents and incorrect voltage polarities. While excessive currents (if not sustained) do not particularly affect an electrical-mechanical contact device, they might destroy a solid-state contact device. The latter device, however, can be designed so excessive current conditions cannot occur even when a short circuit occurs. In the same way, DC switches, which might be destroyed by a wrong polarity connection, can be designed to not operate on reverse polarity and will not be damaged.

Other semiconductor devices. A photoelectric sensor is an electrical device that responds to a change in the intensity of the light falling upon it.

An LED (light emitting diode) is a solid-state semiconductor, similar electrically to a diode (except it emits a small amount of light when current flows through it in a forward direction). LEDs can be built to emit green, yellow, red, or infrared light; the latter being invisible to the human eye.

A phototransistor is the most widely used optical element in photoelectric sensors. It offers the best trade-off between light sensitivity and response speed compared to photo-resistive and other photo-junction devices.

You use photocells for greater sensitivity to visible wavelengths, as in some color registration and ambient light detection applications.

Photodiodes are reserved for applications requiring extremely fast response time or linear response over several magnitudes of light level change.

Modulated LED sensors. Unlike their incandescent equivalents, LEDs can be turned "ON" and "OFF" (or modulated) at a high rate of speed, typically at a frequency of several kHz. This modulating means you can tune the amplifier of the photo-transistor receiver to the frequency of modulation and amplify only light signals pulsing at that frequency. In our process control industry, we call the modulated LED light source of a photoelectric sensor the transmitter (or emitter), and the tuned photo-device the receiver.

An ambient light receiver is a commonly-used, non-modulated photoelectric device. Red-hot metals and glasses emit large amounts of infrared light. As long as these materials emit more light than the surrounding light level, you can reliably detect them with an ambient light receiver.

Ultrasonic sensors emit and receive sound energy at frequencies above the range of human hearing (above about 20 kHz). We categorize ultrasonic sensors by transducer type, which is either electrostatic or piezoelectric.

Electrostatic types can sense objects up to several feet away by reflection of ultrasound waves from the object's surface. Piezoelectric types generally sense at only shorter ranges.

Remote photoelectric sensors contain only the optical components of the sensing system. Circuitry for system power, amplification, and output are all at another location, typically in a control panel. Thus, remote sensors are generally smaller and more tolerant of hostile sensing environments than self-contained sensors.

Self contained photoelectric sensors contain the optics along with all the electronics. Their only requirement is a source of voltage for power. The sensor itself does all of the work, which includes modulation, demodulation, amplification, and output switching.

Optical fiber. There are many sensing situations where space is too restricted or the environment too hostile for even remote sensors. For such applications, photoelectric sensing technology offers optical fiber as a third alternative in sensor technology. These are transparent strands of glass or plastic used to conduct light into and out of such areas. Fiber light guides, used along with either remote or self-contained sensors, are purely passive components of the sensing system. Since they contain no electrical circuitry and have no moving parts, they can safely pipe light into and out of hazardous sensing locations and withstand hostile environmental conditions.

<http://m.ecmweb.com/content/correspondence-lesson-3-process-controls>

Correspondence Lesson 5: Safety And Troubleshooting

[Paul Rosenberg](#)

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Jun 01, 1999

Obviously, safety is paramount when you're working on control systems (or the machines under their control). This is true for many reasons. Aside from potential injury, accidents can directly and indirectly affect both you and your company. **Personnel safety.** On whatever control systems you design, install, or maintain, it's critical you incorporate guards, shields, monitoring devices, and interlocks

Obviously, safety is paramount when you're working on control systems (or the machines under their control). This is true for many reasons. Aside from potential injury, accidents can directly and indirectly affect both you and your company.

Personnel safety. On whatever control systems you design, install, or maintain, it's critical you incorporate guards, shields, monitoring devices, and interlocks in every applicable location. You should worry about the safety of everything and everyone associated with it.

The first level of maintaining operator safety is to stop the motion of any moving mechanism having the potential to injure any portion of a worker's body. Here, you connect the safety device to the control circuit like a second stop button. That is, you locate it in a series portion of the motor control circuit, just like the usual stop button. Obviously, this must be a normally closed (NC) device. When the device senses a worker's presence (or any other object) in the danger zone, the control circuit opens, shutting down the operation. But pay attention here: Some processes don't shut down quickly. In such cases, you must have other plans. One example is a plugging circuit, which momentarily reverses current to the motor, stopping it quickly. The type of switched device depends on the manner of activation.

When an operator approaches a moving mechanism, he or she could:

- Break the light beam of a photoelectric switch;
- Move a gate that activates a limit switch; or
- Move off of a footpad that activates a pressure switch.

Other common types of devices used for safety circuits include ultrasonic switches, proximity switches, and push buttons with oversized activators.

There are two primary methods for using electrical controls in safety circuits:

- Circuits that activate a guard, which prohibits the operator from reaching into the danger zone at any time (machine stopped or running).
- Circuits that immediately stop the machine when any portion of the operator's body is in the danger zone.

There are three types of safety control techniques. One is installing a multiple stop device in series. A second is installing a redundant stop device in case of failure of one device. The third is installing an interlocked control circuit, which inserts an NC contact from one device's controls into the control circuit of another. This ensures device A will not operate while device B is energized.

You usually design safety circuits specifically for the shape and geometry of the machine, where the operator is in relation to the machine, the speed of the mechanism, and other factors you can determine only after watching how the operator moves and works. As you can see, safety is a continuous process you must constantly refine and improve. The way people use machines changes over time. The way they receive materials, how they're loaded and unloaded, and what their duty cycles are may change.

During the design stage, you should think about the operation of the machine, but you may not be able to foresee some of the fine operational details that will occur through years of use.

When you're installing a control system, you'll spend a great deal of time in the areas where operators use these control systems. Hopefully, you'll gain some familiarity with the people who run the machines. Here's where you may be able to notice safety issues during the design stage.

When maintaining a control system, you'll notice changes to it as well as the machine itself, its process, and the operator working with your system. In any case, and at whatever time you see a potential safety problem, it's your responsibility to take care of it.

Facing reality. The safety of systems we install is a serious issue, affecting peoples' bodies and lives. Machines are dangerous, and no matter how hard we try to eliminate all accidents, they still happen. If you are to be involved with machine controls, it is critical to you (for your own personal benefit) to consider the real situations you face.

This issue, more than any other we'll cover in this course, deserves plain-spoken coverage.

The first common safety problem many of us face is that of management not wanting to spend the necessary money. Let me begin by saying that almost any manager in the world would be sickened by one of their employees being maimed or killed. Safety problems that you may have with management are not going to be that they don't care about

people being maimed; they will be that they do not understand the gravity of the situation. Production pressure is foremost in their minds, and “adding a guard to a machine” just doesn’t stick out.

That means it’s your responsibility to get the point across. If you design, install, and maintain control systems, you are in a highly-responsible positions—you’ll have to be ready to make a stand.

If you notice an acutely unsafe situation (capable and likely to cause injury at the present time), it’s your responsibility to eliminate it immediately. It is your job to make an intelligent decision quickly. If you have to shutdown a machine or assembly line, do it. It is your responsibility to everyone who works on that equipment to keep them safe. If taking such an action will result in your job being threatened, you must do it anyway. You are responsible, and you must be able to exercise that responsibility. If you cannot or will not—find a different position.

If you run into a situation where there is a concern, but that you don’t think is likely to cause injury right away, try to eliminate it as quickly as you can, without creating a disturbance in your company’s operation. If it requires extra resources, go to your manager and explain the situation. In most cases, the manager will tell you to get the job done right away. In the rare instance where the managers do not want to do the work, explain it again, and try to make them understand the seriousness of the matter. If they still refuse, it is your obligation to quit your job immediately. If you can’t stand up to this type of choice—do a different job.

Safety programs. Don’t rely on the current fad of safety programs—in general, they’re designed and promoted by consultants and quasi-governmental activists who end up getting a piece of the action somewhere along the line. Any program that gives you a template of exactly what you should inspect, write down, and report is suspect. Not because they are telling you to do wrong things, but because they are telling you how to think. Take whatever good ideas they have (and they do have some), and make the reports that your employers ask you to; but never allow someone else to do your thinking for you.

Take whatever good ideas your safety plan has, and follow your instructions, but never, ever, let the “plan” substitute for your own good sense and diligence.

What if? Although few of us will ever come face-to-face with this, that fact is that accidents can and do happen to people who work on our machines.

This is an ugly issue, but an important one, especially if it happens to you. There are three critical time periods surrounding any accident you are involved with:

1. Preparation before the accident occurs.
2. During the emergency situation.
3. Afterward.

Before the accident occurs: If the possibility of an accident exists, as it does in our business, it makes sense to prepare yourself for an accident. Face the fact that such a thing could happen. It is easy to ignore such an ugly subject as a serious accident, but you must not, for two reasons:

First, because turning away from the subject will prevent you from recognizing some preventative actions. Once you decide that accidents can happen, but that you won’t let them happen, you will be more effective at preventing accidents.

Secondly, because facing the subject head-on will get you prepared to get through the trauma of an accident, should one really occur. If you can honestly say (to yourself) that you did consider the possibility as well as you could, you’ll fare much better.

Bear this in mind—You will never be able to prepare perfectly for safety. You do not have perfect knowledge. Neither can you spend all of your time and energy on accident prevention only—that is not realistic. But you should do enough preparing now, so that if something bad does occur, you won’t have reason to be angry with yourself. Once in a while, take a walk through your area with the goal of spotting possible trouble spots. Most likely, you will not find any. But an occasional tour will help you spot anything that is a real problem.

Don’t invent problems, and don’t expect to eliminate every possible risk—that’s impossible. Your goal is to keep your systems free of reasonably avoidable risk.

In all your analysis of safety, you will be making judgements as to what is an acceptable risk and what is not. Your judgement is not perfect, nor should you expect it to be. But it should be reasoned. Beyond that you can do no more.

During an accident: If an accident does occur, you must make sure that you take care of first things first. In many ways, facing an industrial accident is very similar to facing a combat situation: a mixture of horror, shock, and disorientation. In such a situation, your perceptions of time and distance are altered, and your focus narrowed to a sort of tunnel-vision.

Under such circumstances, you will need to keep your actions as simple as possible. Do the really important things first—shutting down the machines, calling the paramedics, and administering first aid (or finding someone who can). After that, you can start to think about other things. Keep it very simple.

Few of us can handle such things very well, and most of us get physically ill during or afterwards. You cannot help this—it's simply the way that human beings are.

Again, should an accident occur, keep it very simple, and take care the few really important things that only you can do (such as shutting down the electrical equipment). After you finish that, you can take care of other things. And expect to experience some sort of physical and psychological shock—that is just what happens to us. Immediately after the accident, let yourself wind-down and come back to normal slowly—it will probably take several hours.

Afterward: After an accident, your problems will be emotional. You will almost certainly feel guilty in one way or another. Again, this is normal. Your goal at this point is not to let feelings of guilt get out of hand. Think about it reasonably, and accept no unearned guilt. Guilt is so deeply conditioned in most of us that it can be a very powerful force. Think about it rationally, and not just emotionally.

If you feel bad about the accident for very long, get some stress counseling. Getting counseling does not mean that you are weak—it means you've been through some really tough stuff, and you have sense enough to get some help from someone who has been through it before. Getting counseling does not mean that you are weak - it means that you've been through some really tough stuff, and you have sense enough to get some help from someone who has been through it before. Take the help.

The safety of the mechanism. You should also consider safety at points physically away from the actual mechanism but ones that still affect the movement of the mechanism. Examples include the circuit breakers that control the electrical power to the machine and the push button controls for the machine. When maintenance personnel work within the danger area of a machine, they should always disconnect the power and lock the control device with a key. A person responsible for the safety of the workers in the area usually holds this key, which also unlocks and reactivates the power.

Under the law, an employer must provide employees with a safe place to work and rules for them to perform their work safely. The federal government enacted the Occupational Safety and Health Act, which created The Occupational Safety and Health Administration (OSHA) as an agency within the Department of Commerce. This group is responsible for monitoring the safety of workers and prosecuting any company violating the agency's policies. Besides OSHA, other organizations are also involved in worker safety, such as insurance companies, industry-specific agencies, and local and state government agencies.

Machine safety needs. Many machines designed today are capable of damaging other machines. For example, in complex manufacturing work cell designs, many independent machines move in and out along controlled paths that intersect the paths of other machines. If any one of the machines stops or malfunctions, others could crash or become entangled with the defective one. This type of industrial accident would not involve an employee, but damage to expensive equipment could stop a company's entire production line.

Today's articulated machines, such as robots and coordinated machines such as machining centers, need protection from themselves and other machines. For example, in machining cells, robot arms frequently pass through the line of travel of adjacent machines.

Machine safety in a coordinated and programmed manufacturing system involves the use of detectors and programmed safety sequences. Examples of common detectors include:

- Zero-speed switches (for detecting stopped motors);
- Over torque switches (for detecting jammed parts);
- Proximity switches (to detect over-travel);
- Flow switches (to detect loss of lubricating oil);

- Pressure switches (to detect jammed hydraulic actuators); and
- Temperature switches (to detect overheating).

Safety sequences are very difficult to put into place, because each machine should work independently. Therefore, when one machine joins a coordinated work process with others, there's sometimes a need for a master controller. This controller is not only responsible for the correct sequencing of each machine in the process pattern, but also for the recognition of a problem, determination of the severity of the problem, and the best corrective action. The corrective action is in the form of a programmed sequence of moves given to each machine in the process. The sequence involves the following actions:

- Stop all machines;
- Evaluate the current position of each machine;
- Begin to retract each machine to a home position;
- Notify the operator of the detected problem;
- Identify the defective machine and source of the problem; and
- Wait for a restart command from the operator.

Troubleshooting. Always turn off all power and use lockout/tagout procedures in any situation where you must come in contact with the circuit or equipment. Make sure no one but you can turn on the equipment.

Use only well-designed and maintained equipment to test, repair, and maintain electrical systems and equipment. Use appropriate safety equipment such as safety glasses, insulating gloves, flash suits, hard hats, insulating mats, etc.

Effective troubleshooting starts with analysis of the problem. Breaking it down into sections limits the size of the job. Considering categorizing the following areas:

- Electrical or electronic;
- Mechanical;
- Fluid power;
- Pneumatic; and
- Personnel.

In many cases, the problem may be a combination of two or more of these areas. For example, problems in the electrical contacts or the mechanical operator may cause a limit switch to not function properly.

As long as people operate machines, problems will arise that do not respond to the usual form of troubleshooting. These problems may stem from misunderstanding, lack of cooperation, or lack of knowledge of the machine. Whatever the cause, you should handle them carefully and diplomatically so you quickly can return the machine to its intended job.

You can further separate problems into physical location or type of operation. For example, you may find a problem localized in only one section of a complex machine. This immediately eliminates the rest of the machine as a possible trouble source. Success in troubleshooting frequently lies in the ability to segregate the problem area from other unrelated circuitry.

Here are the common problem spots.

Blown fuses. You must eliminate the reason for the overload and replace the fuse with the proper type.

Loose connections. There could be dozens of connections on a given machine. Each of these spots may be a source of trouble. A loose connection in a power circuit can generate local heat, which spreads to other parts of the same component, other components, or conductors. One example of where direct trouble can arise is in thermally sensitive elements, which can be overload relays or thermally operated circuit breakers.

Faulty contacts. Such components as motor starters, contactors, relays, push buttons, and switches apply here.

Problems in the NC contact are of the most difficult to locate. A contact may look closed, but still not conduct any current. Check any contact that has had an overload through it for welding. Weak contact pressure, dirt, or an oxide film on the contact surfaces will prevent it from conducting.

Many times, you can clean contacts by drawing a piece of rough paper between them. Use only a fine abrasive to clean contacts, and do not file them. Most contacts have a silver plate over the copper. If you destroy this by filing, the contact will have a short life. If a fine abrasive will not clean them, it's better to change them out.

Another problem occurring with double-pole, double-break contacts is cross-firing. That is, one contact of the double break travels across to the opposite contact, but the other remains in its original position. If you're using both the NO and NC contacts in the circuit, a malfunction of control may occur.

Incorrect wiremarkers. This problem usually appears on the builder's assembly floor or in reassembly in the user's plant. The error can be difficult to locate, as a cable may have many conductors running some distance to various parts of the machine.

Combination problems. Typically, these problems can be electrical-mechanical, electrical-pressure (fluid power or pneumatic), or electrical-temperature.

The greatest problem is not always indicative of which aspect is at fault. It may be both. It's usually faster to check the electrical circuit first; however, you must check both systems. For example, few solenoid coils burn out due to a defect in the coil. Most solenoid troubles on valves develop from a faulty mechanical or pressure condition that prevents the solenoid plunger from seating properly. This condition causes the solenoid to draw excessive current. The result is an overload or burned-out solenoid coil.

Low voltage. If no immediate indication of trouble is apparent, one of the first checks to make is the line and control voltage. Due to inadequate power supply or conductor size, low voltage can be a problem. This problem generally shows up more on starting or energizing a component, such as a motor starter or solenoid.

Heat is one result of low voltage you may not notice immediately in the functioning of a machine. As the voltage drops, the current to a given load increases. This produces heat in the coils of the components (motor starters, relays, solenoids), which not only shortens the life of the components but may also cause malfunctioning. For example, heat can cause moving metal parts with close tolerances to expand to the point of sticking.

Grounds. There are many locations on a machine where a grounded condition can occur. However, there are a few spots in which grounds occur most often, including:

- Connection points in solenoid valves, limit switches, and pressure switches. Due to the design of many components, the space allowed for conductor entrance and connection is limited. As a result, a part of a bare conductor may be against the side of an uninsulated component case.
- *Raceway openings.*
When pulling conductors through conduits or into pull boxes and cabinets, you can scrape or cut the conductor insulation. If you don't eliminate sharp edges or burrs on freshly cut or machined parts, cuts and abrasions can occur.
- *Loose strands.*
The use of stranded wire greatly reduces many problems in machine wiring. However, you must be careful when placing a stranded conductor into a connector: You must use all the strands. One or two unconnected strands can touch the case or a normally grounded conductor, creating an unwanted ground. Even if the ground condition does not appear, you reduce the current-carrying capacity of the conductor.

<http://m.ecmweb.com/content/correspondence-lesson-5-safety-and-troubleshooting>

Correspondence Lesson 6: Distributed Intelligence Controls

[Paul Rosenberg](#)

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Microchip-based electrical control products account for an increasing share of the electrical controls market. How we use electrical controls affects everyone: from the specifier to the electrical contractor to the maintenance electrician. It also affects how you interface with your local utility company, especially in the area of coordinated load management. Computer-based control technologies continue

Microchip-based electrical control products account for an increasing share of the electrical controls market. How we use electrical controls affects everyone: from the specifier to the electrical contractor to the maintenance electrician. It also affects how you interface with your local utility company, especially in the area of coordinated load management.

Computer-based control technologies continue to evolve from central intelligence and control to distributed intelligence and control. Our first generation of computer-based electrical controls focused around a central controller, which received all information, made every decision, and oversaw every operation. Similarly, as computer systems evolved into networks, so have control systems.

In this lesson, we'll look at distributed intelligence control schemes: BACnet and LonWorks. The HVAC industry developed BACnet, which is widely used in many of the more complex "Intelligent building" systems. However, space prohibits us from covering this system. LonWorks is a control network (like a data network), but it operates at lower speeds and increases reliability.

Echelon developed the LonWorks system with a goal to establish a commodity solution to the problems of designing and building control networks. Currently, Echelon has at least 1000 users (mostly large manufacturing and utility companies) for its 100+ products. It currently uses LonWorks for process control, building automation, engine control, elevator control, life safety systems, power distribution controls, and similar applications. Many other firms, such as Ameritech, use LonWorks in test facilities before adopting it as a standard.

The LON (Local Operating Network) concept. The LonWorks communications structure is similar to that of a LAN because a number of processors continually exchange messages. But, because the LonWorks system operates under a different set of rules than LANs, they are termed Local Operating Networks, or LONs.

LON technology offers a means for using distributed systems that perform sensing, monitoring, control, and other applications. It also allows intelligent devices, such as actuators and sensors, to communicate with one another through an assortment of communications media. LON technology supports distributed, peer-to-peer communications. Individual network devices communicate directly with one another without requiring a central control system.

Purpose distinguishes LONs from LANs. You design LANs to move data (such as documents, images, and databases) among computers, shared disks, and printers. You view a LAN's performance in terms of its throughput, or the amount of data transmitted and received over the network (measured in megabits transmitted per sec).

You design a LON to move sense and control messages. These are typically short messages that contain commands and status information to trigger actions. You view LON performance in terms of transactions completed per sec and response time.

The critical factor in LAN technology is data speed: in LON technology it's the assurance of correct signal transmission and verification. Control systems do not need vast amounts of data, but they do require correct send and receive messages.

How the system works. An advantage of LonWorks networks is their ability to communicate across different types of transmission media such as powerlines, twisted pair control wiring, coax cable, optical fibers, via radio waves, etc.) in a single network.

For example: A forklift approaches a conveyor belt to pick up a pallet. As it approaches, the conveyor activates to position the pallet properly for loading, then turns off after the pallet is loaded. At the same time, a record enters in the inventory control system to track movement of the pallet from the warehouse to the shipping department.

In this application, the forklift contains a node (a single control location) that uses radio frequency for communication with the conveyor belt node. In turn, the conveyor belt node uses twisted pair media to communicate with power actuators and interfaces to the inventory control system.

Another application might be this: An intruder enters a secured area of a building after hours. A motion detector (connected to other motion detectors with twisted-pair wiring) sends a message to the powerline-based lighting system to illuminate the intruder's area and sends a message by radio frequency to sound the alarm at the security entrance to the building. The security gate receives the message sent over the powerlines by the motion detector and seals off the area so the intruder can't exit.

Although we can accomplish the two examples above with our current home control systems, we can't do them nearly as well. The LonWorks system is almost error-proof.

As all of us who have worked with the most popular home control systems know (referring primarily to X-10 based systems), they require a lot of troubleshooting. There are all sorts of filters, bridges, and associated changes required. The LonWorks system works as reliably as a hard-wired control system. Double redundancy and verifications are built-in.

The Neuron chip. The basic component of the LonWorks system is a special computer chip, called a Neuron chip. Along with a power supply and communications transceiver, this chip makes a fully functional control node. Fig. 1, on page 52, shows the basic LonWorks node. Notice each control location has a physical ID, timer, computation unit (a small CPU), device controllers, power supply, and I/O ports. This makes the node a sophisticated computer. Moving to the center of this drawing, notice the node contains a data storage section and communications transceiver for sending and receiving control signals.

The Neuron chip, which is approximately 1/2-in. square, differs from standard computer chips in that it has several different parts performing several different duties at once (Fig. 2, on page 54). It's a processor and device controller, as well as a memory chip with built-in EEPROM memory. When installing Neuron chips, or even after they are in service, you must use special tools to program them.

Transceiver modules. Like power supplies, you must add transceiver modules to the chips for them to operate correctly with other types of equipment. Since these chips can accommodate virtually every type of communications media, no one type of transceiver works every time. Therefore, basic transceiver functions are built into the chip, and you must connect separate modules to it to complete the transceiver.

The most common types of transceiver modules are twisted pair, power line transmission, and RF transmission.

One of the more important types of transceiver configurations is the power line module. Sending data safely over power lines can be difficult. Echelon's methods uses a spread-spectrum transmission (100 kHz to 450 kHz), error correction, and a variety of other techniques to reach transmission speeds of 10kbits/sec.

Distributed intelligence. One important thing to remember about the LonWorks system is: There is no central controller. Instead of using central intelligence, this system uses distributed intelligence.

When one component in a LonWorks system activates or detects a certain condition (temperature, proximity, light, time, etc.), it informs the system. The installer programs each node with all the information it needs to function. (The programming is modifiable at any time.) You can change, add, or remove parts of the system without major reprogramming. In effect, each node has its own brain and can communicate with every other part of the system.

Message passing. There are a number of trade-offs between network efficiency, response time, security, and reliability. LonWorks defaults to the greatest degree of safety and verification for all communications over the LON network.

The LON TALK protocol offers four basic types of message service:

1. The most reliable service is acknowledged, or end-to-end acknowledged service, where you send a message to a node or group of nodes and expect individual acknowledgments from each receiver. If you don't receive an acknowledgment from all destinations, retry the transaction. The number of retries and the time-out are both selectable. The network CPU generates acknowledgments without intervention of the application. You use transaction IDs to keep track of messages and acknowledgments so the application does not receive duplicate messages.
2. An equally reliable service is request/response, where you send a message to a node or group of nodes and expect individual responses from each receiver. The application on the receiving side processes the incoming message before generating a response. The same retry and time-out options are available as with acknowledged service. Responses may include data, so this service is particularly suitable for remote procedure call or client/server applications.

3.The next reliable is unacknowledged repeated, where a message is sent to a node or group of nodes multiple times expecting no response. You'd typically use this service when broadcasting to large groups of nodes, in which the traffic generated by all the responses would overload the network.

4.The least reliable method is unacknowledged, where you send a message once to a node or group of nodes expecting no response. Typical uses range from when the highest performance is required, network bandwidth is limited, and the application is not sensitive to the loss of a message.

Design. The design approach with LonWorks is: to define the required system nodes; define the connections between nodes; and write application programming (also called code) for each node. With the proper design, the nodes become generic building blocks, applied in various ways to accomplish tasks (to control the lighting on many different buildings using a variety of communications media, etc.). The connection and configuration of each node determines their tasks. Because hardware, software, and network design are independent in a LonWorks-based system, you can program a node's functions without concern about the specifics of the networks.

<http://m.ecmweb.com/content/correspondence-lesson-6-distributed-intelligence-controls>

Correspondence Lesson 7: Motor Controls

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We use motors more than any other electrical device. They vary in size from specially designed medical motors that are less than 1 in. long to gigantic industrial units of several thousand horsepower. In between are hundreds of different types of motors for thousands of different applications. To better understand the rules regarding the application and wiring of motors, as well as the control details,

We use motors more than any other electrical device. They vary in size from specially designed medical motors that are less than 1 in. long to gigantic industrial units of several thousand horsepower. In between are hundreds of different types of motors for thousands of different applications.

To better understand the rules regarding the application and wiring of motors, as well as the control details, we have categorized concerns into four sections.

Mechanical safety. We ensure the motors themselves are not a source of danger. For instance, we would not want to install open motors in areas where children may go; it's too easy for them to stick their fingers into operating motors. It's also often necessary to put a clutch on a motor, to avoid possible injury to a machine operator.

Mechanical stability and operations. Motors have mechanical stresses placed upon them. One of the primary forces is vibration, which unfortunately loosens bolts and screws. This also affects the equipment the motor operates and surrounding items.

Electrical safety. Motors should not become the source of an electrical shock or fault. They should not cause problems to the electrical system.

Operational circuits. The circuits on which you install the motors should operate continually and correctly. Motors place unusual demands on electrical circuits. They can cause large starting currents. (Fully loaded motors can draw starting currents of four to eight times their normal full-load current; in some circumstances even higher.) They also put inductive reactance into electrical systems. Because of the high currents some motors draw, they overheat electrical circuits more commonly than many other types of loads.

The basics. The operation of electric motors involves not only current and voltage, but also magnetic fields and their associated characteristics. All electric motors operate by using electromagnetic induction, which is the interaction between conductors, currents, and magnetic fields. Any time an electrical current passes through a conductor, it causes a magnetic field to form around that conductor. This is a law of physics. Conversely, any time a magnetic field moves through a conductor, it induces (causes to flow) an electrical current in that conductor. By manipulation of these two laws, in combination with magnetic attraction and repulsion, you can operate a motor. The operation of

electric motors is such that, by intelligent use of electromagnetic induction, electricity turns into physical force, causing the motor to turn.

Here is the step-by-step operation of an electric motor. An electrical current flows through the motor's windings, causing a strong magnetic field to form around the windings. This magnetic field attracts the rotor (the part in the center of the motor that turns; the shaft is at the center of the rotor) and moves it toward the magnetic field, causing the initial movement of the motor. Various means of rotating the magnetic field perpetuate this movement. Various types of motors do it differently, although the most common method is by using several different windings and sending current to them alternately, causing magnetic strength to be in one place one moment, and another place the next. The rotor then follows these fields, causing continuous motion.

Motor controllers. The simplest controller is the branch-circuit protective device. With this, you can safely control motors of 1/8 hp or less that you would normally leave running. A simple "controller" is a cord-and-plug connection for portable motors of 1/3 hp or less. Controllers must have horsepower ratings no lower than the horsepower rating of the motor they control, except in these two cases.

- Stationary motors 2 hp or less and 300V or less can use a general-use switch that has an ampere rating at least twice that of the motor it serves. You can use general-use AC snap switches on AC circuits to control a motor rated 2 hp or less and 300V or less; having an ampere rating of no more than 80% of the switch rating.
- You can use a branch-circuit inverse-time circuit breaker rated in amperes only (no horsepower rating). Unless a controller also functions as a disconnecting means, it does not have to open all conductors to the motor. If you supply power to a motor by a phase converter, you must control the power in such a way that, in the event of a power failure, power to the motor cuts off and cannot reconnect until you restart the phase converter. Each motor must have its own controller, except when a group of motors (600V or less) uses a single controller rated at no less than the sum of all motors connected to the controller. This applies only in the following three cases:
 - If a number of motors drive several parts of a single machine;
 - When a group of motors is protected by one overcurrent device, as specified elsewhere; and
 - Where the group of motors is located in one room, within sight of the controller. A controller must be capable of stopping and starting the motor, and interrupting its locked-rotor current.

The disconnecting means must be within sight of the controller location and within sight of the motor, except in the following two situations:

- If the circuit is more than 600V, you can place the disconnecting means out of sight of the controller; as long as the controller has a warning label indicating location of the disconnecting means to be locked in open position.
- You may locate one disconnecting means next to a group of coordinated controllers on a multi-motor continuous process machine. The disconnecting means for motors 600V or less must be rated at least 115% of the full-load current of the motor being served. A controller operating motors more than 600V must have the control circuit voltage marked on the controller. You must provide fault-current protection for each motor operating at more than 600V. (See NEC Sec. 430-125(c).)

All exposed live parts must be protected. (See Sec. 430-132 and Sec. 430-133 if necessary.)

Motor control installation requirements. Motor control circuits tapped from the load side of a motor's branch-circuit device controlling its operation are not considered branch circuits. You can protect them with a supplementary or branch-circuit protective device. Control circuits not tapped this way are signaling circuits: You must protect these following Art. 725.

You should protect motor control conductors usually with an in-line fuse and in accordance with Column A of Table 430-72(b), except:

- If they extend no further than the motor controller enclosure, you may then follow Column B of Table 430-72(b).
- If they extend further than the motor controller enclosure, follow Column C of Table 430-72(b).
- The primary side of the transformer protects control circuit conductors, taken from a single-phase transformer that has only a 2-wire secondary. However, the primary protection ampacity should not be more than the ampacity shown in Table 430-72(b) multiplied by the secondary-to-primary voltage ratio (secondary voltage divided by primary voltage).

- When the opening of a control circuit would cause a hazardous situation (as would be the case with a fire pump, etc.), the control circuit can tap into the motor branch circuit with no further protection. Control transformers must be protected according to Article 450 or Article 725, except:
- Protect control transformers that are an integral part of a motor controller and rated less than 50VA by primary protective devices, impedance limiting means, or other means.
- If the primary rating of the transformer is less than 2A, you may use an overcurrent device rated at no more than 500% of the primary current in the primary circuit.
- By other approved means.
- When the opening of a control circuit would cause a hazardous situation (as would be the case with a fire pump, etc.), you may omit protection. When damage to a control circuit would create a hazard, you must protect the control circuit (by raceway or other suitable means) outside of the control enclosure. When one side of a motor control circuit is grounded, the circuit must be arranged so that an accidental ground will not start the motor. You must arrange motor control circuits so they will shut off from the current supply when the disconnecting means is in the open position.

Overload relays. The motor starter is similar to the contactor in design and operation. Both have one feature in common: contacts operate when the coil energizes. The important difference is the use of overload relays on the motor starter.

The magnetic motor starter has three main contacts in the form used most frequently. These contacts are normally open. You use this arrangement in the starting of 3-phase motors. Almost all industrial, institutional, or large commercial facilities in the U.S. use 3-phase power. A few areas (such as upstate New York and a few others) use 2-phase, 4-wire power. In such cases, you can order four normally open contacts for the starter. Like the power contactor, magnetic motor starters are available in many sizes. The smallest units are approximately the size of a relay, and the largest starters can take up most of a room.

Motor starters can close and open the contacts that connect the motor to the source of electrical power. Unfortunately, it is not always possible to control the amount of mechanical load applied to the motor. Therefore, the motor may be overloaded, resulting in damage. This is why we add overload relays to the motor starter.

The goal is to protect the motor from overheating. The current drawn by the motor is a reasonably accurate measure of the load on the motor, and thus of its heating. Thus we call this protective device an overload device.

Most overloads today use a thermally responsive element. That is, the same current that goes to the motor coils (causing the motor to heat) also passes through the thermal elements of the overload relays.

After mechanically connecting the thermal element to a normally closed contact, any excessive current flowing through the thermal element for a long enough time period will trip the contact trips open. This contact connects in series with the control coil of the starter. When the contact opens, the starter coil de-energizes. In turn, the starter power contacts disconnect the motor from the line.

A motor can operate on a slight overload for a long period or at a higher overload for a shorter period. Overheating of the motor will not result in either case. Therefore, the overload heater element should be designed to have heat-storage characteristics similar to those of the motor. However, they should be just enough faster so the relay will trip the normally closed relay contact before excessive heating occurs in the motor.

The ambient temperature in the location of the motor and starter also has some effect. It's necessary to specify the rating of a given temperature base plus the allowable temperature rise due to the load current. For example, an open-motor rating is generally based on 40DegrC (104DegrF). The motor's nameplate will specify the allowable temperature rise from this base.

Because the motor and starter are usually in the same ambient temperature, the same temperature conditions affect the overload-heater elements. The overloads will open the motor starter control circuit through excessive motor current, a high ambient temperature, or by a combination of both.

The ambient-compensated overload relay operates through a compensating bimetal relay. The relay maintains a constant travel-to-trip distance, independent of ambient conditions. Operation of this bimetal relay is responsive only to heat the motor overcurrent generates.

All overload relays should be trip free. Overload relays are not intended to protect against short-circuit currents. Short circuit protection is the function of fuses and circuit breakers.

Another feature of the motor starter is the auxiliary contact, which is normally used in the control circuit. In some small starters, an additional load contact can be used instead.

Auxiliary contacts may be normally open or closed and have 10A rating.

Reversing starters. Reversing and multispeed motor starters are specialized applications of across-the-line starters. In the reversing starter, there are two starters of equal size for a given horsepower application. You accomplish the reversing of a 3-phase, squirrel-cage induction motor by interchanging any two line connections. You must properly connect the two starters to the motor so the line feed from one starter is different from the other. We use mechanical and electrical interlocks to prevent both starters from closing their line contacts at the same time.

Multispeed starters. Many industrial applications require the use of more than one speed in their normal operation. While many new methods for adjustment of speed are available, the pole changing method is still used. For example, in a squirrel cage induction motor, the speed is dependent upon the number of poles. This is obtained by the design of the stator winding. At 60 Hz, which is standard in the United States, a two-pole motor operates at approximately 3600 revolutions per minute (r/min). A 4-pole operates at 1800 r/min, a 6-pole at 1200 r/min, an 8-pole at 900 r/min, and a 10-pole at 720 r/min. By having two or more sets of winding leads brought out into the terminal connection box, the number of effective poles can be changed. With a change in speed, the horsepower will also change. For example, in a two-speed motor for which the slower speed is one-half the higher speed, the horsepower will also be one-half the horsepower of the higher speed. This means that two sets of overload relays must provide adequate protection.

Two features common to complex motor starters are no-voltage release and no-voltage protection (sometimes seen as low-voltage release and low-voltage protection). No-voltage or low-voltage release means that if there is a voltage failure, the starter will open its contacts. However, the contacts will close again as soon as the voltage returns. No-voltage or low-voltage protection means that if there is a voltage failure, the starter contacts will open, but they will not reclose automatically when the voltage returns.

Each set of STOP/START push buttons may be located remote from any of the other sets to energize or de-energize a motor starter, so long as all the stop buttons remain in series with all circuit current flowing through them. For safety, the START push buttons may have individual locks to prevent unauthorized operation of any units. The use of a pilot light to indicate that a motor is energized is a safety factor along with a jogging feature. You wire a jogging switch like a start switch, but without the parallel holding contacts, so the motor runs only as long as you depress the jog button.

You can sequence a number of motors into operation using the auxiliary contact of the preceding motor starter. This gives a short, fixed-time delay of only the starter closing time. The opening of any of the motor starters through overload will de-energize all the motors in sequence. All the motors will be de-energized through operation of the one common STOP push-button switch.

Reduced-voltage starters. You can divide reduced-voltage motor starters into several different designs: Autotransformer (or compensator), primary resistor, wye (or star) delta, and part winding.

Connecting a motor directly across the line, the resulting current at start condition is frequently 4 to 12 times the full-load rating of the motor, and occasionally higher. This high starting current often causes voltage sags if the power supply is inadequate or the motor is large.

The basic principle of the reduced-voltage starter is to apply a percentage of the total voltage to start. After the motor starts to rotate, switching is provided to apply full line voltage. For example, in the autotransformer type, starting steps may be 50%, 65%, or 80% of full voltage.

At reduced voltage, the torque available from the motor reduces. This may cause concern when the motor is starting under load. Torque varies as the square of the impressed voltage.

For example, starting on the 50% voltage tap, torque will be 25% of normal. On the 65% voltage tap, torque will be 42% of normal. On the 80% voltage tap, torque will be 64% of normal.

Solid-state starters. Many applications require the lower starting torque and smooth acceleration that solid-state systems offer. Typical of such applications are conveyor systems, pumps, and compressors. Solid-state reduced-voltage controllers provide smooth, stepless acceleration of a motor through the use of silicon-controlled rectifiers. By

controlling the conduction of the silicon-controlled rectifiers, voltage is gradually applied to the motor. We sometimes call this a soft start to the motor. An adjustable current-limit feature limits current to 25% to 70% and starting torque to 6% to 40% of full voltage values.

Solid-state reduced-voltage starters are available in voltage ratings of 200V, 230V, 460V, and 575V to 600V. Such starters are available from 1 hp through 1,000 hp at 480V or 600V, and from 10 hp through 300 hp at 208V or 240V.

Variable-frequency drives. One thing that never changes in traditional electrical work is the standard 60 Hz frequency of our AC power. But the rotational speed of an AC motor is a direct function of the frequency of the current it uses. That frequency (as well as phasing and number of poles) determines the speed of the motor's rotating magnetic field.

Equipment, such as boiler fans, large pumps, or conveyors require speed control. Traditionally this was accomplished with wound-rotor motors and large, variable rotor-resistors, or other complex arrangements. These methods were inefficient and expensive, but they did allow for speed control.

Varying the frequency of the current supplied to an AC motor is a much more elegant method than any of the others that have been tried, but it was not something that could easily be done. It was expensive and difficult.

Affordable variable-speed drives are possible because of advances in solid-state electronics. Insulated gate bipolar transistors (IGBT) made it possible to switch large current levels at high-kHz rates, leading to pulse-width-modulated (PWM) inverter drives. Distortion levels are reasonable, units are compact, and operating efficiencies are very high.

PWM inverters do the job by controlling the kHz-chopping rate of a wave; varying its envelope in voltage and fundamental frequency; to control the acceleration and running of the familiar induction motor. The pulse rate can be set high enough to make audible noise of vibration, stimulated in the motor by the VFD, inaudible. You can arrange the range of fundamental frequencies through which the drive accelerates and operates the motor to skip critical frequencies where resonances in the mechanical equipment and structural supports might be excited to damaging levels.

The circuitry of a variable frequency is similar to other system applications of solid-state motor controllers and uninterruptible power supply systems, as used for large computer systems and data centers.

There are two general categories of VFD designs: Voltage source inverter (VSI) and Current source inverter (CSI.) Manufacturers of VFDs seemed to use one style or the other. The two types differ in the manner in which power is passed to the motor after being processed by the drive. The VSI drive treats the motor as a parallel-connected load and controls the overall performance envelope by adjustments to the output voltage of the drive. The CSI drive centers on motor impedance (inductive reactance) since it is driving current through the motor as part of the performance envelope. You must match CSI drives with their motors.

This motor matching is less important for VSI drives. Some use VFDs with motors of unknown origin, particularly in retrofits. However, one source should specify, purchase, and install VFDs with their associated motors whenever possible. Keep track of the thermal budget for the operating motor, and even for rooms housing large groups of drives.

<http://m.ecmweb.com/content/correspondence-lesson-7-motor-controls>

Correspondence Lesson 8: HVAC Controls

[Paul Rosenberg](#)

Paul Rosenberg, Datacom Consulting Editor

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Controls for heating ventilating and air conditioning (HVAC) cover a broad range of products, functions, and sources of supply. We define control as the starting and stopping or regulation of heating, ventilating, and air conditioning. Our concern here is control devices and systems to control larger commercial and industrial HVAC systems, not residential heating and cooling, except in a few cases

Controls for heating ventilating and air conditioning (HVAC) cover a broad range of products, functions, and sources of supply. We define control as the starting and stopping or regulation of heating, ventilating, and air conditioning. Our concern here is control devices and systems to control larger commercial and industrial HVAC systems, not residential heating and cooling, except in a few cases where residential controls crossover into light commercial controls.

The application of HVAC controls starts with an understanding of the building and HVAC systems, and the use of the spaces to be conditioned and controlled. The type of HVAC system determines the control sequence. Several types of control products such as pneumatic, electric, analog electronic, or electronic direct digital control (DDC) can then do the basic control sequence.

The way buildings are used determines the benefits you can obtain from additional controls.

Background. At one time, draft dampers (followed by thermostat control of the dampers) controlled heating. The use of mechanical stokers for coal firing required another step in the use of control. When oil burners were introduced, the concept of combustion safety control became necessary. This involved the sensing and proof-of-flame in the proper time sequence of introducing draft, fuel, and ignition.

The use of steam and hot water radiators led to the concept of zone control and individual room control (IRC). Forms of zone control included closed loop control using zone thermostats and open loop control with outside conditions setting the rate of heat delivery to the zone. Both of these forms of control were used to regulate the delivery of heat. The means of regulation included the following: Valves to control the flow of steam or hot water, controlling pumps to circulate hot water, and controlling boiler operation. When IRC was used the central supply was maintained and radiator valves were controlled by room thermostats.

The use of fans to deliver ventilation as well as heated air was controlled by dampers, which varied the source and volume of air. The typical control of unit ventilators was by pneumatic controls and included the following features: minimum outside air, discharge air, low-temperature limit, and thermostats with lower night settings activated by compressed supply pressure level. The increased usage of air conditioning led more complex control sequences in larger systems to central monitoring and control.

The development and use of computers and microprocessors has caused great changes in the HVAC controls industry. Minicomputers were installed on jobs to collect data to provide centralized control. Then, microprocessors were used for remote data-gathering panels to gather data and provide direct digital control. Computers are now used as on-site central controllers with operator interfaces and as computer assisted engineering (CAE) tools in the design of system programs, databases, and documentation. Microprocessors are still used in remote data gathering, yet also in small unit controllers and in smart thermostats.

HVAC systems. HVAC systems are classified as either self-contained unit packages or as central systems. Unit package describes a single unit that converts a primary energy source (electricity or gas) and provides final heating and cooling to the space to be conditioned. Examples of self-contained unit packages are rooftop HVAC systems, air conditioning units for rooms, and air-to-air heat pumps.

Central systems are a combination of central supply subsystem and multiple end use subsystems. End-use subsystems can be fan systems or terminal units. If the end use subsystems are fan systems, they can be single or multiple zone type. With central systems, the primary conversion from fuel such as gas or electricity takes place in a central location, with some form of thermal energy distributed throughout the building or facility.

There are many variations of combined central supply and end use zone systems. The most frequently used combination is central hot and chilled water distributed to multiple fan systems. The fan systems use water-to-air heat exchangers called coils to provide hot and/or cold air for the controlled spaces. Another combination central supply and end use zone system is a central chiller and boiler for the conversion of primary energy, as well as a central fan system to delivery hot and/or cold air. The multiple end use zone systems are mixing boxes, usually called VAV boxes. The typical uses of central systems are in larger, multistoried buildings where access to outside air is more restricted. Typically central systems have lower operating costs.

Besides packaged unitary and central systems, there are a variety of special-purpose systems. These include the following:

1. Heat pump cycles on chillers that use rejected heat or tower cooling.
2. Thermal storage.

3. Cogeneration of electricity and heat.

Basic control. Basic control regulates the amount of heating or cooling necessary to meet the load in conditioned spaces. Minimum outside air needed for ventilation is provided whenever a space is occupied. When outside air temperature is a suitable source for free cooling, it's controlled as needed at values greater than the minimum.

The approach in packaged unitary equipment is to control the generation of heating or cooling by space thermostats. The approach in central systems is to control the delivery of heating and cooling by the end use zones to match the load in the space. The supply is controlled to match the load imposed by all the zones. A typical method of doing this is for room thermostats to control zones, and discharge controllers to control central supplies. Discharge temperature controllers control the rate of primary conversion (chillers or boilers), and pressure controls determine the delivery rate of the pumps or fans distributing the central supply. In many cases there are multiple boilers and/or chillers and pumps, which are put on or off line as necessary to provide proper capacity. Those online are modulated as necessary to meet load needs. The controls to put units online and off-line would normally be applied to meet the system needs.

Supervisory control. The role of supervisory control is to control the scheduling and interaction of all the subsystems to meet building needs. Supervisory control systems have many names; each used for a particular emphasis. Among the names their acronyms are the following:

1. BAS: Building automation system.
2. EMCS: Energy monitoring and control system.
3. FMS: Facility management system.
4. EMS: Energy management system.
5. BAS: Building automation system. (The most generic of these terms.)

DDC (direct digital control), is sometimes used to describe everything a computer or microprocessor-based control system does. The original use of the term providing closed-loop control of local loops by a digital computer or microprocessor.

We implement direct digital control in stand-alone panels in intelligent data-gathering panels that are the remote panels building automation system. Energy management programs originally in the central computer of a building automation system are now placed in remote data-gathering panels; or even in stand-alone DDC controllers. This has led people to use DDC to describe all microprocessor-based control systems' functions.

Energy management application programs are different than local loop control and are named for their specific function, such as start or demand control. The considerations of which energy management application programs should be used rely upon the type of building and HVAC system. For instance, optimum start-stop programs are not appropriate for a hospital that has 24-hour operation. Load reset of supply temperatures is appropriate for systems that supplying heating and cooling simultaneously, such as reheat systems or hot and cold deck mixing box systems.

Optimizing. The concept of optimizing control is not only to control space conditions, but also to do it in a manner that minimizes the energy and costs when different forms of energy are available. An optimizing strategy is generally to improve the efficiency of primary supply equipment or to reduce the losses of energy in end-use systems. The sizing of equipment is to meet maximum loads, but the equipment is usually run at less than maximum load. This means that the part load characteristics of the equipment determines the efficiency in meeting a given load.

When there are multiple chillers or boilers, an optimizing strategy would be to choose the most efficient equipment that has the capacity to meet the load at any given time. Also, with some types of end use systems, energy wasted by bucking heating against cooling can be minimized by resetting supply temperature levels to be no more than is necessary to meet a given load condition. Another way to optimize is to use the thermal storage of a building to make use of energy stored at low cost and used when needed. Moving heat from one area of a building to another can be an optimizing opportunity as well.

These optimizing principles are used for specific types of HVAC. The variable in all of these circumstances is the amount of heating or cooling load and the control action to make some change in the way a load is supplied. This process has led to the use of the terms load reset and dynamic load control to describe this general approach to optimizing control. The selection of the most efficient combination of chillers to supply a cooling load has been called optimized chiller selection.

History of HVAC controls. Before World War II, the main suppliers HVAC controls in commercial buildings were companies that promoted pneumatic controls. The predominant idea at that time was that controls for commercial buildings were too complicated to sell over the counter and had to be installed and supervised by the controls manufacturer. This concept included having branch offices with installers and service people.

Electric control systems for commercial buildings were modulating type controls. They were sold on a supervised basis. When several other companies entered the commercial controls market with electric and electronic controls, some of their distribution was through distributors and branches. Some of the newcomers, who started with electric and electronic controls, expanded into pneumatic controls either by their own development or by association with foreign companies.

When computer-based supervisory control systems came to market, some larger companies with computer-based products entered the HVAC controls market; but eventually gave up. As international business developed and companies became multinational, some foreign-based controls companies expanded into the U.S. markets directly or through associations with smaller U.S. control companies. During the 1970s, some small companies evolved with limited product lines for energy management functions. When DDC became accepted, some small companies developed microprocessor-based DDC controllers and supervisory systems.

The full line control companies that started out as major players currently remain as major players but with more competitors that have limited systems. Some major HVAC systems manufacturers have acquired or developed control capabilities. They market packaged HVAC systems with controls and supervisory control systems. Some companies provide products for specific applications. The selection of a source of supply should consider the life cycle needs and costs as well as the track record of suppliers.

Room thermostats. The mounting of room thermostats and room humidistats has been the subject of much discussion, and for many years the industry standard has been for the thermostat to be mounted near the door of a room 5 ft from the floor. The problem is that if the room is full of children, the thermostat is not controlling the temperature where the occupants are.

It's important to study the location of the room thermostat or humidistat as to the effect of conditions at the thermostat. Remember, the thermostat responds only to what is going on at its location. If there is a ceiling diffuser blowing air at the location where the thermostat is mounted, there is going to be cycling of the system.

Sometimes installers and others are concerned about the way thermostats and humidistats are mounted on the wall; that is, whether they should be mounted in a horizontal or a vertical position. Generally, aside from writing on the unit's face, either horizontal or vertical mounting is fine. There is, however, one important exception: when the thermostat is electric and has a mercury bulb switch contact. This type is common in residential and commercial buildings and requires the thermostat to be mounted a certain way. Some of these require the installer to use a level to mount it properly.

Room thermostats and humidistats are devices that control automatic valves and dampers in a control system. These devices have built in sensors as well as moving parts that control the device. An example is a pneumatic thermostat that has a bimetallic sensor and a relay. Usually the complete package is under one cover on the wall, and all action takes place at the thermostat. There are, however, sensors that are mounted under the cover in the room that have no actuators (relays) under the same cover. They usually transmit the temperature information to another device at a remote location that does the controlling with relays, and so on. Normally this principle is used in electronic control systems involving a wire wound resistor mounted under a cover that reads the temperature in the space and transmits that information to an electronic controller in an equipment room.

Often, there is confusion with the terms thermostat and sensor. The concept of a sensor under a cover in the room is new and came about because of the advent of electronic control systems. Room sensors are used with other control systems and are sometimes called transmitters. In the case of pneumatic controls, the transmitters use a sensor and a special relay that transmits a pneumatic air signal proportional to the medium being sensed. An example is a transmitter under a room thermostat cover that transmits an air signal based upon the temperature being sensed in the room.

The transmitter may look like a thermostat, but it does no controlling by itself: it depends upon a receiver controller in a different location to take the action on the controlled device. The dials of these devices are only used for calibration, and are not moved once they are. These transmitters come in standard ranges and send out an air signal based on the medium being sensed. An example is a room transmitter with a range of 30 DegrF to 80 DegrF that

sends out a signal of 9 psi the temperature being sensed is 55 DegrF. In this case, the transmitter produces a signal of 8-15 psi as its sensed temperature goes from a low of 30 DegrF to 80 DegrF. Transmitters are not strictly sensors but are sometimes classified as such, since they are devices that do not do the controlling themselves.

Dampers. Automatic dampers have two classifications: The parallel blade, and the opposed blade. The parallel blade types were the first ones used.

The air that is being controlled can be considered an incompressible fluid at pressures below 12 in. of water. Above that, compressibility should be considered. Gases (air) can bend so that the volume will not be affected, and may not be controlled at all. Air can easily stratify in duct. Therefore, a damper can be considered a poor control device at best. At the same time, dampers can be as good at control as valves, provided they are sized properly.

Parallel versus opposed blade dampers. Parallel blade dampers tend to bend the air during the first few degrees of rotation as they go from full open to closed, and thus do little controlling in the first 20%-30% of movement. They bend the streams, rather than modulate them. But bending there are instances where this bending of air is useful, as when mixing air streams.

Opposed blade dampers are usually used where better control of the airstreams are desired and we want to prevent large amounts of stratification in the ductwork. Some dampers are not used for control or to maintain comfort are used for safety. These are the fire and smoke dampers.

Fire and smoke dampers. Fire dampers are put in the ductwork to stop the spread of fires, and to confine any fires to one area of a system. As such, they need to be made of rugged material that can withstand heat. They are seldom pivoted in the middle like an automatic damper, although they can be similar. They are almost always held in the open position by a linkage system that can be fused and is designed to melt and close when the temperature reaches about 165 DegrF. The closure is accomplished by springs or weights, and the dam must never be of the type that can be easily opened during a fire situation. In all cases, the dampers must meet the requirements of such organizations as the National Fire Protection Association (NFPA) and Underwriters Laboratories (UL). The locations of the dampers are clearly spelled out in most codes that apply to a particular type of system.

Smoke dampers, on the other hand, are not required by all codes. They are manufactured by vendors that supply automatic dampers, and in some cases are used as both control and smoke dampers. With these dampers, as with fire dampers, there are codes that apply, but they usually are not as stringent as those for fire dampers. Smoke dampers to stop the propagation of smoke and the resulting panic in event of a fire. Generally, they are involved with the central control and monitoring systems.

Manual Dampers. Manual dampers are an important part of the overall system, in particular the ability to balance the system and assist the control system to work better. They are used primarily to balance constant volume systems correct amount of air is distributed to the various places at the rate. They can be of the splitter type or the usual closure type. Splitter dampers try to split the air and redirect it to various sections ductwork. Balancing dampers try to apportion the air to the different sections of ductwork so the correct amount gets to the correct p all times.

Static control dampers. Static control dampers are different from the usual control dam in that they maintain the static pressure in the duct work base the action from a static pressure controller. Since static pressure is usually difficult to control in a system, static control dampers need to be the best available for their applications.

Damper motors. Damper motors are not motors in the true sense of the word, but the term has been used and has stuck with the industry. Damper motors are the devices (both pneumatic and electric) that operate the dampers in a system from the control signal of a device.

<http://m.ecmweb.com/content/correspondence-lesson-8-hvac-controls>

Correspondence Lesson 9: Programmable Logic Controllers

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Computer-based controllers (most commonly called programmable controllers or programmable logic controllers, PLCs) are the first controlling system able to harmoniously tie other control systems together. Essentially, PLCs are specially designed computers equipped with input and output (I/O) devices to interface with industrial controls and machines. They are housed in high-strength cases, for enduring

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At the core of the PLC is a central processing unit (CPU), which performs internal calculations. PLC processors are computer microchips, designed to support a fixed number of discrete, analog, or specialty I/O ports. They run programs, perform mathematical functions, and communicate with other processors (if programmed). PLC processors also monitor the status (ON/OFF) of the input devices.

Memory. Memory chips in PLCs serve two main purposes: • Retain information that the processor needs for logical decisions. This part of the memory is sometimes referred to as "storage" or "data table," where the status (ON/OFF) of all the discrete inputs and outputs is stored. Numeric values of timers and counters may also be stored in this memory. • Retain programming instructions. This is often referred to as the "user program memory" and contains ladder-diagram instructions. This section of memory is generally many times larger than the storage or data table. Instructions are in the memory via a programming device, magnetic tape, or systems computer.

This memory may vary from a few thousand bytes to several million bytes. This measurement is expressed in "K" (Kilo) for thousands of bytes or "M" (Mega) for millions of bytes. A byte is a grouping of eight bits of binary information, used in a computer's internal operations.

There are two general classes of memory: "volatile" and "nonvolatile." Volatile memory will lose its contents if power is lost. Therefore, it's necessary to have battery backup power available at all times.

Volatile memory is Random Access Memory (RAM). There are two common types of RAM memory chips: MOS (metal-oxide semiconductor) and CMOS (complimentary metal-oxide semiconductor).

Nonvolatile memory is Read Only Memory (ROM). It retains its information with power loss; thus, it doesn't require battery backup power.

There are several types of ROM:

Programmable Read Only Memory (PROM). The manufacturer generally programs PROM through use of special programming equipment. Once programmed, you cannot erase it or alter it.

Erasable Programmable Read Only Memory (EPROM). You can erase an EPROM chip by the using an ultraviolet light source. After you completely erase the program chip, you can make changes to the program.

Electrically Alterable Read Only Memory (EAROM). An erasing voltage applied to the proper pin of an EAROM chip will completely erase the program.

Electrically Erasable Programmable Read Only Memory (EEPROM). EEPROM, also known as "E-squared prom", is a nonvolatile memory that offers the same programming flexibility as RAM. The program can easily be changed with a manual programming unit.

PLCs come in many different memory sizes. The amount required depends on the user's application. Sizes are generally expressed in kilobytes: 2K, 4K, 16K, and so on. (Note: Due to the quirks of the computer business, 1 kilobyte is actually 1024 bytes, not the 1000 bytes you would expect.)

Many PLCs also come with a special feature called Error Detection and Correction (EDC). EDC ensures that the thousands of bit transfers taking place in the memory are processed accurately, by finding and correcting errors without interrupting the program operations.

While the CPU and memory are enough to handle most calculations and operations, they're not suited to connect to machines or sensors. To allow the processor to work along with the varied machines and mechanical devices, you must use I/O devices.

I/O devices. I/O devices connect many types of machines and sensors. Some I/O devices send or receive digital signals, others send or receive analog signals.

The rating of an I/O device must coordinate with the rating of the equipment or sensors it connects. Working with the PLC manufacturer can help you to properly coordinate pertinent characteristics. It is usually necessary to incorporate special relay devices or transformers to coordinate different voltage or current values. To keep track of details of the manufacturing process, many types of sensors are used with PLC-controlled systems. These devices are the "I" or inputs to the I/O devices. The most common types are proximity sensors and photoelectric sensors (which sense the location of parts during the manufacturing process).

Networking. Networking systems allow many systems and controllers to communicate with each other. The most common systems are Bus, Ring, Star and Mesh networks. These systems are the same ones used for computer networking. A network of coaxial, twisted pair, or optical fiber cables run between the units. This wiring must be used with networking software, which oversees the communications between the various machines. Special integrated circuit cards must also be installed in various machines to accommodate such systems. More industrial automation systems are going to a network setup, due to data-processing advantages. For network wiring, installers must follow the manufacturer's instructions.

Programming. PLC manufacturers developed a language of commands, instructions, and operations, presented in a form similar to the relay ladder-type diagram. However, in drawing the circuit for a programmable logic controller, programmers do not have symbols like those used in electromechanical circuit diagrams (i.e. push buttons, limit switches, and pressure switches). Programming devices use relay contact symbols for component switches. Each action is represented by a specific instruction. The instruction tells the processor to do something with the information stored in the data or user table (see Fig. 2, on the original article's page 52).

Programming language adopted by PLC manufacturers uses basic ladder logic symbols. For more complex symbols, such as timers, a generic rectangle is used to store the parameters of the device. Whether the input switches are normally open or normally closed will affect the state of the logic to be programmed in the PLC. For example, a normally closed stop switch in a motor control circuit is necessary to interrupt the flow of current to the motor controller. However, in a PLC circuit the switch could be either programmed open or programmed closed. The important condition is the state of the logic in the PLC ladder diagram.

In industrial applications where the PLC will replace the relay logic circuit, generally the operator controls and process switches will remain as they are wired. Therefore, the PLC logic will need to reflect the normal condition of the operator or process switches.

Control signals. In electromechanical control, a contact is closed to energize a circuit or open to de-energize a circuit. In solid-state control, these closed and open contact concepts are referred to simply as 1 and 0, respectively. In PLCs, the inputs and outputs are always such as to allow a signal to pass or to prevent it from passing.

Control depends on initial and continuing information from the machine and/or the process. This information is gathered from inputs such as pushbutton switches, limit switches, transducers, and other control devices. An input module then communicates the status of the discrete inputs to the processor. The signal from the input must be prepared for access to the processor. Field-supplied voltages from the input may cover a range of 24V to 240V AC/DC. The input signal must be converted to a low-level DC logic voltage. This voltage varies with different manufacturers. If the input is alternating current, a rectifier converts it to direct current, and a resistor reduces the voltage. If the input signal is analog (as it would be with many temperature, speed, and pressure sensors), the signal is converted to a binary through an analog to digital converter.

Signal isolation. Because of power quality concerns, incoming signals are isolated before they enter the processor. Optoisolators, reed relays, or transformers can perform this task. Optoisolators, the most common of the three, operate by changing the electrical signal into a light signal, sending the light signal to a second circuit where it is changes back into an electronic signal. The optoisolator transmits the signal while eliminating any electrical current that could pass stray voltages.

Troubleshooting. It's difficult to offer a complete and comprehensive approach to troubleshooting the PLC, since each manufacturer takes a slightly different approach to the design of the unit. However, one area generally applies to all units: the peripheral devices. This hardware at least is the same in most cases. It consists of components supplying input information for motion, pressure, and temperature, and components controlling all outputs, such as solenoids,

relays, motor starters, visual indicators, and alarm systems. The area between the inputs and outputs is the responsibility of the central processing unit.

In most cases the PLC manufacturer will supply visual indication of all incoming and outgoing signals. However, there are several areas in which the use of a meter to check volts as well as an ohmmeter to check continuity is useful.

One manufacturer of PLCs has provided diagnostic monitoring I/O modules. They provide the troubleshooter with information to quickly identify the possible source of a problem. In practice, the module actually "learns" the operation of a process. When the operator is satisfied the process is correct, the monitoring module observes the PC and the system hardware.

In this article, we've covered only the basics of programmable controllers. If you are new to the use of these controllers, you will have to familiarize yourself with specific types of controllers, the control devices that will be used with them, and have a clear idea of your control objectives.

Sidebar: Electronic Troubleshooting

The electrician doesn't need to be an electronics technician to locate and repair problems associated with solid-state equipment. The following hints will help you locate common problems associated with solid-state equipment.

- Actual test voltages and other measurements should be recorded when the equipment is in good working order. They can then be compared to measurements during troubleshooting. This will speed up the process.
- Look for obvious problems by visual inspection (physical damage, overheated terminations, broken wires, etc.).
- Measure supply voltage and current. Is it within the limits of the equipment?
- Check all circuit breakers and fuses. If circuit breakers or fuses are open, close them only if the fault is cleared.
- High temperature is the leading cause of failures of electronic equipment. Equipment should be placed in a well ventilated area so heat can escape.
- Transient AC voltage peaks or voltage surges in control circuits may last only milliseconds, but can cause problems in the control circuit. A digital multimeter with peak-hold function can be used to determine whether a transient AC peak problem exists.

<http://m.ecmweb.com/content/correspondence-lesson-9-programmable-logic-controllers>

Correspondence Lesson 10: Home Controls

[Paul Rosenberg](#)

Paul Rosenberg, Datacom Consulting Editor

Nov 01, 1999

So far in this course, we've limited our coverage to technologies for the commercial, industrial, and institutional sectors. In this lesson, however, we will take a look at a variety of residential control technologies. While these technologies were the province of electronic enthusiasts for many years, electrical contractors, home designers, and homebuilders are now showing a healthy interest in

So far in this course, we've limited our coverage to technologies for the commercial, industrial, and institutional sectors. In this lesson, however, we will take a look at a variety of residential control technologies. While these technologies were the province of electronic enthusiasts for many years, electrical contractors, home designers, and homebuilders are now showing a healthy interest in this market.

We use residential automation technologies to control three main systems: security, entertainment, and energy management. You can also combine them with a variety of sensors and inputs to create an infinite number of control schemes that can:

1. Monitor a certain area of the home (nursery, backyard) via audio, video, or both.
2. Turn air conditioning or heating on and off (according to schedule or via telephone).
3. Control any electrical device with a handheld controller.
4. Provide a more complex security system than old-style burglar alarms.
5. Set the controller to call a parent at work if the children don't come home at a preset time.
6. Make repetitive phone calls.
7. Send the same fax to multiple numbers automatically.
8. Gently raise the light levels in the room as well as volume levels of a stereo in the morning to replace a screaming alarm clock.

Controllers. The brain of the home automation system, a controller coordinates the sensors and equipment on the system, performs various logic functions, and ensures the desired events take place at the right time.

A simple central controller can control up to 256 separate devices; turning them on/off, or dimming lights, according to a schedule programmed into the unit via the computer. The user issues controller commands, based on the time desired for each operation. Mid-range systems offer the same type of control as the simpler ones; plus they have the ability to react to other pieces of equipment instead of reacting only to a clock. The advanced systems incorporate all these features, plus security, climate control, entertainment, telephone (voice mail and automatic calling), and other built-in abilities.

There are two primary types of home automation control devices: those using centralized intelligence and those using distributed intelligence. The original, and still most popular, home controller is the central intelligence type, which bundles a processor, timer, and various inputs together into one package. Connecting to other pieces of equipment in a star pattern, a typical installation has one controller regulating all of the devices in the home.

Distributed intelligence systems use many controllers rather than one. In effect, each device you want to control has its own controller attached or built in. Therefore, each sends control signals, receives control signals, and responds only in predetermined circumstances.

The controllers used in distributed intelligence systems are special microcomputers, usually contained on a single chip. These chips perform the same functions as the much larger central controllers, but on a smaller scale. They perform only the controlling functions pertaining to the device with which they are associated.

The most basic function of any controller is to listen and speak. Listening refers to the controller's ability to receive and interpret signals from sensors and control devices. Speaking refers to the controller's ability to send signals to the various items connected to it. All residential controllers perform the basic speak and listen functions, and almost all have a built-in timer.

Beyond the basic capabilities of controllers is their ability to perform logical functions. By far the most important of these is the if/then function. This type of logic allows the controller to make decisions based on certain conditions. For example, you could program this type of controller to follow commands, such as:

If the bedroom light turns on, and
if it is a weekday, and
if it's between 6:30 a.m. and 7:00 a.m.,
then turn on the coffee maker.

This capability allows controllers to perform more complex tasks than simply turning things on and off at preset times. This also requires the controller to do a much better job of listening, because it must know what devices are in use at all times.

While controllers have great capabilities, you must program them in advance to ensure proper execution of functions. They must also have a database containing designations of the devices you want to control (these are generally called

addresses), communication information for sensors, and one or more interfaces with means of communications such as copper wire, radio signals, or others.

You generally program a controller using a PC. The simple controllers connect to the computer only when being programmed, and run independently thereafter. Others require connection to the computer at all times. You must program distributed intelligence control chips with a portable, handheld programming tool.

This process is not very difficult, although it can be a bit tedious, as the programmer must define a large number of addresses, inputs, outputs, and logical functions. Once complete, however, extensive programming should never again be necessary.

Communicating. Components of home automation systems communicate in five ways:

Electronic signals over electronic cables. Used in almost any configuration and frequency, these signals allow for almost any type of signal; even data transmission.

This arrangement uses dedicated copper wires run between the various pieces of equipment. This is usually in the form of an 8-wire telephone type cable. While having excellent transmission characteristics, the use of copper conductors (also called hard wiring) is more expensive and difficult to install than other methods. This is especially true in existing home installations. In such cases, running new cables through existing walls is cost prohibitive.

Electronic signals sent through power lines. A powerline carrier system performs communications functions less expensively than hard wiring, but not as effectively. This term implies a building's existing power lines carry the communications signals; providing the signal at all places on the power wiring system.

By sending control signals over the existing wires in a building, no new control wiring is necessary. This saves a great deal on material and labor costs. Instead of running multiple control wires from a special controller to all various controlled items, you just tie the controller and control devices into the power-wiring network. However, most of these systems are limited. They can send the proper signals to turn something on/off, or dim lights; nothing more. While this is good enough for most types of control installations, it's not enough for complex systems requiring data transmission.

These send a variety of 121 kHz signals into the power system. The controller assigns a code for each of the devices in the home to be controlled. Once the devices are set to this code, they will sense the 121 kHz signal, and respond when they recognize a signal for their code. Since the 121 kHz signal is only about 4V, no other equipment should have any reaction to it.

Radio waves. Rather than communicating over copper wires, some systems are set up with radio waves. These are typically termed RF systems. RF simply means radio frequency, which are electromagnetic waves generated at radio frequencies; operating at different frequencies, and usually at lower power levels than commercial radio.

Security systems commonly use wireless systems. These units generate frequencies and coding of radio signals and send instructions, which are decoded and acted upon only by encoded devices. Annual battery replacement and higher initial cost are drawbacks.

One practical advantage of RF signals is they can pass through walls and most other structures.

Infrared light. Some types of systems can communicate with coded pulses of infrared light. We commonly call these IR systems. IR stands for infrared. Like RF systems, these systems require devices that can transmit and receive the special signals. Unlike RF signals, however, infrared signals cannot pass through walls.

All television, videocassette recorders, and stereo remotes use IR light pulses to communicate with the appliances they control. The power levels of all these remotes are fairly low. The only significant difference between IR and RF systems is the frequency of the light pulses they emit.

Optical fiber. Optical fiber signal transmission is the finest communications medium available. Optical fiber is necessary when you must transmit large amounts of video.

Interfaces. When combining two or more communication methods, some method of cross-communications is necessary. We commonly call such devices interfaces. The simplest types of controllers communicate only via powerline carrier signals; thus, combining them with devices using other communication means requires a separate interface device operating as a go-between. The more advanced controllers frequently have one or more interfaces built in.

Some widely used interfaces are:

- *Burglar alarm interface.* This device receives input from a security system and sends out a prearranged series of signals upon activation.
- *Universal modules.* This device is a power relay that's turned on or off by a control signal. It operates as a maintained or momentary contact relay, and ties home control systems to many types of electrical and electronic devices.
- *Telephone responders.* A telephone responder is an interface between the telephone system and home automation system. It receives phone calls, gives access to the automation system if a special code is entered, and can control between 12 and 256 different devices, depending on the brand.
- *Transceivers.* The most common of these devices receive RF signals, decode and interpret them, and send the appropriate powerline carrier signals through the home's power lines. They plug directly into a wall outlet.
- *IR/RF links.* These devices connect home automation systems with TVs, stereo devices, and VCRs. You can also use them to control such equipment from remote locations. For instance, even though a stereo system may be in the home's living room, this link allows you to control it from the bedroom. In this instance, an interface in the bedroom receives the IR signal from the remote, translates it into RF, and transmits the RF signal throughout the home. Another interface in the living room receives the RF signal, translates it back into IR, and transmits the IR signal to the stereo.

Sensors and actuators. If the controller is the brain of a home automation system, then the sensors and actuators are its eyes and muscles. Sensors pick up a stimuli, interpret or measure them, and relay the information to the controller. The many types of sensors include thermostats, electrical sensors, light sensors, humidity sensors, proximity sensors, magnetic sensors, and so on.

Actuators are devices that cause action. These are the parts of the system that make something happen. These include motors, valves, solenoids, or anything else that causes an action.

With few standards in the home automation industry, don't take compatibility for granted. Verify voltages, current levels, and signal protocols before connecting devices to controllers.

<http://m.ecmweb.com/content/correspondence-lesson-10-home-controls>

Correspondence Lesson 11: Building Systems Controls

Paul Rosenberg, EC&M Data Consulting Editor

Dec 01, 1999

After a half-hour commute by train, Mary arrives at the office to begin her workday. As she approaches the building's front door, a smart security system identifies her and unlocks the doors. After she passes through the entrance on the ground floor, an intelligent identification system senses her entry and energizes her personal workspace on the 20th floor. The system turns on her office lights,

After a half-hour commute by train, Mary arrives at the office to begin her workday. As she approaches the building's front door, a smart security system identifies her and unlocks the doors. After she passes through the entrance on the ground floor, an intelligent identification system senses her entry and energizes her personal workspace on the 20th floor. The system turns on her office lights, starts her computer, pulls up her electronic mailbox, and adjusts the local temperature based on her personal settings.

Although this example is fictitious, it's not that far from reality. Intelligent buildings with the same capabilities, as well as many more innovative features, are just around the corner. But keep in mind, "intelligent building" means different things to different people.

To an architect, an intelligent building may be one that's energy efficient and flexible; using materials such as raised floors and movable interior walls for quick low-cost renovations. Building managers will likely dream of digital control systems and the ability to maintain temperatures within a few degrees of the thermostat setting. He or she

might also want such a building to constantly monitor exterior conditions and automatically adjust for rate of air exchange and energy savings.

Still others may talk about Personal Environment Systems; where workers have the ability to customize individual work areas for temperature, acoustics, and fresh air ratios. IBM and Digital Equipment Corp. look at this concept as a logical extension to processing opportunities, while control companies such as Honeywell and Johnson Controls focus on traditional markets with new digital solutions.

The primary features of an intelligent building include: full communications and computer network infrastructure; building security system; energy- saving equipment and controls; flexible work areas; and environmental control systems.

Selling. The demand for intelligent buildings is growing steadily, although people tend to specify types of equipment and services (since there is no standard list of equipment that makes up an intelligent building). Systems now in demand are those that give the building and its tenants greater flexibility, increased productivity, and expanded services. People tend to buy specific communications, control, energy management, and computer network systems; not some unknown package of systems sold under the tag "Intelligent Building." Your job is to tie these systems together so they become transparent to the occupants. Your first order of business should be to understand what the customer is willing to pay for.

Distributed or central intelligence. One of the key questions regarding an intelligent building is how the system will handle computing. One method is to have a central computer handle all control functions. Another is to use separate computers for various systems. Typically, a combination of the two methods works best.

For example, it's common for an office building to supply base services and for the tenants to pay for supplemental services. The building provides basic security access, fire alarm systems, and telephone and communications cabling from the point of service to every occupied space. From this point, it's up to the tenant to provide his or her own telephone system as well as supplementary security and fire protection equipment. However, for a truly intelligent building, you must tie supplemental security and fire equipment in with the central building systems.

Communications methods. Of all the pieces making up an intelligent building, none is more important than the communications system. Beyond traditional cabling like twisted pairs, coaxial, and fiber-optic, you can find almost any type of communications media in modern buildings, including: wireless infrared links (IR); radio frequency (RF) signals; microwave transmissions (usually from one building to another; not inside one building); and laser links (used in the same way as microwave links).

In most cases, communications within an intelligent building includes a mix of technologies, with each media moving signals from one place to another, or from one system to another. The signals may represent a phone call, computer file, security signal to open a door, thermostat calling for more heat, or window receiving a signal to "go-opaque."

Wiring. The distributed wire network is the primary link between all individual control systems. In most instances, it's a pair of wires looped throughout the facility. Someone must supervise the network for "loss of integrity," which is necessary for the UL listing of the life-safety systems.

There's also a design issue: the necessity for redundant cabling and using separate conduit located some distance from the initial cable. Life-safety systems don't require this redundancy for UL compliance, but consulting engineers and suppliers often consider it to be a good engineering practice. Manufacturers of integrated systems think the cost of this redundant network is not excessive in relation to the amount of additional safety it brings to the system.

For nonlife safety systems, especially for renovations or retrofits, you may have an opportunity to use existing power lines as the communication distribution network. This alternative allows easy system upgrade or installation with minimal disruption to walls or ceilings. However, manufacturers caution you to check the electrical distribution system for power line integrity before proceeding with final design.

Fire alarm. All office buildings must have a UL-approved fire-alarm system that's fully capable for stand-alone operation. All cabling from the main or distributed controllers must be supervised per UL standards. Note: This is standard for a fire alarm system. When tied into the computer, the fire alarm panel will have all of its operations overseen by the computer. Nonetheless, it must have the ability to function independently.

This type of system, supported by the CPU and the "hot backup" redundant CPU, provides an extremely reliable and safe fire-alarm system. It also allows for excellent monitoring and control, as well as flexibility for expansion or

modification. Most importantly, this arrangement should be acceptable to any code authority. The power feed to the fire alarm panel is required to come from an emergency circuit, according to code.

HVAC control. You should control all HVAC equipment, from air handling devices to pumps, in an intelligent building by an energy management controller that communicates with a central computer. Standard software allows for monitoring status of the equipment, operation of equipment based on preset conditions and maintenance scheduling. This controller is the main point of control for firefighters' override capability. Most manufacturers have a built-in software check system that provides a certain level of supervision.

Many governing authorities in the United States accept the use of the HVAC controller as a code-approved firefighters' override system. Before design progresses too far, you should contact the local code-governing body for approval. An emergency power circuit must power such a controller, or any periphery device, which controls life-safety related equipment.

Work areas. As the number of automated workstations increases, housing the wires and fiber-optic cables that energize, connect, and control the various systems becomes a more vital consideration.

Systems contractors must find a way to install the equipment and its wiring wherever they can find a space. In an intelligent building, you should plan both horizontal and vertical paths for communication and control wiring from the outset with a provision for future changes. For horizontal distribution, twisted pairs of copper wires are still common while you can use fiber-optic cables for vertical and inter-building communication.

Tying it all together. Integrating various computer systems can be difficult; unless they all accept the same software and interconnections. An industry-wide effort to establish appropriate protocols for computer-aided environmental controls and energy management systems enables a "host" system to accept new subsystems.

For example, certain facilities management systems under development can control several subsystems, including HVAC controls, lighting/daylighting interface, fire alarm and life safety/fire management system, security, computer data, and communications systems; all manufactured by different companies. So although the intelligent building industry is getting better, significant challenges remain.

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DMM: Demystifying Maintenance Maladies

Ed Shen, Fluke Corporation

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Master troubleshooters often solve electrical mysteries with nothing more than a professional-quality digital multimeter (DMM) and their knowledge of how to use it. Let's look at how the masters do it. At 9 p.m., Plant Maintenance Manager Joe Brewster ends a long day by tucking his kids into bed. At 9:01, his phone rings. He recognizes the plant controller's voice immediately. "Joe, I've got 16 people

Master troubleshooters often solve electrical mysteries with nothing more than a professional-quality digital multimeter (DMM) and their knowledge of how to use it. Let's look at how the masters do it.

At 9 p.m., Plant Maintenance Manager Joe Brewster ends a long day by tucking his kids into bed. At 9:01, his phone rings. He recognizes the plant controller's voice immediately. "Joe, I've got 16 people standing around here because our computers aren't working right. The monitors jump around and..." Have you been in a similar situation? Has an electrical mystery taken a 25 hr chunk out of your day? Let's look at how others used their DMMs to solve similar problems.

Mystery No. 1: Case of the unstable computer screen. A large office building had an unhappy tenant. In one of the rooms, monitors near a particular wall had jiggling screen images. Moving the power cord to a different receptacle showed no improvement. But, moving the computer to a different room caused the problem to disappear.

An experienced troubleshooter investigates the problem. He suspects a magnetic field, since such a field (if strong enough) can deflect the monitor's electron gun. With everyone watching, he sets up his DMM to detect an AC magnetic field. He uses a length of small gage wire to make a 30-turn coil the size of his fist. He then connects the coil to the DMM's voltage input jacks and sets the function switch to "AC volts." Holding the coil horizontally at the height

of the monitor, he moves slowly along the wall, watching the reading. At first, the DMM shows a noise level reading with the last digit flickering. Then, at one point on the wall, the reading rises rapidly to about 3 mV. It drops again as the coil moves farther along. The troubleshooter marks the point of maximum reading on the wall, and repeats the process with the coil held vertically. The vertical readings are similar to the horizontal ones.

The troubleshooter explains something inside the wall; or on the other side of it; is generating a stray field. The stray field induces a small voltage in the sensing coil, which shows up as a reading on the DMM. A quick look at the building plans reveals an electrical room behind the wall. This room holds a service panel.

Behind the panel cover, the source of the stray field is obvious. All branch circuit conductors enter from the top of the panel. These conductors, tightly bound together within the panel, extend to the bottom of the panel where they make a "U-turn" to go back up to reach the circuit breakers. This configuration (a "service loop") allows for easy rearrangement of conductors and circuit breakers without any "wire stretching." In this case, the loop is too long; the concentration of conductors generates much more field than a proper loop would. How does the owner resolve this? An electrician shortens the conductors (and dresses them properly). This reduces the stray magnetic field to a point where it no longer causes a problem.

Mystery No. 2: Case of the arcing light switch. A library attempts to do its part for energy conservation by retrofitting all fluorescent light fixtures with electronic ballasts. On the one hand, they are successful. Their power consumption drops, and their quality of light improves. On the other hand, they aren't so successful. Whenever someone turns on any of the wall-mounted light switches, a blue arc greets them from behind the wall plate. Consequently, the library staff leaves the lights on; thus eliminating the energy savings.

The troubleshooter uses a DMM to measure the steady-state current in each branch circuit. He carefully inspects these switches for any signs of overheating. The running currents measure low, well below 10A, and there are no signs of overheating. The next step is to measure the inrush current when the switches are on and compare the results to the switch ratings. Over the phone, the switch manufacturer says the switches, with a continuous rating of 25A, passed tests at 40A.

To measure the inrush current, the troubleshooter sets his DMM's recording function to capture current impulses as short as 1 millisecond. The inrush current measures 70A. Repeated tests give the same results. This high value is hard to believe for a lighting circuit. So, the troubleshooter brings in a storage oscilloscope to verify the DMM's results.

A detailed circuit analysis reveals the ballast circuit has a diode-capacitor input configuration that draws high current for a short time when the capacitors are charging. Contact bounce in the switch interrupts the inrush current and causes an arc. Modifying the ballast circuit is not an option, so the library needs some other solution. What it does is replace the switches with ones designed to minimize contact bounce. The library also uses an enclosure that prevents arcing from being visible to the operator.

Mystery No. 3: Case of the missing voltage peak. You've probably dealt with buildings the owners use for purposes not planned for in the original design. Here's a case of a manufacturing facility that began life as a large commercial office building. The factory's operation involves diagnostic machines with significant amounts of computer memory.

The test department manager complains of repeated problems seemingly related to line power. Small loads turning off and on cause memory resets that result in lost test time. Several electricians try but fail to correct the problem. So, the manager calls in the local master troubleshooter, who listens carefully to the test manager's story and suggests the investigation start where incidents of the problem occur most frequently. When the troubleshooter and an electrician arrive, DMMs in hand, they see multiple power strips and extension cords feeding various nonlinear loads. Potential overloads of branch circuits?

The troubleshooter asks the electrician to measure the AC voltage at a nearby receptacle using his average-responding DMM. This test gives a reading of 118V. The troubleshooter then repeats the measurement using a true-rms DMM that displays 115V. Theory says an average-responding DMM will read 11% high if the waveform is a square wave. Even though difference in readings isn't 11%, it's more than enough to arouse suspicion about "flat-topping" distortion.

Next, the troubleshooter sets his DMM for "Peak Min-Max" to capture the instantaneous peak of the AC voltage. The display reads 135V peak. The correct value for a true sinewave would be about 163V peak ($\text{RMS value} \times 2 = 115\text{V} \times 2 = 230\text{V}$). This shows a clipping of 27V off the peak of each line cycle. A digital storage oscilloscope verifies the DMM's results. What's causing this flat-topping distortion? The answer: A high level of nonlinear loads drawing large

peak currents through a series of long feeder runs. The 27V missing from the line voltage peaks results in an undercharge condition in the equipment power supply capacitors. This makes the memories vulnerable to the slightest sag in line voltage. The solution involves installing stepdown transformers near the load and raising the feeder voltage from 208V to 480V.

Mystery No. 4: Case of the unlabeled conductors. Imagine this situation: The job is an electrical upgrade to a historic building (built in 1925) with "knob and tube" wiring. The apprentice removes all the two-wire receptacles, and your job is to replace them with GFCIs. Because of age and discoloration, it's impossible to tell which wire is neutral and which is hot. The usual method to sort this out involves energizing the circuit and using a capacitive voltage sensor (tick-tracer) to identify the hot conductor. But, your tick-tracer has a dead battery, and the only thing you have available is your DMM. To complicate matters, the nearest grounded water pipe is 50 ft away.

Can your DMM substitute for a tick-tracer? Think of it this way. The capacitive voltage sensor works as a series-connected, capacitive-coupled circuit using your body as a capacitively coupled ground conductor. Your DMM can work in the same mode because of its high input impedance.

Try this procedure: Energize the circuit and set your DMM to "AC volts." Verify you have full voltage between the conductors. Next, lay the DMM and black test lead on the floor. Stretch out the black lead to maximize its capacitive coupling to ground (earth). Take the red lead and alternately connect it first to one conductor and then the other. For best results, use an alligator clip and take your hand away. The side giving the highest reading is the hot conductor. Typical readings might be 0.2V on a neutral wire and 16V on a hot wire. You won't get full line value because the voltage is dividing between the DMM's input impedance and the capacitive impedance between the black lead and earth ground. The test method assumes you have a grounded (at the service) neutral.

<http://m.ecmweb.com/content/dmm-demystifying-maintenance-maladies>

Troubleshooting Residential Submersible Pump Systems

[David Herres](#)

David Herres, Master Electrician

Dec 01, 2008

Practical tips for applying job-specific troubleshooting techniques to your next submersible pump project

Why is it that residential deep well submersible pump system malfunctions are notoriously hard to diagnose? For one, a pump/motor assembly suspended 10 feet from the bottom of a 300-foot well brings new meaning to the word. The symptoms also have a troublesome way of overlapping so that precise diagnosis can be elusive at first. Invariably, however, persistence and logic prevail for skilled electricians.

Why is it that residential deep well submersible pump system malfunctions are notoriously hard to diagnose? For one, a pump/motor assembly suspended 10 feet from the bottom of a 300-foot well brings new meaning to the word "inaccessible." The symptoms also have a troublesome way of overlapping so that precise diagnosis can be elusive at first. Invariably, however, persistence and logic prevail for skilled electricians.



In this article, we'll take a look at the 3-wire 240V single-phase submersible pump system for drinking water applications, typically set between 50 feet and 300 feet below grade. The pump is fed down a steel well casing that extends through the earth until bedrock is encountered — at which point the rock itself becomes the casing.

The wiring system

A 3-wire system (actually there is a fourth equipment ground conductor that is not counted in the number of wires) implies there is a control box inside the house, containing a large electrolytic capacitor, microprocessor, and other electronics (**Photo 1**). In contrast, a 2-wire system omits the in-house controller so that the capacitor is inside the hermetically sealed underwater motor. Although this arrangement makes for a cheaper initial installation, when it comes to repair, options are limited.

To efficiently diagnose one of these systems, it's necessary to clearly understand the wiring arrangement you're dealing with — at least external to the control box and motor. A 240V branch circuit from a double-pole breaker within the entrance panel is run to an off-the-shelf double-pole pressure switch, which is actuated by low cut-in and high cut-out water pressure inside the pressure tank. This switched power goes to the input terminals of the control box.

The control box output exits the building, is buried in the same trench with the water pipe, and emerges from the ground in metal or PVC raceway where it enters the well cap through a passage designed for the purpose. Inside the well cap, it's usually spliced with twist-on wire connectors, and then goes down along the water pipe to the pump motor, which comes furnished with a pigtailed set of three wires (plus equipment ground). These are connected to the pump cable by means of crimp-on connectors and heat-shrink tubing, which is the same as an underground splice kit. The spliced wires should be routed inside the torque arrester — an expandable rubber cylinder that keeps the pump/motor assembly centered and minimizes counter rotational motion and wire chafing.

The pump cable, which comes from the control box output terminals, consists of red, yellow, and black (plus green) twisted conductors. Because they are not jacketed, you must run the indoor segment in raceway. As for the outdoor portion, you can direct bury it, as far as the NEC is concerned. However, state and local well installation regulations may call for UL-listed gray RNMC or similar protection to the wellhead.



Many people assume that the well cable is made up of two hot legs with a neutral; however, that is not the case. The red is start, the black is run, and the yellow is common. One of the functions of the control box is to energize the red for a short period of time so that the pump motor can get up to speed, after which black is switched online.

One large manufacturer offers the motor and control box, which are used by many pump manufacturers. The pumps are bolted onto the motors (with matching splined shaft). However, one major manufacturer makes its own motor and control box, which are not interchangeable. As for the others, motors and control boxes may be interchanged as long as horsepower, phase, and voltage are a match.

The great advantage in the universal control box is that the cover contains the electronic components so you can replace them without doing any rewiring.

Stepping through a diagnosis

In troubleshooting a nonfunctioning submersible pump system, you must strive to diagnose and repair the problem without unnecessarily pulling the pump out of the well or digging up the buried line.

Let's say a homeowner has a drilled well with a submersible pump located 500 feet from the house set at a depth of 300 feet. The complaint is that when a faucet is turned on, there is no water. The homeowner says the system has been in place for several years, and there has never been a problem. Because it has rained a lot all summer, lack of water in the well can be ruled out. There is a good probability the pump has failed due to sand preventing the impeller from turning or because the motor has seized up. But because pulling a pump that is set at 300 feet is a fairly difficult task, other possibilities are pursued first.

It is a 3-wire system, and all electronics are mounted inside the control box cover. After verifying with a voltmeter that there is power to the input terminals — indicating house wiring and pressure switch are functional — a new cover of the correct horsepower rating is acquired. The cover contains a capacitor, relay, microchips, and other electronic components (**Photo 2** on page C26). When it's snapped in place, it supplies power to the pump motor via 3-wire (plus equipment ground) underground pump cable. Unfortunately, this procedure did not cause water to flow to the pressure tank, so other avenues must be pursued.

At this point, it's time for the “divide-and-conquer” troubleshooting technique. The idea is that when a system consists of a number of components connected serially — and the endpoint is not receiving power — the most efficient strategy is to perform a test at the midpoint to narrow the focus, not necessarily spatially but in terms of probable causes. The bad half can be tested in turn at its midpoint, allowing the technician to methodically zero in on the problem location. This technique is especially useful when working on large complex machinery or wiring systems.

In the case of the pump system, it is decided to take measurements at the wellhead. The well cap is removed. The three wires at that point are frequently spliced with twist-on wire connectors, which can be removed to access test points. If not, the conductors can be cut with power locked out and later reconnected using silicone-filled twist-on wire connectors. Although it's found that voltage is present, a clamp-on ammeter reveals no current flow with the wires reconnected. Thus, an underground line fault is eliminated. Before pulling the pump, go back to the control box and take additional resistance and current measurements, referring to the information printed inside the control box.

With a good helper and power locked out, you can pull a pump-motor assembly in about an hour. Now is a good time to visually check the cable for abrasion or other damage. Separate the motor from the cable, and do meter tests to the cable to further isolate the problem. Let's assume the cable turns out to be in good condition. If the earlier clamp-on ammeter test had indicated excessive current flow — and there is no line damage — it looks like the motor has one or

more shorted windings, is seized, or the pump will not turn. Before giving up completely on the motor, consider the possibility that the motor pigtail has acquired a short circuit, possibly due to lightning damage. This wiring harness has a plug-in connector right on the motor and can be unplugged after removing the protective metal channel. If that is not the problem, unbolt the pump from the motor, and see if the motor shaft will turn by hand. If it is seized, you must replace the motor. This is because it's a sealed unit; therefore, it's not possible to rebuild it. If it will turn, then you can conduct a brief dynamic test.

Another piece of advice is to make sure you have a good equipment ground connected. You should clamp the motor to a large, dry piece of plywood. Do not touch the metal case while the motor is energized. Fire it up very briefly, because it is designed to be run submerged in water for cooling purposes.

If the motor runs, there is the possibility that the pump has become sand bound. It is a simple matter to take it apart and clean it, reassemble the unit, and retest. You can inspect the pump impellers and replace them if worn or damaged. If it runs and meter readings are good, reset the pump a few feet higher so the problem does not recur.

Usually, the foregoing procedures will result in a successful repair. If not, you'll have to retrace your steps, and see if you made a false assumption or misread data somewhere along the line. It's also possible there is partial damage to the cable that did not show up during your visual inspection. This can be misleading and cause you to go around in circles. Another curve ball that may be thrown your way is when there are two or more faults. Finding and correcting one of them will not restore normal operation. The answer to both of these problems involves using patience, persistence, and perception. You have to go over the whole system, keep isolating stages, and apply logic to each one.

Beyond the basics

One mistake many beginning electricians make is they think they're going to find the problem in 5 minutes. If the above described procedure doesn't pinpoint the problem, then you'll need to turn to other troubleshooting techniques.

Suppose the readings at the well cap indicate there was no fault in the cable going down inside the well casing, and the pump motor was good. Ohm readings from the wellhead back to the control box — with wires disconnected at the wellhead and control box cover removed — indicate a short between yellow and black conductors. Then you know the fault is in the underground portion of the circuit, which is buried at a depth of 4 feet, for example. At this point, you know some digging is required.

Because backhoe time is costly, careful planning is essential. What is the minimum digging required to locate the fault? Can cutting the wire to take readings at various points along the cable be avoided? If you use the divide-and-conquer method again, it's possible to dig at the midpoint of the underground portion of the line and take readings to ascertain which half of it contains the fault. Then repeat this process by digging at the midpoint of the faulty half of the line — further narrowing down the problem segment. This limits the amount of digging you'll have to do to locate the exact point of failure. As for cutting the wire to take measurements, that is not necessary. Apply a voltage in series with a heavy load and use a clamp-on ammeter to perform the test.

You might also consider what some view as an even better method — one that involves a more sophisticated test and less digging. Take ohm readings from both ends to locate the short. If the reading at the house is double the reading at the wellhead, then you know the short is one-third the distance, or about 166 feet, from the well.

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<http://m.ecmweb.com/ops-amp-maintenance/troubleshooting-residential-submersible-pump-systems>

Troubleshooting Variable Speed Drives

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May 01, 1999

Fast, effective troubleshooting combines intuitive and logical thinking with special training, the right test instruments, and "SMARTS." Troubleshooting electrical equipment is always a challenge. As electrical equipment incorporate more complex electronics, the challenge becomes increasingly formidable. More and more time is required to troubleshoot this equipment; certainly, increased knowledge is

Fast, effective troubleshooting combines intuitive and logical thinking with special training, the right test instruments, and "SMARTS."

Troubleshooting electrical equipment is always a challenge. As electrical equipment incorporate more complex electronics, the challenge becomes increasingly formidable. More and more time is required to troubleshoot this equipment; certainly, increased knowledge is needed to do this task correctly and efficiently.

This certainly is the case with variable frequency drives (VFDs), which consist of a complicated combination of electric power components and sensitive electronic circuits. Furthermore, VFDs are directed by precision, low-voltage, low-current digital or analog circuitry that can be quite sensitive to outside disturbances. Most importantly, proper VFD operation is greatly dependent on many exterior factors such as the load itself, input voltages, surges, dirt, heat, moisture, and so on.

Thus, if you want to be an effective troubleshooter of VFDs, you must apply systematic common-sense thinking and troubleshooting techniques. Another essential factor is training.

Applying SMARTS

The secret to troubleshooting VFDs is using SMARTS, an acronym for a proven method used in troubleshooting numerous VFD installations over the last 20 years. SMARTS stands for Safety, Manuals, Application, Readings, Talk, and Symptoms.

This technique can be applied to all kinds of VFDs and other electrical equipment as well. It's practical, economical, effective, and fast.

Safety. This is the first step in all troubleshooting; safety for yourself, your fellow employees, and the equipment. Take as many test readings as possible with the power off. However, many measurements and test procedures will have to be done on energized equipment. Be sure to install warning signs, tape off the test location, and follow lock-out procedures.

It's important to be aware that both AC and DC voltages are present within the VFD. DC voltage levels will be equal to the peak voltages of the AC input. For example, a 480V drive will have a DC voltage of 676V (480×1.41). The DC voltage used in a VFD is stored in capacitors that are capable of holding a lethal charge for several minutes. Before attempting any internal hands-on work on the VFD control enclosure, you should first take a voltage reading of its capacitors with a verified working meter. Remember, your life could depend on this simple test.

With today's modern microprocessor designs on VFDs, it's very easy to program in a set of operation parameters that could inadvertently prove to be fatal. A motor designed to operate at 3600 rpm at 60 Hz can be made to operate at 7200 rpm when the VFD is programmed to supply a 120 Hz output. Imagine what would happen to boxes moving on a conveyor at twice the conveyor's rated speed. Make sure you understand the results of any changes or adjustments made.

The best advice is to read and follow all safety cautions on the drive and in the supplied manual.

Manuals. It's impossible to troubleshoot and repair a VFD without first studying the manufacturer's manual. With on-board diagnostics available on most units, you need the manual to decode these fault messages as well as to identify the probable causes for the respective faults.

The manual also provides you with test procedures and troubleshooting guidelines that will save many hours of unnecessary downtime. A good practice is to request additional manuals or copy the ones you already have. Many hours of lost downtime can be attributed to looking for the equipment manual.

Application. What is "application?" It's the VFD, the motor, and the load connected to the motor. Don't forget the external controls that control the VFD itself, such as interlocks, push buttons, digital controllers, etc.

The VFD will operate only when and how it's told to operate. At the same time, it will respond to any dramatic changes, such as incorrect or faulty load and power demands, by shutting down in an effort to protect itself.

The majority of problems with VFD applications (fans, pumps, conveyors, etc.) have nothing to do with the VFD control console itself. As such, you should troubleshoot the VFD controller last and troubleshoot the application first.

Readings. As part of your troubleshooting process, you'll need to collect readings of the drive. The most common readings needed are as follows.

- * AC input line voltage.
- * AC output load voltage.
- * Line current.
- * Load current.
- * Frequency output to the motor.
- * DC bus voltage.

Today's VFDs use operator interface panels and digital displays. Most of the readings needed for troubleshooting can be read on these displays. In some units, you can even choose one of four different languages.

When using test equipment to collect readings, be sure to verify the meter's capability for the intended reading. The supplied manual will usually indicate what readings are needed as well as the type of meter to use.

Solid-state VFD components are easily damaged by overvoltages and fault currents. Therefore, they may be checked early on prior to sequential, logic procedure. Also, be sure that insulation resistance readings are not taken on a motor with the drive connected and power removed.

It's good practice to log the readings on paper as you troubleshoot. This will help you from having to take the same readings over again.

Talk. That's right, talk. Talk to the operator of the equipment; talk to other mechanics working in the area; talk to anyone that might know anything about the equipment or an unusual activity near it. As a detective (troubleshooter), your job is to get the facts and get the equipment back up and running.

Here are some questions that you should ask. When did it stop working? Has anything else also stopped? Have there been any recent changes or additions to the load? Is this a random failure or has this been happening for some time?

The better your understanding of the events surrounding a failure, the more accurate and faster your troubleshooting will be. Ask yourself, "Am I troubleshooting symptoms or the actual problem itself?"

Symptoms. This last step is especially critical for effective troubleshooting of VFDs. It's vital that you learn to separate symptoms from the actual problem.

For example, a fuse blew; is this the problem or a symptom? Actually, the blown fuse is a symptom of the true problem. A fault of some type has caused the fuse to blow. Replacing a blown fuse without troubleshooting the cause will not correct the problem and, in many cases, will create even greater problems.

Another example: the drive trips in an overload condition. Is this the problem or a symptom of an undiscovered problem?

Collect all your symptoms and carefully review them. Let the symptoms as well as your readings and interviews with operators point you in the right direction.

Actual case history

Location: Baltimore Museum of Art, Baltimore, Md.

Application: Numerous VFDs for an HVAC system in new building addition.

Type of loads: Fans, pumps.

Problem 1: A 60-hp chilled-water pump VFD shutdown.

Problem 2: Only one standby chilled water pump remaining. The museum needs to maintain precise control of temperature and humidity to preselected constant levels to protect art work.

Getting down to the business of troubleshooting, we instinctively began thinking in terms of SMARTS.

Safety. Even with many years of experience, we always start troubleshooting with these simple questions. What are the dangers in troubleshooting this equipment? Am I safe in what I am doing?

Manuals. Guess what? The pocket for the manual was on the VFD controller door but the manual was gone. As such, a twin unit VFD next to the inoperative VFD had to be shutdown and its manual removed. (With the digital displays and error codes available for diagnostics, it's essential to have the VFD manual.) The fault code on the display indicated that control power had been lost and recommended checking control inputs, fuses, and the control transformer. But before taking readings, we reviewed the application.

Application. Sure we're electrical people, but you can't fix the piece of equipment if you don't know what it's doing or supposed to do. If you're not sure of the application, ask. Get someone to help you. Try to think of the things you'll need to know. For example, how is the drive controlled: auto, manual, local, or remote? What and where are exterior sensors and limit controls? Anything special about the load?

Readings. At the VFD control panel, the manual was reviewed and the suggested readings, as outlined, were taken.

As the readings progressed, we found a blown control fuse on the secondary side of the control transformer. When you see a blown fuse, a little voice in your head always seems to say, "Go ahead, change it; it probably got weak and needed to be changed. Your problems will be over." Don't do it. A fuse is a protective device, and when it blows, it means a problem exists and the fuse cleared a potential over-current or fault-current condition.

Using a meter in the resistance test mode, we checked for shorts in the control circuit, but did not see any dead faults or unusual resistance values. As such, the fuse was replaced and the VFD was powered up. Boom! It blew again.

Now, it made sense to start tracing wires from the transformer. In doing this, we started asking some questions.

Talk. "So, when did this unit actually go down? Did you lose any other equipment? Anybody do any work nearby?" We were told that some painters had just finished painting the mechanical room just prior to the drive shutdown.

Although there appeared to be no relationship, we asked where they were painting. The painting included areas right over the VFD itself. Looking at the top of the VFD, we saw where pieces of cinder-block from the painting prep work had fallen down onto the enclosure. Behind the VFD a piece had fallen into the cooling fan and jammed the fan blade, causing the fan motor to short out. This fan motor operated at 120V and was directly wired to the control transformer. End of the story? Not yet.

Symptoms. Everything pointed to the fan motor as being the culprit. But after disconnecting it from the transformer and further testing, a fault still showed up.

We continued troubleshooting and discovered that someone had made repeated fuse changes and faults had damaged an additional gate-firing transformer.

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<http://m.ecmweb.com/content/troubleshooting-variable-speed-drives>

Basic Electronic Troubleshooting Techniques

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You can troubleshoot problems with electronic equipment by using a few basic test instruments and a little common sense

As an industrial electrician, you'll encounter many complex electrical circuits and drives that will break down as they age. Chances are, you won't be an expert in repairing these devices, but you can take some practical steps to solve problems and help get your systems operational again in a timely manner. Let's start with the motor drive. Solid-state electronic AC motor drives are becoming more

As an industrial electrician, you'll encounter many complex electrical circuits and drives that will break down as they age. Chances are, you won't be an expert in repairing these devices, but you can take some practical steps to solve problems and help get your systems operational again in a timely manner. Let's start with the motor drive.

Solid-state electronic AC motor drives are becoming more common within industrial plants. They control a wide variety of devices like pumps, conveyors, air handlers, chillers, machine tools, mixers, and a host of other devices once designed to run at constant speed or be powered by DC. Since failure in these devices can be often attributed to the rectifier section, you'll need a fundamental understanding of transistors, diodes, silicone controlled rectifiers (SCRs), and insulated gate bipolar transistors (IGBTs).

Pulse width modulated (PWM) inverter drives are the most prevalent type of AC drives (**Fig. 1** [here](#)). The AC line voltage is converted to DC (in the converter section) and then reconstructed back into a variable frequency and a variable voltage output. Changing the frequency varies motor speed, and motor torque is maintained by keeping the ratio of volts-to-frequency at a constant.

Troubleshooting motor drives. Because most failures occur within the power sections instead of the circuit boards, they aren't very difficult to troubleshoot. The typical plant maintenance technician will rarely see enough failures to build up any proficiency in repairing circuit boards.

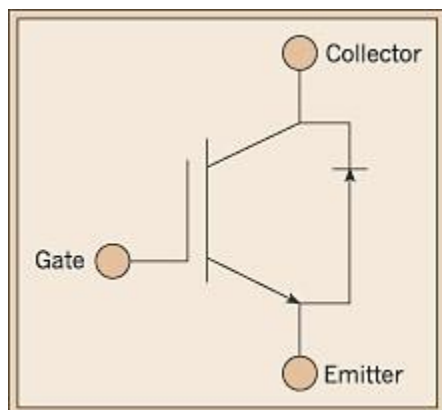
Effective troubleshooting on a variable frequency drive (VFD) requires a methodical approach. The classic divide-and-conquer method, taught by most technical schools, is effective when knowledge of the equipment is limited. A good troubleshooter will first isolate the box or section that isn't passing the signal and then work on it.

So how can you quickly and efficiently troubleshoot a dead VFD? Remember to always put safety first. The capacitors within the power section can maintain a dangerous charge even after the power is removed.

First make sure that the capacitors are discharged before putting your hands into the power section. With the power off, begin checking the power sections of the drive. Then, place your digital multimeter (DMM) in the diode check mode. Find the positive DC bus (sometimes this may be brought out to a terminal), place the negative (black) lead from your DMM on it, and then check each incoming phase in turn with the positive (red) lead. You should read a diode drop of about 0.6V on each phase. If it reads open, then the charge resistor is open and needs to be replaced. This is a common source of many problems.

Next, place the DMM's positive lead on the negative bus and the negative lead on each incoming phase in turn as you did before. You should read a diode drop, not a short or an open. Place one DMM input lead on the positive bus and the other on the negative bus. On this measurement you should read the capacitor charging rather than a short.

To check the inverter section, place the positive DMM lead on the negative bus and the negative lead on each output phase. You should read a diode drop because diodes are connected across each output transistor. Again, you shouldn't read a short. Check the remainder of the inverter section by placing the negative lead of the DMM on the positive bus, checking each output phase again with the positive lead of the DMM. You should read a diode drop again and not a short. If you read OPEN from either of these checks then the bus fuse is most likely open. If no problems are present within the power section and the unit still won't function, it's either improperly connected or programmed or has a bad circuit board.



Newer PWM drives use IGBTs in the driver sections of the output, and are much less likely to fail. These devices perform like a metal-oxide semiconductor field effect transistor (MOSFET). When the voltage at the gate exceeds the threshold voltage, the device turns on. If the voltage applied to the gate contact is less than the threshold voltage V_{th} , then the device is turned off (**Fig. 2**).

Visual PLC troubleshooting techniques. Most PLCs incorporate light emitting diodes (LEDs) in their design, which offer a good source of diagnostics. They can provide valuable information about the wiring, and input/output (I/O) modules within the unit. Typically, I/O modules have at least one LED indicator; input modules normally have a power indicator, while output modules usually have a logic indicator.

A lit power LED on an input module indicates that the input device is operating and its signal is present at the module. However, this indicator by itself can't isolate malfunctions to the module. Consequently, some manufacturers provide an additional diagnostic indicator known as a logic indicator. If a logic LED is lit, the logic section of the input circuit has recognized the presence of the input signal. If the logic and power indicators don't match, then the module is unable to correctly transfer the incoming signals to the processor. This indicates a module malfunction and most likely points to the problem area.

The output module's indicator functions in a similar fashion to the input module's indicators. When on, the logic LED indicates that the module's circuitry has acknowledged a command from the processor to turn on. In addition to the logic indicator, some output modules incorporate either a fuse indicator or a power indicator, or sometimes both. A blown fuse indicator displays the status of the protective fuse in the output circuit. The power indicator displays that power is being applied to the load. Similar to the power and logic indicators in the input module, if both LEDs aren't on simultaneously, the output module is malfunctioning — again pointing to the probable problem area.

As you can see, LED indicators greatly assist the troubleshooting process. With power and logic indicators, you can immediately pinpoint a malfunctioning module or circuit. Although they can't diagnose all problems, they serve as a good first round indicator of a system malfunction.

Troubleshooting the PLC inputs. If the field device connected to an input module doesn't seem to turn on, a problem may exist somewhere between the line connection and the terminal connection to the module.

First place the PLC in standby mode so the output isn't activated. This will permit you to manually activate the field device. A limit switch can usually be manually closed to achieve this result. When the field device is manually activated, the module's power status indicator should turn on, indicating the power link is working properly. If this occurs, then the wiring most likely isn't the root of the problem.

Next, analyze the reading of the PLC's input module. Place the PLC in its test mode. The device should read its inputs and execute its program, but not turn on its outputs. If the PLC reads the device correctly, then you know the problem isn't located in the input module. If it doesn't read the device correctly, then the module could be defective. However, several causes are possible. First, the logic side of the module may not be operating correctly. Second, its optical isolator may be blown. Third, one of the module's interfacing channels could be damaged. In any case, you'll need to replace the module.

If the module doesn't read the field device's signal, then further tests are required. Bad wiring, a faulty field device, a faulty module, or an improper voltage between the field device and the module could be causing the problem. First, measure the voltage to the AC input module. Your DMM should be in voltage measuring mode and should display the voltage that powers the module. If the voltage is present and at the correct level, you know you have a bad input module because it's not recognizing the applied voltage. If the measured voltage is 10% to 15% below the specified signal voltage, then the problem most likely is in the source voltage to the field device. If no voltage is present, then the wiring is broken or shorted or the field device is dragging it down. Check the wiring connection to the module to ensure that the wire is properly secured at the terminal or terminal blocks. You can also perform an insulation check on the wiring to look for shorts and/or damaged insulation. Be sure that the system is de-energized first before conducting this test.

To further locate the problem, confirm that voltage is present at the field unit. With the device turned on, measure the voltage across it using your DMM. If no voltage is present on the load side of the unit (the side that connects to the module), then the input device is probably defective. If there is power, then the problem is in the wiring from the input device to the module. In this case, the wiring must be inspected and tested to find the problem.

Troubleshooting PLC outputs. The first step in troubleshooting the outputs is to isolate the problem to either the module, the field device, or the wiring.

First check that the source of power to the output module is at the specified level. This value should be within 10% of the rated value. In a 120VAC system, for example, it should be between 108VAC and 132VAC. Examine the output module. If the fuse is blown, check its rated value to be sure the correct fuse was installed in the first place. Also, check the output device's current specifications to determine if the device is pulling too much current.

If the module's output status indicator fails to turn on despite receiving the instruction to turn on from the central processing unit, it's faulty. If the indicator does turn on and the field device doesn't activate, then check for voltage at the output terminal to be sure that the switching device is, in fact, operational. If no voltage is present, then you should replace the module. If voltage is present, then the problem lies in the wiring or the field device. At this point, make sure the field wiring to the module's terminal or to the terminal block has a good connection and that no wires are broken. This can be accomplished in the same fashion as described earlier.

When you finish checking the output module, check to see that the field device is functioning correctly. Check the voltage coming to the field device while the output module is on. If voltage is present, but the device doesn't respond, then the field device is probably defective or damaged.

One trick for checking the field device is to test it without using the output module. Remove the output wiring and connect the field device directly to the power source. If the field device doesn't respond, then it's faulty. If the field device responds, then the problem lies in the wiring between the device and the output module. Check the wiring, as noted earlier, looking for broken wires, shorts, worn insulation, and oil or grease on the connection points and along the wiring route.

Troubleshooting the CPU. PLCs also provide diagnostic indicators that show the status of the central processing unit (CPU). These indicators include such display messages as POWER OK, MEMORY OK, and COMMUNICATIONS OK.

You should first check that the PLC is receiving enough power from the transformer to supply all the loads. If the power received is in accordance with specifications and the PLC still isn't working, check for a voltage drop in the control circuit or for blown fuses. If these conditions are all proper, then the problem lies in the CPU. Most likely, the diagnostic indicators on the front of the CPU will display a fault in either memory or communications mode. Should one of these indicators be lit, it's highly likely that the CPU needs to be replaced.

Troubleshooting the input and output sections of a motor drive can be easy when approached logically and addressed section by section. All you need to do is measure volts and ohms with a DMM. As with motor drives, the most practical method you can use to diagnose PLC input/output malfunctions is to isolate the problem to either the wiring, the module, or the field device. When these systems have both power and logic indicators for you to view and interpret, then module failures become very easy to recognize and isolate.

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<http://m.ecmweb.com/content/basic-electronic-troubleshooting-techniques>

Troubleshooting PLCs

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Once you get over the "black box" syndrome, PLCs are actually easier to troubleshoot than traditional hard-wired control systems. Programmable logic controllers (PLCs) have become important building blocks for automated systems. Because they have constantly increased in capability while decreasing in cost, PLCs have solidified their position as the device of choice for a wide variety of control tasks. What

Once you get over the "black box" syndrome, PLCs are actually easier to troubleshoot than traditional hard-wired control systems.

Programmable logic controllers (PLCs) have become important building blocks for automated systems. Because they have constantly increased in capability while decreasing in cost, PLCs have solidified their position as the device of choice for a wide variety of control tasks.

What is a PLC? In brief terms, a PLC is a digital electronic device that contains a programmable (changeable) memory in which a sequence of instructions is stored. Those instructions enable the PLC to perform various useful control functions like relay logic, counting, timing, sequencing, and arithmetic computation. These functions usually are used to monitor and control individual machines or complex processes via inputs and outputs (I/Os). I/O modules connected to the PLC provide analog or digital electronic interfaces to the external world. The PLC reads inputs, processes them through a program, and generates outputs. [ILLUSTRATION FOR FIGURE 1 OMITTED].

One reason for the popularity of PLCs is their high reliability in harsh industrial environments; occasionally, however, things do go wrong and troubleshooting becomes necessary. Those unfamiliar with PLCs often fear troubleshooting a device that appears to be a mysterious "black box" (see side bar on page 22), but in fact today's PLCs actually are very open systems that lend themselves to relatively easy diagnosis.

The internal operation of a PLC can be monitored via a handheld programmer, terminal, or personal computer, and many indicator lights are provided for I/O troubleshooting.

Upfront assumptions

The intent of this article is to cover only the basics of PLC troubleshooting, so there are some limitations on what will be discussed. First, it's assumed that the PLC system under analysis was operating correctly at some time in the recent past, so the problems of program debugging and wiring errors that are more typical of a startup situation will not be addressed. It's also assumed that the PLC is programmed using some form of ladder-logic and not a higher-level language, and the discussion is limited to the most common I/O module types, namely those that support digital and analog inputs and outputs.

Divide and conquer

The first step in PLC troubleshooting is to decide if the problem is internal to the processor or in the I/O system. It seems to be natural to assume that most malfunctions of PLC systems are due to processor problems, but in fact the opposite is true. Experience has shown that more than 80% of all PLC malfunctions can be traced to problems with I/O modules or field equipment. Furthermore, it's relatively easy to determine whether a problem is located in the processor or in the I/O system because each type of problem has a unique signature.

Problems that can be localized to a specific I/O module or even a specific input or output device are usually external, while internal problems normally result in large groups of failures, globally erratic behavior, or even total failure of the PLC system.

Let's look first at the possible causes for internal problems.

The first thing to check is the integrity of the PLC's power and ground. Visually inspect the power and ground wiring, looking for loose, corroded, or otherwise questionable connections. The integrity of the ground can be electrically checked by measuring the voltage between the PLC ground terminal and a known ground. Using a digital meter set on the lowest scale, both the AC and DC voltages should be zero.

The power supply also can be tested electrically. If the PLC processor has an AC power source, check the input voltage; it should be within the manufacturer's recommended range. PLC processors actually operate on DC power, so that also must be checked. Measure each of the outputs of the DC power supply and check if the voltages are within the recommended ranges.

Also check the DC supplies for AC ripple. This can be done using a digital meter set on a low AC range, and the value measured should be well below the manufacturer's specifications. Excess ripple has drastic effects on the operation of the microprocessors and memory devices typically found in PLC processors.

The final power check is to measure the voltage of any batteries in the system. Battery power is often used to prevent a PLC from losing its program during power outages, and battery voltages should be within recommended values.

Other causes for erratic processor behavior are electro-magnetic interference (EMI) or radio frequency interference (RFI). Try to correlate the erratic behavior with an external EMI or RFI event like a large motor starting, arc welding in the area, lightning strikes, or even the use of handheld radio transmitters. Although they may seem harmless,

handheld radios commonly used by maintenance personnel emit powerful RF radiation and can seriously disrupt the operation of unprotected electronic equipment.

Long-term solutions to EMI and RFI problems usually involve improvements in power conditioning, grounding, and shielding.

Power, grounding, and interference problems all can cause the corruption of the PLC memory, so the next step is to verify that the program is still correct. All PLCs have some method for doing this, most of which involve comparing the program in the PLC with a backup copy on tape or disk.

Verify the program with the backup, and reload the program if problems are encountered.

Keep the program backups up-to-date and safely away from temperature extremes, high humidity, and EMI and RFI exposure to ensure they will always be usable.

Troubleshooting inputs and outputs

Now on to the more common problem of troubleshooting inputs and outputs. The primary goal of I/O troubleshooting is to find out why the internal status of the PLC (what the PLC thinks is happening) does not agree with the external situation (what is actually happening). The first thing that must be done is to determine the relationship between physical I/O modules and the I/O instructions in the PLC program. This is done by using the addressing scheme for the particular PLC that you are working on, and this scheme differs from one manufacturer to another. Somewhere in the documentation there will be an explanation of how to determine which physical I/O point a specific program address is connected to and vice versa. Once this scheme is understood, each problem can be isolated to a single I/O module and a program monitoring device (usually a handheld unit, terminal, or personal computer) can be used to check the internal status of the input or output in question.

Troubleshooting digital input modules

The function of a digital input module is to determine the ON/OFF status of a signal or signals in the external world and communicate that information to the PLC processor. Most digital input modules detect changes in voltage levels, and they are available with various AC, DC, or universal ratings, with universal modules typically accepting a fairly wide range of either AC or DC signals. A typical AC input channel is shown in Fig. 2. Note that the figure shows indicator lamps on both the power and logic sides of the circuit; many modules, however, have only one or the other of these. If only one indicator is present, it's important for troubleshooting purposes to determine where it is connected. If the threshold unit on an active input has failed, for example, a power-side indicator would be ON while a logic side indicator would be OFF.

The power to drive PLC inputs usually is not supplied by the input module, so it's important to find out where that power comes from. There are two types of inputs: Isolated and nonisolated. Troubleshooting differs depending on which type you are dealing with. Each channel on an isolated input module is electrically separated from the others and may have a different source of power. On the other hand, one side of each input channel on a non-isolated module is connected to a common reference. Both module types are shown in Fig. 3.

Determine if the power for the input in question is present, as faults in field wiring and devices can blow a fuse, trip a breaker, or cause some other power disruption. If input power is not present, determine and rectify the cause of the failure before proceeding.

If input power is present, connect a voltmeter across the input as shown in Fig. 3, actuate the input device in the field, and measure the voltage at the PLC input to determine if it changes adequately when the field device changes state. If it does not, the field device or wiring are most likely at fault. If a proper voltage change is observed, the power and/or logic indicators on the module should change when the voltage does, and the addressed location in the PLC, when monitored with the programming device, also should change state. If the indicators do not properly reflect the state of the input, replace the input module.

If the input module is working properly but the PLC still is not registering the input internally, the problem lies in the system used to communicate input information from the module to the processor. Consult the manufacturer's documentation to determine how to troubleshoot this equipment, which may include an I/O rack, back plane, communication module, and cabling.

Troubleshooting analog input modules

Instead of monitoring the on/off status of an input, analog inputs measure the actual value of a voltage or current and communicate it to the processor. Analog input modules are available in many DC voltage and current ranges, and basic troubleshooting is almost identical to that for digital modules.

First determine if the input is isolated or nonisolated, and determine the source of power and verify that it is present.

Next, change the voltage or current level generated by the field device, verify that the change is reflected at the input module terminals, and verify that the content of the address associated with the input reflects the voltage or current change.

There are two additional complications introduced by analog modules, however. First, there usually is no indication on the module to reflect the level of the input, so an external meter must be relied upon. Second is the scale problem: You must determine what range of voltage or current the module is designed to measure, and what numerical scale is associated with that range in the PLC. An input with a 1-5 VDC range may be expected to generate a change from 0 to 1000 in a PLC register, for example. Just determining if the number changes when the input does is not enough. A good approach is to adjust the external voltage or current to minimum, half scale, and maximum values, and to observe the PLC register to determine if corresponding changes have occurred. In the previous example, 1 VDC should generate 0 in the PLC register, 3 VDC should generate 500, and 5 VDC should generate 1000. If the field device cannot be easily manipulated in this manner, it can be temporarily replaced for troubleshooting purposes by a signal transmitter. The signal transmitter can be connected directly to the input module, and if the module does not respond correctly it should be replaced. If it does respond properly, the problem most likely is in the field device or wiring.

Field wiring can be tested by temporarily replacing the field device with a signal transmitter and observing the reaction at the PLC to signal changes.

Troubleshooting digital output modules

Output modules are designed to cause some change in the external world in response to an instruction in the PLC processor. Digital outputs will often be used to perform tasks like starting motors, turning on indicating lights, and energizing solenoid valves. Many different digital output module types are available, with the most common varieties being DC outputs that rely on transistors as switching devices, AC outputs that rely on triacs, and universal outputs that use relay switching. A block diagram for a typical output channel is shown in Fig. 4 (on page 28). Both power and logic indicators are shown once again, but as in the case of digital inputs, only one or the other may actually be present.

The power to drive PLC outputs, like inputs, is usually not supplied by the module, so it's important to find out where that power comes from. Once again, there are isolated and nonisolated modules, and troubleshooting differs depending on which type you are dealing with. [ILLUSTRATION FOR FIGURE 5 OMITTED]

Again, the first step in troubleshooting is to determine if the power for the output in question is present and to restore that power if it is not. There is a further complication to troubleshooting most output modules, because they typically contain a fuse to protect the output switching device [ILLUSTRATION FOR FIGURE 4 OMITTED]. Faults in field wiring and devices can blow that fuse, so its condition must be verified before proceeding. Many modules are equipped with a "fuse blown" indicator that shows which channel or module has a blown fuse. These fuses may be accessible from the front of the module, or the module may have to be removed or even disassembled in order to gain access to them.

Once power has been verified and the fuses checked, the procedure for troubleshooting digital outputs is somewhat the reverse of that for digital inputs.

First the programming device must be hooked up to the PLC, and the address that is associated with the output in question must be determined. The output then can be "forced" ON or OFF internally in the PLC, and the module can be observed for a reaction. If the indicators on the module do not reflect the forced condition, change the output module. If the module is working properly but still does not react to the forcing, the problem again lies in the communication between the processor and the module, and the manufacturer's documentation is your best source for troubleshooting information.

If the indicators are observed to be reacting to the forced state, measure the voltage across the output device [ILLUSTRATION FOR FIGURE 5 OMITTED] to see that it's changing as the state of the output changes. If the voltage is changing but the device is not reacting, the problem is in the output device.

If the voltage is not changing, the problem can most likely be found in the field wiring. If the field wiring is in doubt, it can be temporarily disconnected and a test load can be connected to the module. If the test load operates properly, the problem lies in the field wiring or field device. It's important that a test load be used as opposed to just disconnecting the field wiring. Because they leak a small amount of current in the OFF state, the voltage at most solid-state outputs will not change a large amount as the output device is switched with no load. A properly sized resistor, solenoid valve, or relay coil provides a good test load.

Troubleshooting analog output modules

Analog outputs are used to generate a variable voltage or current typically used to perform tasks like throttling the speed of a variable speed drive, adjusting the position of a control valve, or driving a panel indicator. As with input modules, analog output modules are available in many DC voltage and current ranges. Usually, there is no indication on the module to reflect the level of the input. As such, you must determine what range of voltage or current the module is designed to produce and what numerical scale is associated with that range in the PLC. An output with a 4-20 mA DC range may be expected to react to a change from 0 to 1000 in a PLC register, for example.

A good approach to testing analog outputs is to "force" the number in the PLC register associated with the output in question to minimum, half scale, and maximum values, and to measure the voltage or current generated at the output. In the previous example, a 0 in the appropriate PLC register should generate 4 mA at the output terminals, 500 should generate 12 mA, and 1000 should generate 20 mA. If the field wiring or field device are in doubt, they can be temporarily disconnected and replaced by a test load. If the proper currents or voltages are not measured at the test load, the analog output module should be replaced. A properly sized resistor, typically between 250 and 1000 ohms, is usually used as a test load in analog circuits.

SUGGESTED READING

EC&M Articles:

"Knowing the Basics of PLCs - Part 1," October 1995, p. 20.

"Servicing PLC 120V I/O Modules," November 1995, p. 24.

"Troubleshooting PLC Circuits by Testing Thumbwheel Switches," December 1995, p. 20.

"Understanding PLC Networks," January 1996, p. 22.

Books:

George L. Batten, Jr., *Programmable Controllers: Hardware, Software, and Applications*, McGraw-Hill, 1994.

Gilles Michel, *Programmable Logic Controllers: Architecture and Applications*, John Wiley and Sons, 1990.

L.A. Bryan and E.A. Bryan, *Programmable Controllers: Theory and Implementation*, Industrial Text Co., 1988.

RELATED ARTICLE: TERMS TO KNOW

Address: a numbered location (storage number) in the PLC's memory to store information.

Analog input: A varying signal supplying process change information to an analog input module.

Analog output: A varying signal transmitting process change information from an analog output module.

Central processing unit (CPU): An integrated circuit that interprets, decides, and executes instructions.

Input module: A component of a PLC that processes digital or analog signals transmitted from field devices.

Output module: A component of a PLC that controls field devices.

Program: One or more instructions or statements that accomplish a task.

Programming device: A device used to tell a PLC what to do and when it should be done.

RELATED ARTICLE: A PLC IS NOT A "BLACK BOX" . . . OR IS IT?

While working as a student engineer on a coop assignment in a hardboard siding manufacturing plant, I had the opportunity to participate in the design and installation of the first PLC-based control system ever installed in the plant. This project represented a major commitment for the company to a new technology, and my boss and I were determined to make it a success. We went to great lengths to alleviate fears of the new technology amongst our

maintenance electricians, who questioned the replacement of well-understood relay logic with an electronic "black box." We repeatedly explained the advantages of the PLC, including the simplicity of wiring, the usefulness of indicator lights on input and output modules for troubleshooting, and the similarities between ladder-logic programming and relay control.

Just when we thought the situation was under control, we received a call from the electricians assigned to the project announcing that the "black box" had arrived. Upon repeating our reasons for believing that the PLC was a useful, flexible controller and not a mysterious black box, we were told to come and take a look for ourselves. When we did, we discovered the reason for the renewed concern. Unpacking the new PLC had revealed a rectangular metal box . . . painted black.

Ryan G. Rosandich, Ph.D. is Assistant Professor, Engineering Management, University of Kansas Regional Center.

<http://m.ecmweb.com/content/troubleshooting-plcs>

Troubleshooting Tips for Industrial Control Systems

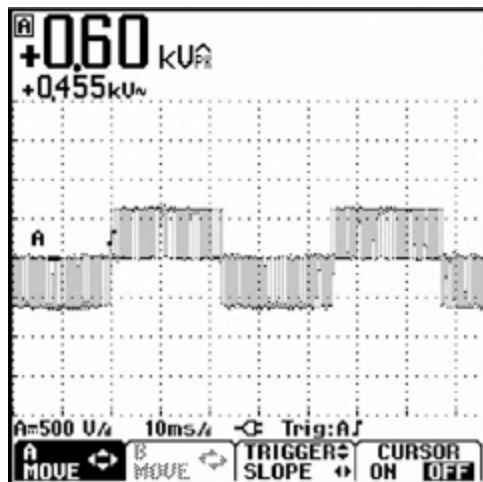
Hilton Hammond, Fluke Corp.

Feb 01, 2012

Analyzing electrical maintenance problems with a DMM and digital oscilloscope

Analyzing electrical maintenance problems with a DMM and digital oscilloscope

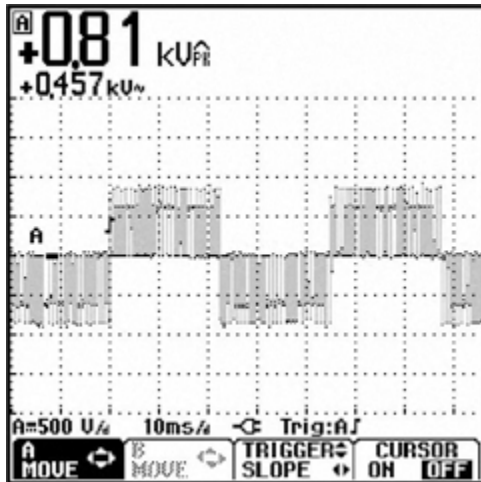
What if a variable-frequency drive (VFD) on a critical system has a fault indication, the "error" LED on a DC power supply has turned on, or a conveyor belt has started behaving erratically — most notably when the conveyor speeds up? You need answers fast, but it's unclear where the problem lies.



The first reaction for most electricians would be to reach for their digital multimeter (DMM) to start the troubleshooting process. However, that's not always the right move. As this article will demonstrate, the use of a digital oscilloscope can help speed up the troubleshooting process for industrial control systems and be more effective in pinpointing the exact nature of the problem.

Analyzing an undervoltage condition with a VFD

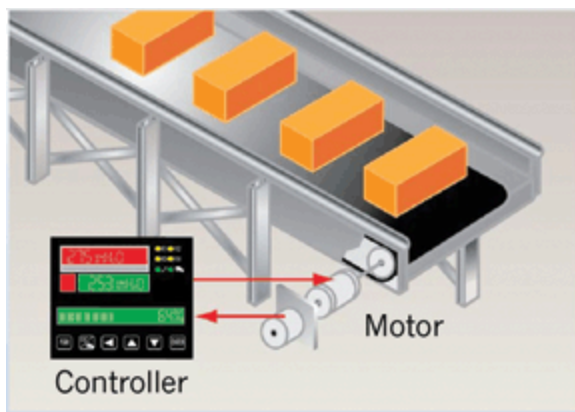
The smooth operation of many industrial processes requires that VFDs be kept in good working order. When a fault indicator comes on, it's important to diagnose and fix the problem quickly.



For example, let's say a VFD on a critical system has a fault indication. The error code on the device is "F4," which means the drive has detected an "undervoltage" condition. At this point, the drive has shut down. But is the failure in the drive, the motor, or a distorted supply voltage? You can check the line input voltage with a DMM, but depending on the type of distortion, a voltage reading alone may not reveal whether a problem exists.

Your next step should be to check the VFD DC bus voltage. Is it directly proportional to the peak of the input line voltage? Check for any distortion or error in peak amplitude of the line voltage, which can cause an over- or undervoltage error.

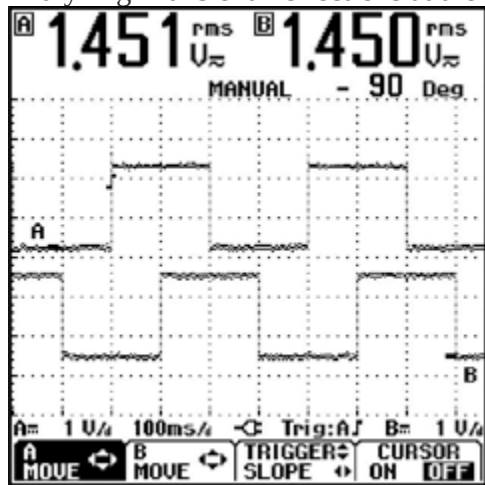
Let's say when you check the drive's DC bus voltage, you find that it's about 20% under the nominal value. That's a problem, but do you replace the drive controller, the motor, or both? You really need more information before you can proceed.



An oscilloscope can give you a clearer picture. At the power input of the drive, connect the scope to phases L1, L2, or L3 and the ground lead to neutral, and take a look at the waveform. Let's say you do this, and it reveals an almost flat-topped sine wave. What does this mean?

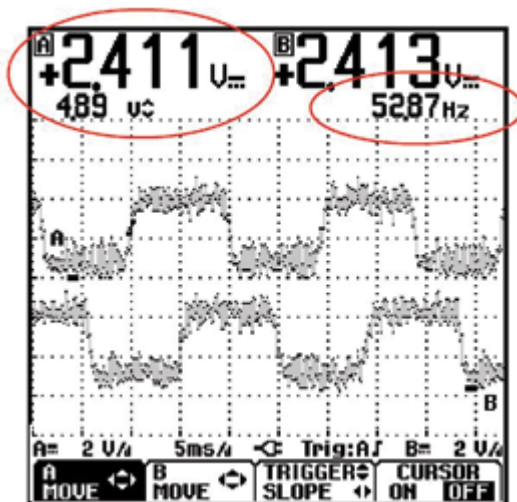
Flat-top waveforms are often caused by nonlinear loads attached to the same feeder circuit. This confirms the problem doesn't lie with the VFD or the motor. The oscilloscope graphically shows not just amplitude, but also any distortion, disturbance, or noise that may be affecting the waveform, giving you the information you need to accurately diagnose and solve problems.

Analyzing Transient Reflections at the output of a VFD



VFDs control the speed of machinery and processes by varying the width of (i.e., “modulating”) the voltage pulses they send to motors. These voltage pulses, which are essentially square waves, have fast-rising edges that can cause transient voltage reflections in the cable between the VFD and the motor. Many factors affect the size of the reflected waves, including cable length, motor load, the surge impedance of the cable and motor, and the rise time, spacing, and magnitude of the drive pulses. Over time, these reflected wave voltages can place too much stress on cable and motor insulation, which can cause either to fail.

Making voltage measurements with a DMM can reveal excessive voltage at the motor terminals when transients are present. To check the voltage, connect the multimeter at the motor terminals between phases L1, L2, or L3. Although you can use a DMM to measure the peak voltage of a pulse width modulated (PWM) signal, the limited bandwidth of the typical DMM makes measuring pulses with short rise times problematic. Therefore, the readings may not provide the complete picture of the presence — or full amplitude — of any reflections or transients. Unfortunately, even the best DMM can’t show you what the waveforms actually look like.



Oscilloscopes display a signal as a line that moves across the screen from left to right. The higher the voltage, the higher the signal appears on the oscilloscope screen. Measurements made with an oscilloscope reveal not only the presence of excessive voltage at the motor terminals, but also give you a clear picture of size, location, and duration of any voltage transients.

To take a reading with an oscilloscope, connect the scope to the motor terminals between phases T1, T2, or T3, just as you did with the multimeter. An ideal PWM signal with normal peak voltage will look like the waveform in **Fig. 1**. If

high-voltage transients are present from voltage reflections in the cable, then the waveform will look like it does in **Fig. 2**. These voltage reflections are often caused by improper cabling.

Analyzing a Rotary Encoder that has been Operating Erratically

A rotary encoder converts the position of a rotating shaft into a series of digital pulses. In this example, the pulses are monitored by a controller that uses them to control the speed and position of the conveyor system to which the encoder is attached (**Fig. 3**). The conveyor belt has started behaving erratically — most notably when the conveyor speeds up — but it's not clear if the problem lies with the rotary encoder or the controller that operates the belt.

Measurements with a DMM can reveal the voltage, frequency, and duty cycle of the signal from the rotary encoder. So you break out your DMM and make these measurements. When you're finished, you realize that all of the values appear to be normal. Because the controller's programming hasn't been modified and has worked well for years, chances are that's not the problem. Looks like it's time to break out the oscilloscope again.

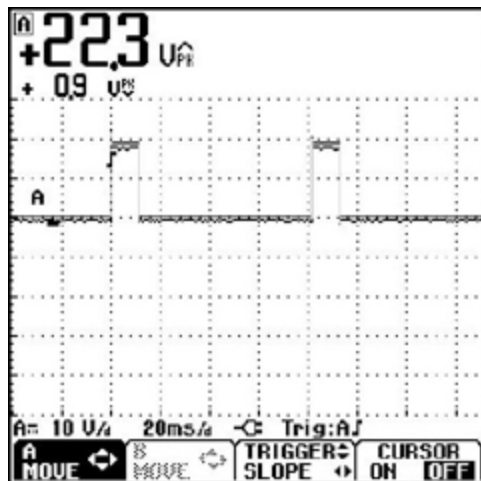
An oscilloscope connected to the output of the rotary encoder can display the signal's average voltage, peak voltage, and frequency, just like a DMM. However, as noted in the previous section, an oscilloscope can also display the voltage change over time. A good digital output from the encoder will look like the waveform shown in **Fig. 4**.

But what if the signal waveform is jagged and electrically “noisy” like the one shown in **Fig. 5**? This is a clear sign of trouble that can arise when there is inadequate shielding on a signal line. Upon inspection of the cabling, we find the cable shield of the controller wasn't fastened down properly. This ends up being a simple problem to fix.

Continue to Page 2

Analyzing the Erratic Operation of a Mechanical Proximity Switch

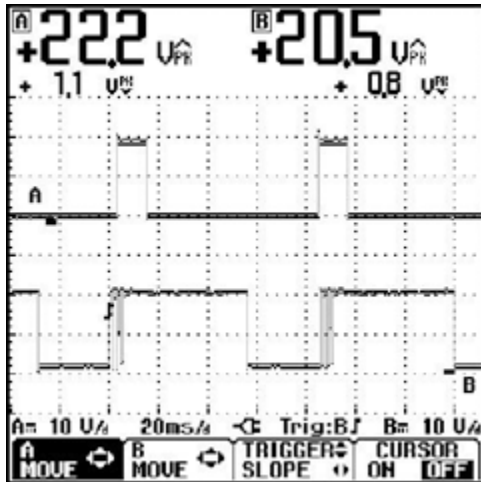
Mechanical switch contacts deteriorate over time, because they are subject to mechanical wear. In addition, arcing can cause pitting of the contact surfaces. Usually, these problems arise slowly and gradually worsen before the switch completely fails.



Troubleshooting a mechanical switch that has failed completely (i.e., a “hard fault”) can often be done with just a visual inspection or with simple measurements made with a DMM. It's the gradual or intermittent switch failures that usually cause real headaches.

In this example, a failing mechanical proximity switch is causing erratic operation in a conveyor system. After connecting a DMM to the output of the conveyor belt controller (which receives its input from the proximity switch), we see the “voltage peak max” and “voltage peak min” readings show the switch is turning on and off as expected. If you have a DMM with frequency measurement capability, you could also see the rate at which the switch is changing state. However, none of this information helps diagnose the real problem.

A modern digital oscilloscope can give you a lot of the same numerical information as a DMM, including the pulse voltage and frequency information noted above. It also allows you to see switch on/off state timing and time relationships between a source and output signal (i.e., the switch and the controller output).

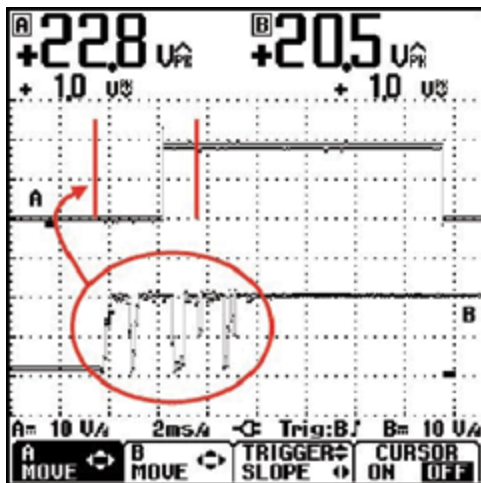


After connecting the oscilloscope to the output of the controller, it reveals nothing out of the ordinary. The pulse waveforms are well-formed and free from electrical “noise” (**Fig. 6**). We then connect the oscilloscope so it captures the signals from the proximity switch on the first channel and the output pulse from the controller on the second channel (**Fig. 7**). When we examine the result, it’s immediately clear that something is wrong. The bottom trace (the output from the proximity switch) is not stable in relation to the top trace (the controller output).

“Zooming in” on the signal (**Fig. 8**) reveals that the output from the proximity switch (lower trace) isn’t changing from OFF to ON in a single transition (red circle). Instead, faulty switch contacts are bouncing on and off for about 5 msec before the output signal stabilizes. The controller is unable to read this jagged voltage correctly, so its output varies broadly (over the time range between the red bars in the top trace). That is what’s causing the erratic behavior.

Analyzing a Power Supply with an Intermittent Failure

The DC power supply unit (PSU) is one of the most critical components in any automation or process system. If a PSU suffers a “hard failure,” you simply replace it and move on. But what if the problem is intermittent, or the problem returns a short time after you replace the failed PSU with a new one?



In this example, let’s say the “error” LED on a DC power supply has turned on. Your troubleshooting task is to determine if the problem is the power supply, the input supply voltage, or a load change on the demand side of the unit.

Using a DMM, you measure the line input voltage, and see that it looks fine. Next, you check the DC output voltage. Again, everything looks good. At this point, you decide to swap out the PSU with a known good replacement unit and hope for the best. However, when you come back 2 hr later, you see that the fault indicator LED is lit again. What should you do now?

It’s time to check the PSU’s input and output

Motor and Motor Control Troubleshooting Techniques

Mark Toland, Buzzell Electric Works

May 01, 2005

Diagnosing your problem starts with using your senses

Got a motor down? No matter how urgent the problem and motor failure can cause some very urgent problems always develop a plan of attack before you start repair work. Gather up any background material you may have on hand and then use your senses to inspect the site. Then conduct resistance, fuse, and motor tests to help pinpoint the cause of the failure. Initial inspections Spend a few minutes with

Got a motor down? No matter how urgent the problem — and motor failure can cause some very urgent problems — always develop a plan of attack before you start repair work. Gather up any background material you may have on hand and then use your senses to inspect the site. Then conduct resistance, fuse, and motor tests to help pinpoint the cause of the failure.

Initial inspections. Spend a few minutes with the operator and get as much history as you can on the failed motor. Find out if anyone else recently attempted repairs or modifications. If so, what did they do and when? Sometimes an inexperienced person can unwittingly create additional problems while trying to help. Knowing what has already been done can shorten troubleshooting time.

Armed with background information, you're ready to start your inspection process. Begin by removing power to the motor and starter — following approved and accepted lock-out/tag-out procedures — and disengaging the motor. Then after recording the motor nameplate information, turn the shaft to determine if it rotates freely. Your basic senses can also help you determine a great deal about what's wrong. Listen carefully for unusual noises (such as scraping), smell for burned insulation, and feel for excess heat. Now look at the motor starter for loose connections and hot spots indicated by discoloration. Check all fasteners, including the mounting hardware, and re-tighten. This completes your initial pre-measurement inspections. Now it's time to begin some testing.

Resistance of line and load circuits to ground. Manually engage the starter and measure the resistance through its contacts. You should read 0.09 ohms or less. Disengage the starter and inspect the contacts closely. You'll need to use your megohmmeter to ground test line and load circuits at the starter. This will effectively identify the resistance to ground of the starter, line circuits to the disconnect, and load lines to the motor and starter windings.

Generally, AC devices can safely operate at not less than two megohms to ground, and DC devices can safely operate at not less than one megohm to ground. But be careful: before ground testing, make sure you disconnect any electronic controls because they can be destroyed by misapplied high-voltage test equipment.

The resistance you see will depend on the horsepower of the motor. For example, a 50-hp motor should ideally show 0.05-ohm resistance. Measurements between phases should be roughly equivalent. Given the variety of motors in use, you may not have access to the specific engineering data for each one. However, the exact measurement value is less important than the balance between phases. While you can't expect identical readings, they should be close.

For single-phase and DC applications, the resistance measurement is suitable for identifying open circuits and for historical use.

Fuse checks. Check each fuse for continuity (one fuse per phase). Remember that a blown fuse is a symptom, not a cause. Don't assume that replacing a fuse will fix the motor. By doing so, you'll run the risk of causing even more damage and downtime.

Then check the fuse holders for spring tension. Repeated fuse replacement can “expand” the fuse holder (usually a clip) and compromise the spring tension necessary for good contact. Pinhead-sized arc damage anywhere on the clip, fuse, or surrounding areas is evidence that the fuse holder has lost tension. Inspect breakers and busses for overheating and arc damage. Inspect the cord caps (connectors) for correct circuit installation and tightness.

Line to line checks. Now you're ready to check voltage value and balance on the line (supply) side of the fuseholder. You're testing line to line, so operating voltage doesn't have a path to ground. If you find an unreasonable voltage imbalance between any two phases, that's a problem. A 5% voltage unbalance is normal and reasonable.

But don't assume the supply is healthy if you get good readings — even the weakest electrical connection will pass a voltage reading without applied load.

Motor junction box. By this point you've made preliminary determinations of where the problem exists. Now it's time to open the motor lead junction box and check the connections inside. Even if you found nothing wrong during the preliminary testing, you should still check the motor connections — many motor failures result from poorly installed wire nuts or insufficiently insulated connections grounding inside the junction box or shorting together.

If during your testing you recorded low ground readings or open readings at the load side of the starter, your next step is to test stator winding phase resistance and resistance to ground. This will help you determine whether the discrepancy is in the motor or in the line circuit. To conduct the tests, break the motor connections and test first in one direction (the motor) and then in the other (the supply).

For the motor, test the stator winding resistance phase-to-phase and phase-to-ground. If you find a phase-on-phase short, the motor needs evaluation for a rewind or for replacement. If you find a phase-to-ground short, a motor shop may be able to do an in-place repair for a large motor. However, you might end up needing to rewind or replace it.

For the supply, test first with the motor disconnect open and locked out. Use your megohmmeter to test the insulation resistance to ground. If the wiring fails that test, look for problems in the connections or in failed conductor insulation. Note: A stator phase resistance test will identify a bad motor but isn't an absolute verification of a good motor. Final load voltage and current tests verify a good motor. You never know when you'll find a motor that tests well but has a thumb-sized hole blown from the coils.

Final tests and procedures. Once you've corrected all identified problems and the motor is installed and aligned, you're almost done. Energize the motor and test the controls and overall system operation. While the motor is running, record the operating voltage and current, check the balance, and verify that your measurements are within the nameplate specifications. Then apply the sense tests again:

- Listen for unusual noises.
- Smell for smoke or hot insulation.
- Feel the motor for excessive heat or vibration.
- Look for possible obstructions.

Now engage the load. Apply the four sense tests again. Once you're sure you've addressed all of the problems with this motor, it's time to go home.

Toland is the foreman at Buzzell Electric Works in San Francisco. This article was prepared for and supported by Fluke Corp.

Sidebar: Tools You Need

- **Megohmmeter**
Minimum 500V output with high impact rating.
- **Clampmeter**
Must measure AC and DC and feature multiple jaw sizes.

- **General DC power source**

An example is a 35A bridge rectifier configured with a 120V power cord and test leads.

- **Series field DC power source**

A 9V battery will suffice.

- **Magnetic compass**

Used to quickly identify field coil polarities.

- **Digital multimeter**

Minimum 600V, CAT III true RMS rating, with low resistance (0.01 ohms or lower) featuring a cycle function and capacitance test.

<http://m.ecmweb.com/content/motor-and-motor-control-troubleshooting-techniques>